NewsAction: paper figures and tables (preview)

Every number is a cell of artifacts/NA-14/tables; captions from captions.yaml.

1 Question and decision models

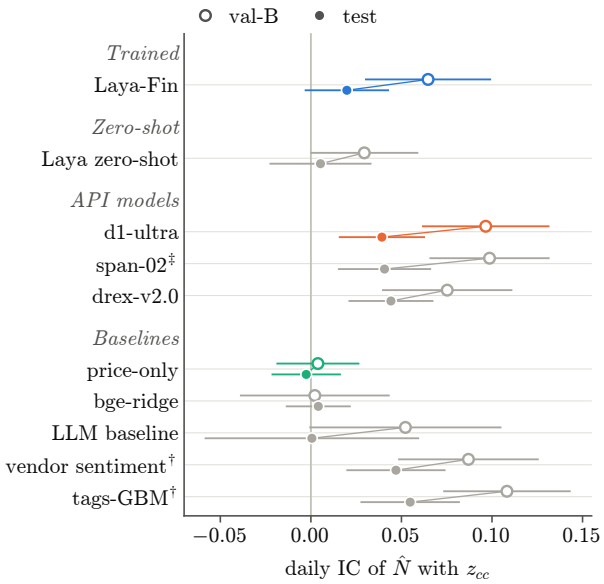


Figure 1: Every validation IC that is positive falls on test. Daily IC of the pre-open news reading $\hat{N}$ with the close-to-close move z_{cc}, on val-B (hollow) and on the test period (filled), each with its 95% interval (±1.96 Newey–West SE), for the instruments and the baselines (NA-12). Laya-Fin, the primary instrument, falls from 0.065 to 0.020; d1-ultra, the API comparator, from 0.097 to 0.039; span-02 and drex-v2.0 alike. The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3). The clean benchmarks are about zero in both periods: price-only −0.003 and bge-ridge 0.004 on test; the LLM baseline, on a 2,000-event subsample, 0.001. The highest test ICs belong to the vendor-label baselines (†: tags-GBM 0.055, vendor sentiment 0.047), whose labels were produced retrospectively: a possibly look-ahead-contaminated upper bound, not the benchmark. The anchor is uninformative (near-constant outputs); its ICs are in the appendix's model comparison. ‡ span-02: the S5 canary fired a STOP after the freeze and the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

2 Design: pre-registration, freeze, blinding, masking, ledgers

No main figure or table; this section's are in the appendix.

3 Reading the news: priced at the open

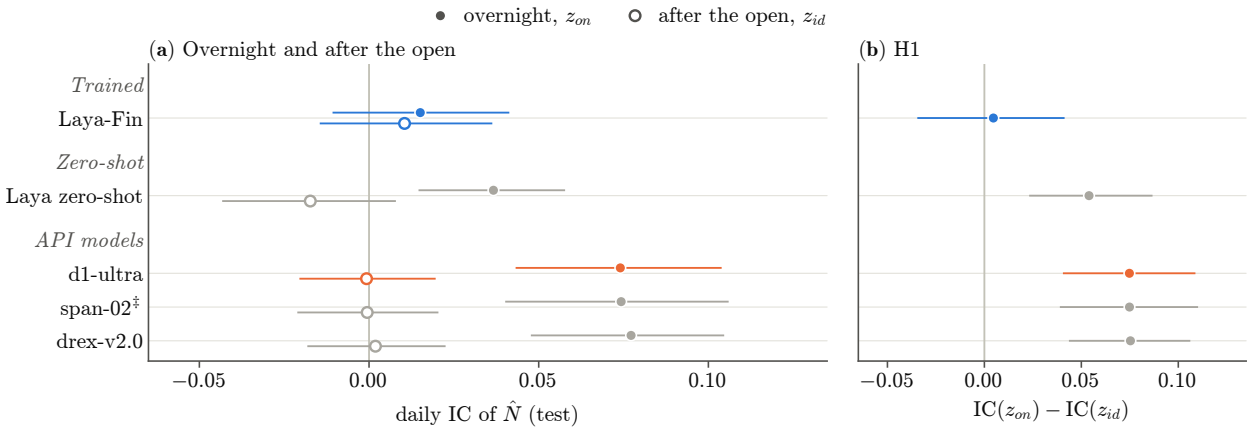


Figure 2: The API models' signal is priced at the open; no model predicts the move after it. (a) Daily IC of $\hat{N}$ with the overnight move z_{on} (previous close to open, filled) and with the move after the open z_{id} (open to close, hollow), test period, with 95% bootstrap intervals (NA-12). (b) H1, their difference, with its bootstrap interval. The API models' IC is all overnight: d1-ultra 0.074 with z_{on} and −0.001 with z_{id}, H1 = 0.075 (p <0.001), model-specific (§1.3); Laya zero-shot's H1 is 0.054. For Laya-Fin, the primary instrument, both ICs are small (0.015 and 0.010): H1 is not supported (Holm p 0.853) and so is H1b, the IC with z_{id} (Holm p 0.853), although on val-B its IC with z_{id} had been 0.054 (t 2.89; Table 2). The anchor is uninformative (near-constant outputs) and is in Table 1. ‡ span-02: the S5 canary fired a STOP after the freeze and the operator chose to proceed (PREREG §15).

Table 1: Primary hypotheses on the test period: none is supported for the primary instrument, Laya-Fin.

	Laya-Fin (primary)				Anchor		
Hypothesis	Estimate	95% CI	p	Holm p	Estimate	p	Verdict (§1.3)
H1	0.005	[−0.035, 0.041]	0.409	0.853	0.004	0.406	not supported
H1b	0.010	[−0.014, 0.037]	0.213	0.853	0.002	0.427	not supported
H2	−0.044	—	0.616	0.853	0.015	0.294	not supported
H3	0.0038	[−0.0008, 0.0085]	0.054	0.269	0.0013	0.017	not supported
H4	0.0012	[−0.0017, 0.0042]	0.216	0.853	−0.0004	0.773	not supported

The tests: H1, $\mathrm{IC}(\hat{N}, z_{on}) - \mathrm{IC}(\hat{N}, z_{id}) > 0$; H1b, $\mathrm{IC}(\hat{N}, z_{id}) > 0$; H2, $b > 0$ and $c < 0$ at D2; H3, ΔIC of J for $y_2 > 0$; H4, ΔR^2 of the news for $|z_{cc}| > 0$.

Test period 2026-01-02 to 2026-07-31, events outside the embargo. Estimates in each hypothesis' direction; p one-sided; 95% CIs and p from a stationary block bootstrap over sessions (block 5, 10,000 draws); Holm over H1, H1b, H2, H3 and H4 at 0.05.

H2: the estimate is $\min(b, -c)$ and $p = \max(p_b, p_c)$ (both must hold), so it has no single interval (—); b and c with their standard errors are in Table 3. Estimates and CIs to three decimals, H3's and H4's to four, since those increments are an order smaller.

Verdict (§1.3): general if Laya-Fin's Holm $p < 0.05$ with the predicted sign and the anchor's own $p < 0.05$ with the same sign; model-specific if only Laya-Fin's holds; otherwise not supported. The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls. Its p is its own, not Holm-adjusted.

Significance: no stars are used; p is one-sided unless stated.

Table 2: Daily IC by target: the API models' signal is in the overnight move, and no model predicts the move after the open.

	Val-B			Test		
Model	z_{cc}	z_{on}	z_{id}	z_{cc}	z_{on}	z_{id}
Trained and zero-shot						
Laya-Fin	0.065	0.045	0.054	0.020	0.015	0.010
	(3.64)	(2.93)	(2.89)	(1.67)	(1.08)	(0.82)
anchor	−0.020	−0.002	−0.011	0.008	0.006	0.002
	(−1.52)	(−0.16)	(−0.75)	(0.83)	(0.50)	(0.18)
Laya zero-shot	0.029	0.047	0.002	0.005	0.037	−0.017
	(1.94)	(2.55)	(0.12)	(0.37)	(3.22)	(−1.25)
API models (model-specific)						
d1-ultra	0.097	—	—	0.039	0.074	−0.001
	(5.38)	—	—	(3.23)	(4.74)	(−0.07)
span-02‡	0.099	—	—	0.041	0.074	−0.001
	(5.83)	—	—	(3.10)	(4.49)	(−0.05)
drex-v2.0	0.075	—	—	0.044	0.077	0.002
	(4.10)	—	—	(3.70)	(5.34)	(0.18)

Mean daily Spearman IC of the pre-open news reading $\hat{N}$ with the close-to-close move z_{cc}, the overnight move z_{on} (previous close to open) and the move after the open z_{id} (open to close), each in typical daily moves, to three decimals; Newey–West t in parentheses, to two; D0 events outside the embargo (val-B 2025-10-01 to 2025-12-31; test 2026-01-02 to 2026-07-31).

Val-B ICs with z_{on} and z_{id} come from NA-11's run of the frozen spec on val-B before the freeze, which covered the trained and zero-shot Laya models only (—: not computed).

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls. The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3). ‡span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

4 Over- and under-reaction, and convergence to the news-implied value

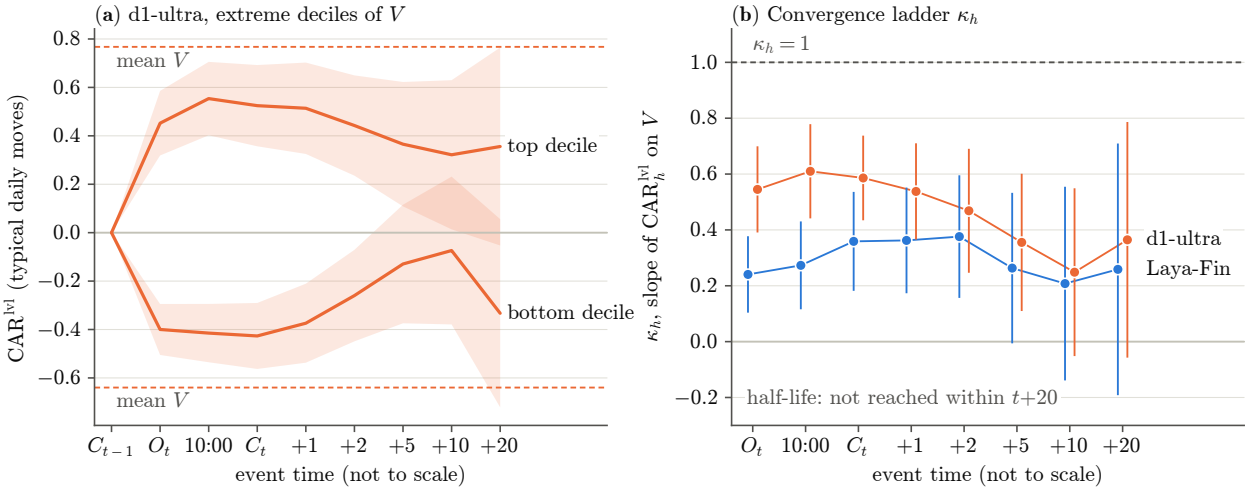


Figure 3: Prices move part of the way to the model's value on day t, then partly fade. Convergence to the news-implied value V (PREREG A1), test period. Event time is not to scale: O_t is the open, 10:00 the D1 decision point, C_t the close of day t and $+k$ the close of session $t+k$. (a) d1-ultra's top and bottom deciles of V: $\mathrm{CAR}^{\mathrm{lvl}}$, the abnormal return from the previous close in typical daily moves, with SE bands (±1.96 event-level SE; descriptive), against each decile's mean V (dashed). The top decile (mean V 0.77) stands at +0.45 at the open, +0.52 at the close and +0.36 at t+20; the bottom decile (mean V −0.64) at −0.40, −0.43 and −0.33. Before the event prices drift in V's direction (not drawn): five sessions earlier the top decile stood −1.19 and the bottom decile +1.54 from the previous close, so part of V reflects momentum. The other deciles, and Laya-Fin's, are in Appendix Figure A7. (b) The convergence ladder κ_h, the slope of $\mathrm{CAR}_h^{\mathrm{lvl}}$ on V with day × tier fixed effects, with 95% intervals from two-way clustered SEs, against full convergence at $\kappa_h = 1$. κ_h peaks on day t or just after and is far from one: Laya-Fin 0.36 at the close and 0.38 at t+2, d1-ultra 0.61 at 10:00; it fades by t+10 (0.21 and 0.25). Laya-Fin's increase from the open to t+20 is +0.02 (p 0.467). The half-life of the gap $U = V - G^{\mathrm{lvl}}$, the first rung at which λ_h reaches 0.5, is not reached within t+20 for any model (∞; Table 4). The convergence tests (H2-C) are not supported after Benjamini–Hochberg. The anchor is uninformative (near-constant outputs) and appears in Figure A7. d1-ultra fails the §8.3 screens and is model-specific (§1.3).

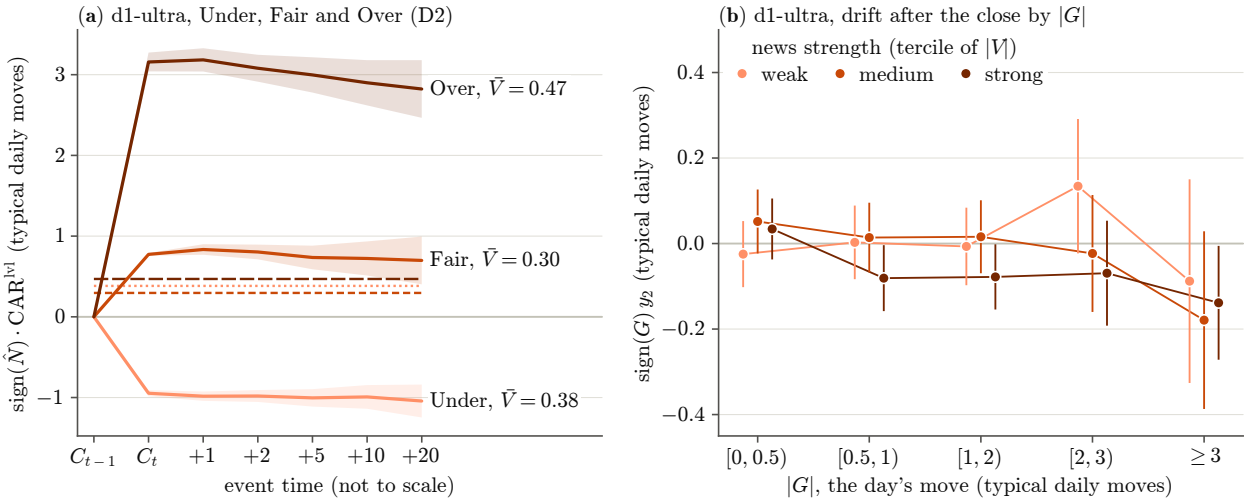


Figure 4: No small-move underreaction; large moves reverse slightly. Over- and under-reaction, test period (NA-12 figure data). (a) d1-ultra's D2 events by RG class, pooled over the tiers and signed in the news' direction (sign($\hat{N}$) · CAR^{lvl}), with SE bands (±1.96 event-level SE), against each class's mean signed V ($\bar{V}$ in the labels; dashed in the class's colour, dotted for Under, short dashes for Fair and long dashes for Over, since the three lie close). The 5,021 Under events moved against the news on day t (−0.95) and never catch up (−1.04 at t+20, against a target of 0.38); the 1,783 Over events give back part of their move, from +3.16 at the close to +2.82. Laya-Fin has only 146 D2 events with $|\hat{N}| \geq 0.5$, too few to draw. (b) d1-ultra's drift after the close, sign(G) y_2 (next open to the close of t+5, in the direction of the day's move G), by bin of $|G|$, one line per tercile of news strength, with 95% intervals. The terciles are on $|V|$, Call A's value on the day's news (NA-12's figure data), not on $|\hat{N}|$. Moves with $|G| \geq 3$ reverse slightly in every tercile (strong −0.14, medium −0.18, weak −0.09); below that the means are small and of both signs, except the strong tercile, which also reverses slightly at 0.5 to 2 (−0.08 and −0.08). There is no small-move underreaction. Laya-Fin's panel is Appendix Figure A8. The pre-registered regression H2 finds, for Laya-Fin, b = −0.04 (p 0.616) and c = −0.015 (p 0.110): not supported (Table 3). d1-ultra is model-specific (§1.3).

Table 3: The over/under regression: the reversal coefficient c has the predicted sign but is not significant, and the news coefficient b is not positive at the close.

Decision point	b	SE	p_b	c	SE	p_c	Events
Laya-Fin (primary)							
D2, close (H2)	−0.044	(0.149)	0.616	−0.015	(0.012)	0.110	17,188
D1, 10:00	0.229	(0.148)	0.062	0.013	(0.012)	0.861	13,203
D0, open	0.278	(0.212)	0.096	—	—	—	13,439
Anchor (uninformative)							
D2, close (H2)	0.748	(1.377)	0.294	−0.015	(0.012)	0.111	17,188
D1, 10:00	1.964	(2.482)	0.215	0.014	(0.012)	0.874	13,203
D0, open	−4.510	(3.236)	0.917	—	—	—	13,439
d1-ultra (model-specific)							
D2, close (H2)	−0.076	(0.031)	0.991	−0.011	(0.012)	0.171	17,188
D1, 10:00	−0.018	(0.032)	0.711	0.015	(0.012)	0.884	13,203
D0, open	0.022	(0.037)	0.277	—	—	—	13,439

$y = a + b\,\hat{N} + c\,G$ with day × tier fixed effects, on the test period: at D2 $y = y_2$ (next open to t+5), $\hat{N}$ from the day's news and G the day's move; at D1 the move after 10:00 on $\hat{N}$ from the pre-open news and the move to 10:00; at D0 $y = z_{id}$ and there is no G, so no c (—). Two-way clustered (session, firm) standard errors in parentheses; p_b tests $b > 0$ and p_c tests $c < 0$, one-sided. H2 is the D2 row of Laya-Fin.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls. The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Significance: no stars are used; p is one-sided unless stated.

Table 4: Convergence to the news-implied value V: prices reach part of V on day t, then partly fade; the gap's half-life is not reached within t+20.

	O_t	10:00	C_t	t+1	t+2	t+5	t+10	t+20	Half-life
Panel A: convergence ladder κ_h (D0 events)									
Laya-Fin	0.24	0.27	0.36	0.36	0.38	0.26	0.21	0.26	—
	(0.07)	(0.08)	(0.09)	(0.10)	(0.11)	(0.14)	(0.18)	(0.23)	—
d1-ultra	0.55	0.61	0.59	0.54	0.47	0.36	0.25	0.36	—
	(0.08)	(0.09)	(0.08)	(0.09)	(0.11)	(0.13)	(0.15)	(0.22)	—
anchor	0.08	−0.02	−0.04	−0.04	−0.03	−0.06	−0.03	0.08	—
	(0.08)	(0.08)	(0.07)	(0.09)	(0.09)	(0.10)	(0.11)	(0.16)	—
Panel B: gap closure λ_h and the half-life									
Laya-Fin, D2	—	—	—	−0.007	—	0.026	—	0.050	∞
	—	—	—	(0.012)	—	(0.026)	—	(0.051)	—
Laya-Fin, D1	—	—	−0.011	−0.024	—	−0.002	—	−0.013	∞
	—	—	(0.013)	(0.020)	—	(0.034)	—	(0.062)	—
d1-ultra, D2	—	—	—	−0.009	—	0.022	—	0.049	∞
	—	—	—	(0.012)	—	(0.026)	—	(0.051)	—
d1-ultra, D1	—	—	−0.013	−0.028	—	−0.012	—	−0.020	∞
	—	—	(0.013)	(0.020)	—	(0.034)	—	(0.062)	—
anchor, D2	—	—	—	−0.007	—	0.027	—	0.052	∞
	—	—	—	(0.012)	—	(0.027)	—	(0.051)	—
anchor, D1	—	—	−0.012	−0.025	—	−0.003	—	−0.013	∞
	—	—	(0.013)	(0.020)	—	(0.034)	—	(0.062)	—

Panel A: κ_h is the slope of $\mathrm{CAR}_h^{\mathrm{lvl}}$, the abnormal return from the previous close to rung h in typical daily moves, on V with day × tier fixed effects; $\kappa_h = 1$ is full convergence. Panel B: λ_h is the share of the gap $U = V - G^{\mathrm{lvl}}$ closed by rung h, at D2 (from the close) and D1 (from 10:00); the half-life is the first rung with $\lambda_h \geq 0.5$ (∞: not reached within t+20). Two-way clustered standard errors in parentheses; t+k is the close of session $t + k$. —: not estimated; λ_h is reported at t+1, t+5 and t+20 from D2 and also at C_t from D1, and the half-life belongs to λ_h alone. κ_h to two decimals, λ_h to three, since it is an order smaller.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls. The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

5 Explaining realized moves

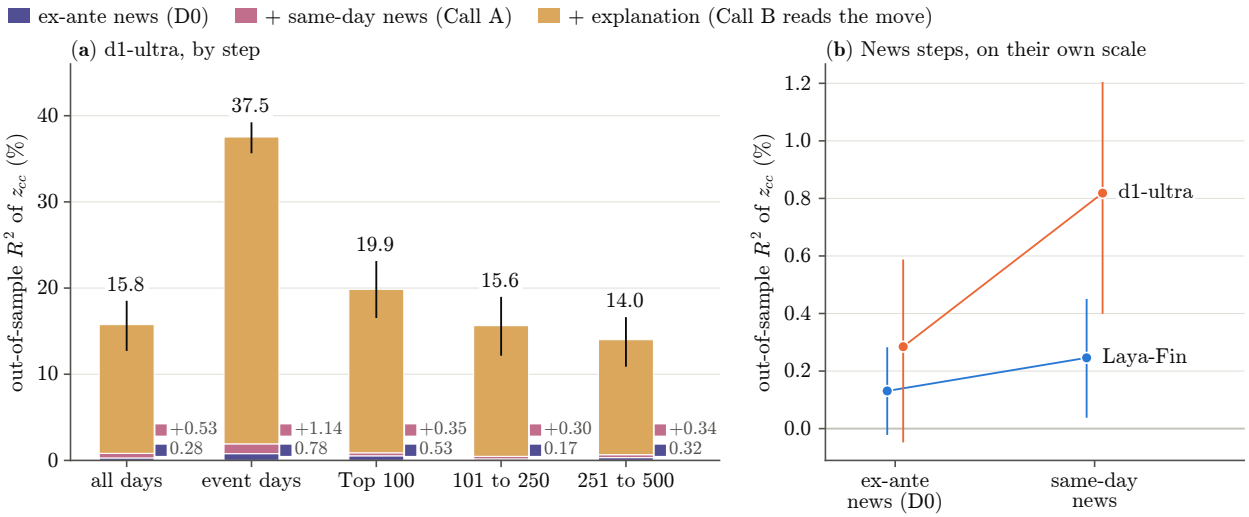


Figure 5: The news explains little of a stock's own move; reading the move explains more. Explaining realized moves at the firm level (PREREG A3, after Kelly–Su–Zhang), test period (NA-12), as out-of-sample R^2 in % of the stock-specific variance. (a) d1-ultra's R^2 of the stock-specific move z_{cc} by cut, stacked by step in the order of the legend: the ex-ante (D0) news, + the same-day news (Call A), + the explanation (Call B reads the realized move, so this step is not a prediction). The total stands above each bar with its 95% bootstrap interval: 15.8% of all covered member-days and 37.5% of event days. The two news steps are too thin to read, so their values stand beside each bar after a swatch of their colour: the ex-ante step's R^2, then the same-day step's increment. For comparison, the market and the sector, Roll's first step, explain 25.3% of total daily variance, a different base. (b) The news steps on their own scale, the zoom that makes them readable: the R^2 after the ex-ante and after the same-day news, with 95% bootstrap intervals. Laya-Fin's same-day news explains 0.25% of the stock-specific variance, the only secondary test that survives Benjamini–Hochberg (H5, q 0.058), and 0.13% was knowable before the open; d1-ultra's steps reach 0.82% and 0.28%. The trained models give no explanation answer; the anchor is uninformative (near-constant outputs; Table 5). d1-ultra is model-specific (§1.3).

Table 5: Explaining realized moves: same-day news explains a small share of the stock-specific variance (H5, supported for Laya-Fin); reading the move explains far more.

	Laya-Fin		d1-ultra		Anchor	
Step	R^2	ΔR^2	R^2	ΔR^2	R^2	ΔR^2
Panel A: the explanation ladder, out-of-sample R^2 of z_{cc} (%)						
+ ex-ante news (D0)	0.13	—	0.28	—	0.01	—
+ same-day news (Call A)	0.25	0.12	0.82	0.53	0.03	0.02
+ explanation (Call B reads the move)	—	—	15.76	14.94	—	—
share knowable before the open (%)	53.17	—	34.78	—	35.63	—
Panel B: the secondary test H5						
H5: same-day news step, $R^2 > 0$	0.25	—	0.82	—	0.03	—
one-sided p	0.010	—	<0.001	—	0.072	—
BH q	0.058	—	<0.001	—	0.358	—

Covered member-days of the test period; out-of-sample R^2 of the stock-specific move z_{cc} (the market and sector removed) by cumulative step, in %; ΔR^2 is the step's increment. Roll's first step, the market and the sector, explains 25.3% of total daily variance, a different base. The explanation step needs Call B's explanation answer, which only d1-ultra gives; it reads the realized move, so it is not a prediction. H5 is the only secondary test that survives Benjamini–Hochberg (q 0.10) for Laya-Fin. —: not applicable (the first step has no increment; only d1-ultra answers the explanation question; the share and Panel B have one value per model). R^2, ΔR^2 and the share in % to two decimals; p and q to three.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls. The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

6 Economic significance

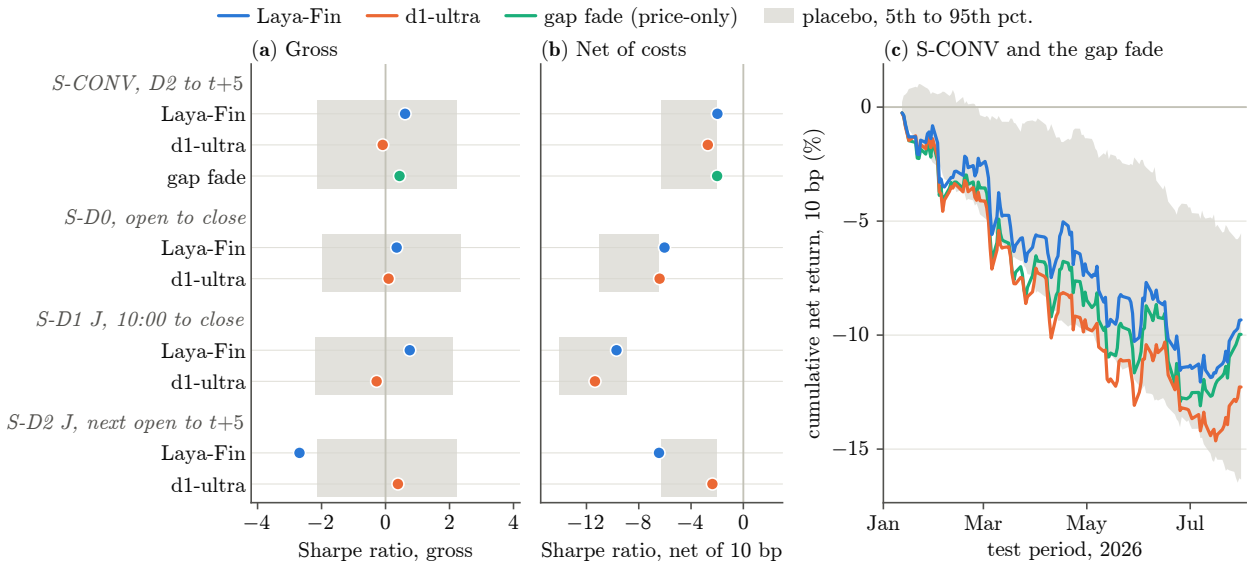


Figure 6: No pre-registered book survives costs and the deflated Sharpe; S-CONV is effectively the gap fade. Strategies on the test period (NA-13). (a, b) Annualized Sharpe ratio of each main book, gross and net of 10 bp per side on both legs and the SPY hedge, at the exit of its hypothesis' horizon; the gray box is the 5th to 95th percentile of the book's 100 random-signal books; (b) has its own scale. Gross, every book lies inside its placebo range (S-CONV's: −2.12 to +2.23); net, every one is negative (Laya-Fin's S-CONV −1.98, gross +0.61; S-D0 −6.03; S-D1 J −9.70). The one-day books lose most to costs: they turn over 4.36 times their capital a day against 0.90 for S-CONV. (c) S-CONV's cumulative net return at 10 bp against the price-only gap fade, with the placebo's 5th to 95th percentile band. S-CONV is effectively the gap fade, because V is small next to G ($U \approx -G^{\text{lvl}}$): the two share 92% of their positions, 99.8% of them on the same side, and their daily returns correlate 0.96. The only books positive net are Laya-Fin's RG books, with 6 to 12 positions over the whole test period, and the anchor's gated S-D2 J held to t+20 (the anchor is uninformative, with near-constant outputs); none clears its placebo or the deflated Sharpe. The highest gross Sharpe, the gated S-D0 held to the close of t+1 (1.88), reaches placebo percentile 93; with the 545 trials in the ledger when NA-13 computed it, the largest deflated-Sharpe probability of any test book is 0.024. Every book is in the appendix's strategy table, the other books' paths and the tier sub-books in Figure A12. Borrow costs are not modelled.

Table 6: Strategies on the test period: no pre-registered book survives costs and the deflated Sharpe ratio.

		Sharpe ratio					
Book	Holding	Gross	Net	Turnover	Break-even (bp)	Placebo pct.	DSR
Laya-Fin (primary)							
S-CONV	D2 to t+5	0.61	−1.98	0.90	2.4	68	<0.001
S-D0	open to close	0.35	−6.03	4.36	0.5	64	<0.001
S-D0, gated	open to t+1	1.88	−0.38	2.31	8.3	93	<0.001
S-D1 J	10:00 to close	0.76	−9.70	4.36	0.7	71	<0.001
S-D2 J	next open to t+5	−2.69	−6.45	0.83	−7.1	3	<0.001
S-D2 RG	next open to t+5	1.69	1.50	0.05	85.1	92	0.024
d1-ultra (model-specific)							
S-CONV	D2 to t+5	−0.09	−2.72	0.90	−0.3	48	<0.001
S-D0	open to close	0.10	−6.41	4.41	0.1	58	<0.001
S-D1 J	10:00 to close	−0.28	−11.35	4.51	−0.3	40	<0.001
S-D2 J	next open to t+5	0.39	−2.37	0.85	1.4	64	<0.001
Benchmarks							
gap fade (price-only)	D2 to t+5	0.44	−2.01	0.90	1.8	66	<0.001
vendor sentiment†	open to close	0.27	−6.24	4.36	0.4	62	<0.001

Annualized Sharpe ratios ($\sqrt{252}$), gross and net of 10 bp per side on both legs and the SPY hedge; no borrow cost. Turnover: dollars traded a day per dollar of capital. Break-even: the cost per side, in basis points, at which the mean net return is zero (gross mean ÷ turnover). Placebo: the percentile of the gross Sharpe among 100 random-signal books on the same events. DSR: the deflated-Sharpe probability at 10 bp with the 545 trials in the ledger when NA-13 computed it. Each family is shown at its hypothesis' horizon; every book is in Table A25. S-CONV ranks by the gap $U = V - G^{\text{lvl}}$; S-D0 by $\tilde{N}$; S-D1 and S-D2 by the judgment J signed by the move; RG books trade the cases the model calls a misreaction (one name a side).

†Vendor sentiment: its labels were produced retrospectively, so its books are a possibly look-ahead-contaminated upper bound, not a benchmark; the clean benchmarks are the price-only gap fade and the random-signal placebo. The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

7 Memorisation and validation-to-test decay

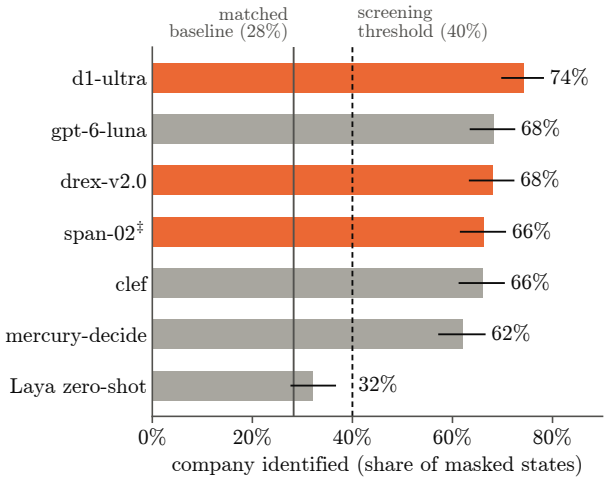


Figure 7: The API models identify the masked company; Laya zero-shot hardly does. Company identification on masked val-A states with activity-matched distractors (NA-11; the §8.3 screen): the share of states in which each model names the company among four candidates, as a bar, sorted, with its Wilson 95% interval and its value at the bar's end, against the matched baseline (a no-reading guess, 28%; solid) and the screening threshold (40%; dashed). The production comparators (d1-ultra, span-02, drex-v2.0) are in the comparator colour, the other API models and Laya zero-shot in gray. The API models name it in 62% to 74% of states, so they fail the screen and are model-specific (§1.3); Laya zero-shot, Laya-Fin's base, names it in 32%. Every model was released after the sample, which ends on 2026-07-31: Laya's open weights on 2026-09-18 (49 days later; the operator's date) and the API models between 2026-10-12 and 2026-10-16 (NA-05 catalog). Identifying a company is not the same as remembering its returns, and the test does not settle that. ‡ span-02: the S5 canary fired a STOP after the freeze and the operator chose to proceed (PREREG §15).

A Decision models

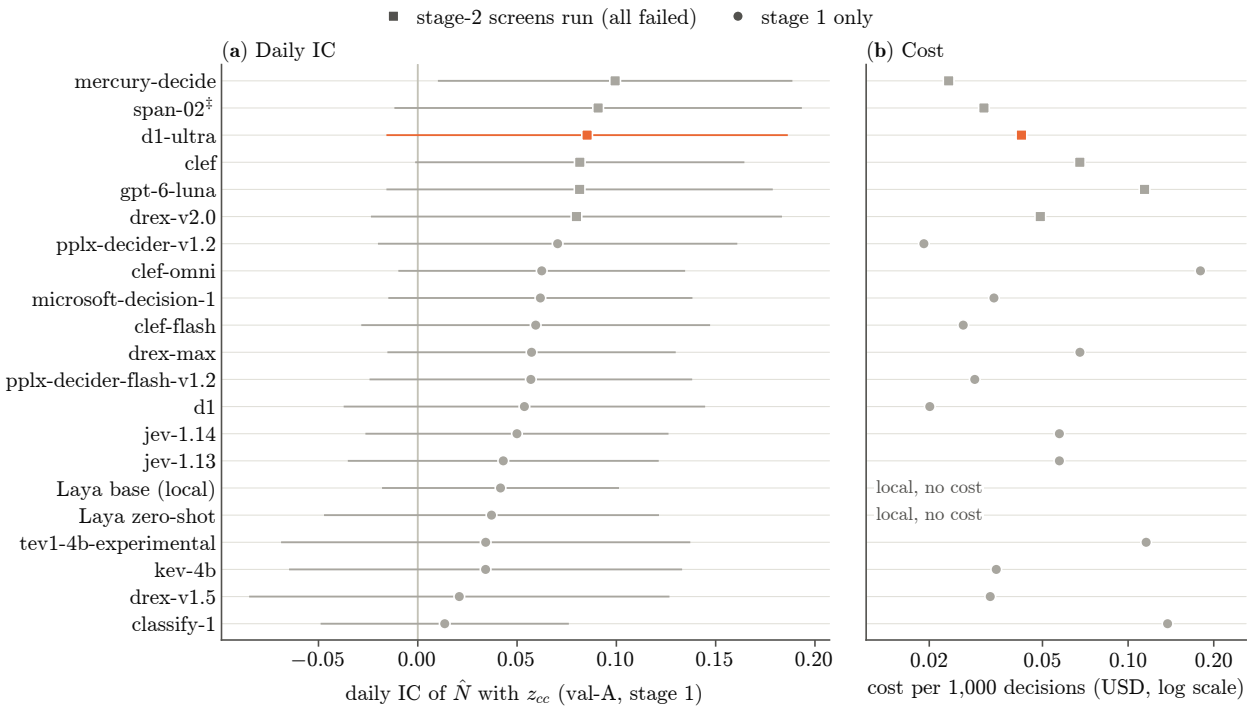


Figure A1: The zero-shot tournament. The zero-shot tournament on NA-06's val-A stage-1 sample: (a) the daily IC of Call A's $\hat{N}$ with z_{cc} and its 95% bootstrap interval, models sorted by IC; (b) the cost per 1,000 decisions (log scale; the local Laya models cost nothing). Every model is in the model confidence set; the highest IC is mercury-decide's (0.099) and d1-ultra's is 0.085. The six models that ran the stage-2 screens (squares) all failed them (Figure 7). ‡ span-02: the S5 canary fired a STOP after the freeze and the operator chose to proceed (PREREG §15).

Table A1: Model comparison: the daily IC with z_{cc} on val-B and test, the cost and the latency.

Model	IC val-B	t	IC test	t	Events	USD/1,000	p50 (ms)
Laya-Fin	0.065	3.64	0.020	1.67	13,748	0.000	—
anchor	−0.020	−1.52	0.008	0.83	13,748	0.000	—
Laya zero-shot	0.029	1.94	0.005	0.37	13,748	0.000	536
d1-ultra	0.097	5.38	0.039	3.23	13,748	0.063	288
span-02[‡]	0.099	5.83	0.041	3.10	13,748	0.028	2,262
drex-v2.0	0.075	4.10	0.044	3.70	13,748	0.045	680
vendor sentiment[†]	0.087	4.39	0.047	3.36	13,748	—	—
bge-ridge	0.002	0.10	0.004	0.45	13,748	—	—
tags-GBM[†]	0.108	6.03	0.055	3.91	13,748	—	—
LLM baseline	0.052	1.93	0.001	0.02	1,977	—	—
price-only	0.004	0.33	−0.003	−0.26	13,748	—	—
Laya-Fin, LLM's 2,000-event subsample	—	—	−0.012	−0.37	1,977	—	—
LLM baseline, its subsample	—	—	0.001	0.02	1,977	—	—
price-only (D2, IC with y_2)	—	—	0.012	1.18	17,188	—	—

—: the source table has no value for the row; the statistic does not apply to it or was not computed.
[†]Vendor-label baseline (vendor sentiment, tags-GBM): the labels were produced retrospectively, so its IC is a possibly look-ahead-contaminated upper bound, not the benchmark; the clean benchmarks are bge-ridge, price-only, the LLM baseline and the random-signal placebo.
[‡]span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A2: The zero-shot tournament on NA-06's val-A stage-1 sample: no model is excluded by the model confidence set.

Model	IC	95% CI	USD per 1,000
mercury-decide	0.099	[0.010, 0.189]	0.023
span-02‡	0.091	[−0.012, 0.193]	0.031
d1-ultra	0.085	[−0.016, 0.186]	0.042
clef	0.082	[−0.001, 0.164]	0.068
gpt-6-luna	0.081	[−0.016, 0.179]	0.114
drex-v2.0	0.080	[−0.024, 0.183]	0.049
pplx-decider-v1.2	0.070	[−0.020, 0.161]	0.019
clef-omni	0.062	[−0.010, 0.135]	0.180
microsoft-decision-1	0.062	[−0.015, 0.138]	0.034
clef-flash	0.059	[−0.028, 0.147]	0.026
drex-max	0.057	[−0.015, 0.130]	0.068
pplx-decider-flash-v1.2	0.057	[−0.024, 0.138]	0.029
d1	0.054	[−0.037, 0.145]	0.020
jev-1.14	0.050	[−0.026, 0.126]	0.057
jev-1.13	0.043	[−0.035, 0.121]	0.057
Laya base (local)	0.042	[−0.018, 0.101]	0.000
Laya zero-shot	0.037	[−0.047, 0.121]	0.000
tev1-4b-experimental	0.034	[−0.069, 0.137]	0.116
kev-4b	0.034	[−0.065, 0.133]	0.034
drex-v1.5	0.021	[−0.085, 0.127]	0.033
classify-1	0.014	[−0.049, 0.076]	0.138

Daily IC of Call A's $\hat{N}$ with z_{cc} on NA-06's val-A stage-1 sample, 95% bootstrap CI. Every model is in the model confidence set; the six best ran the stage-2 screens and failed them (§8.3).

‡span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

B Design

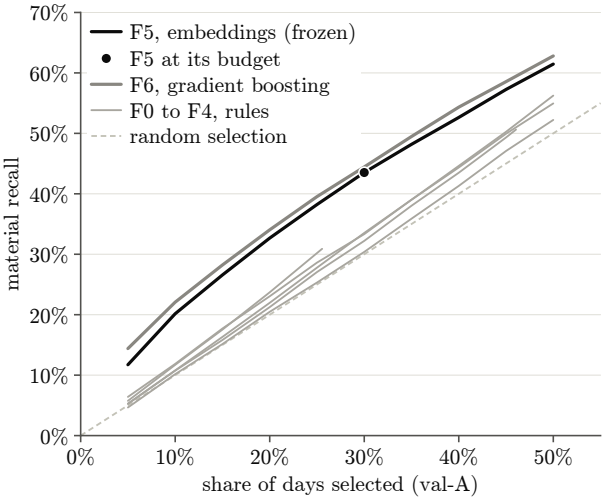


Figure A2: An embedding filter selects material news better than rule filters. Material recall against the share of val-A days selected, per candidate importance filter (NA-03 frontier, budgets from 5% to 50% of days). The frozen filter F5, a logistic regression on bge-small embeddings, selects 30% of days at its budget with material recall 0.435. F6 (gradient boosting) reaches 0.444 at the same budget, a tie within one point that goes to the filter with fewer inputs (§6.8). The rule filters F1 to F4 recall 0.309 to 0.335 there, close to random selection (dashed); F0 keeps every clean article.

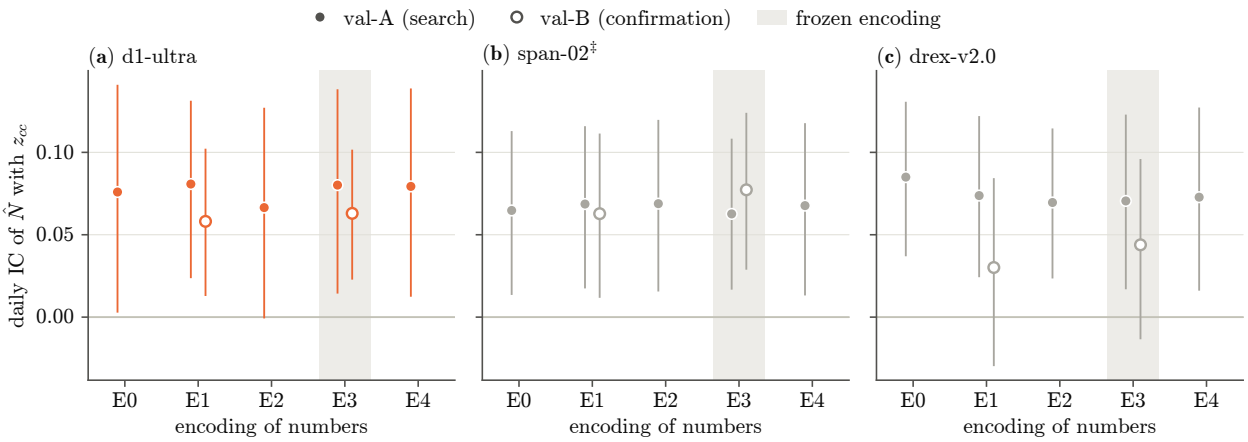


Figure A3: How numbers are written in the state hardly matters. The encoding of numbers in states (NA-07; §8.4): daily IC of $\hat{N}$ with z_{cc} by encoding E0 to E4 on NA-07's val-A sample (filled, 95% bootstrap intervals) and the val-B confirmation of E1 and E3 (hollow), for the three API comparators; the gray column is the frozen encoding. E1 had the best mean val-A rank (1.67) but fell short of E3 on val-B by more than one standard error, so NA-07's rule kept E3 (d1-ultra on val-B: 0.058 with E1, 0.063 with E3). ‡ span-02: the S5 canary fired a STOP after the freeze and the operator chose to proceed (PREREG §15).

Table A3: The importance filters on val-A: material recall at each budget.

Filter, budget	Selected	Threshold	Recall (material)	Recall (delayed)	Precision
F0, 0.05	0.050	1.000	0.046	0.043	0.196
F0, 0.1	0.100	1.000	0.102	0.090	0.214
F0, 0.15	0.150	1.000	0.151	0.138	0.213
F0, 0.2	0.200	1.000	0.205	0.185	0.216
F0, 0.25	0.250	1.000	0.253	0.233	0.214
F0, 0.3	0.300	1.000	0.304	0.288	0.214
F0, 0.35	0.350	1.000	0.359	0.333	0.216
F0, 0.4	0.400	1.000	0.413	0.384	0.218
F0, 0.45	0.450	1.000	0.471	0.435	0.221
F0, 0.5	0.500	1.000	0.522	0.487	0.220
F1, 0.05	0.050	1.000	0.052	0.044	0.222
F1, 0.1	0.100	1.000	0.107	0.091	0.227
F1, 0.15	0.150	1.000	0.158	0.138	0.222
F1, 0.2	0.200	1.000	0.213	0.186	0.224
F1, 0.25	0.250	1.000	0.271	0.236	0.228
F1, 0.3	0.300	1.000	0.322	0.286	0.227
F1, 0.35	0.350	1.000	0.380	0.340	0.229
F1, 0.4	0.400	1.000	0.435	0.394	0.229
F1, 0.45	0.450	1.000	0.493	0.446	0.231
F1, 0.5	0.461	1.000	0.506	0.462	0.232
F2, 0.05	0.050	1.000	0.057	0.043	0.243
F2, 0.1	0.100	1.000	0.117	0.095	0.247
F2, 0.15	0.150	1.000	0.175	0.148	0.247
F2, 0.2	0.200	1.000	0.237	0.208	0.250
F2, 0.25	0.250	1.000	0.302	0.259	0.255
F2, 0.3	0.255	1.000	0.309	0.267	0.255
F2, 0.35	0.255	1.000	0.309	0.267	0.255
F2, 0.4	0.255	1.000	0.309	0.267	0.255
F2, 0.45	0.255	1.000	0.309	0.267	0.255
F2, 0.5	0.255	1.000	0.309	0.267	0.255
F3, 0.05	0.050	1.000	0.053	0.045	0.223
F3, 0.1	0.100	1.000	0.108	0.086	0.227
F3, 0.15	0.150	1.000	0.163	0.130	0.230
F3, 0.2	0.200	1.000	0.220	0.180	0.232
F3, 0.25	0.250	1.000	0.278	0.224	0.234
F3, 0.3	0.300	1.000	0.335	0.270	0.236
F3, 0.35	0.350	1.000	0.390	0.317	0.235
F3, 0.4	0.400	1.000	0.446	0.368	0.235
F3, 0.45	0.450	1.000	0.503	0.423	0.236
F3, 0.5	0.500	1.000	0.562	0.470	0.237
F4, 0.05	0.050	0.431	0.064	0.064	0.270
F4, 0.1	0.100	0.406	0.119	0.113	0.250
F4, 0.15	0.150	0.389	0.177	0.168	0.249
F4, 0.2	0.200	0.375	0.230	0.218	0.243
F4, 0.25	0.250	0.363	0.286	0.275	0.241
F4, 0.3	0.300	0.352	0.333	0.327	0.234
F4, 0.35	0.350	0.341	0.390	0.379	0.235
F4, 0.4	0.400	0.329	0.444	0.424	0.234
F4, 0.45	0.450	0.317	0.499	0.480	0.234
F4, 0.5	0.500	0.304	0.550	0.522	0.232
F5, 0.05	0.050	0.603	0.117	0.052	0.495
F5, 0.1	0.100	0.554	0.202	0.122	0.425
F5, 0.15	0.150	0.520	0.266	0.187	0.374
F5, 0.2	0.200	0.493	0.327	0.253	0.345

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Table A3, continued

Filter, budget	Selected	Threshold	Recall (material)	Recall (delayed)	Precision
F5, 0.25	0.250	0.469	0.382	0.305	0.323
F5, 0.3	0.300	0.450	0.435	0.365	0.306
F5, 0.35	0.350	0.431	0.482	0.423	0.290
F5, 0.4	0.400	0.415	0.526	0.466	0.278
F5, 0.45	0.450	0.399	0.573	0.514	0.268
F5, 0.5	0.500	0.385	0.615	0.559	0.259
F6, 0.05	0.050	0.666	0.144	0.046	0.608
F6, 0.1	0.100	0.554	0.221	0.131	0.466
F6, 0.15	0.150	0.487	0.282	0.198	0.397
F6, 0.2	0.200	0.432	0.340	0.259	0.359
F6, 0.25	0.250	0.391	0.395	0.313	0.334
F6, 0.3	0.300	0.362	0.444	0.367	0.313
F6, 0.35	0.350	0.342	0.495	0.421	0.298
F6, 0.4	0.400	0.325	0.543	0.471	0.287
F6, 0.45	0.450	0.309	0.586	0.519	0.275
F6, 0.5	0.500	0.297	0.628	0.567	0.265

Shares of NA-03's val-A comparison sample of member-days. F5, logistic regression on bge-small embeddings, is the frozen filter at a 30% budget.

Table A4: Encodings of numbers in states: E1 ranked best in the val-A search but fell short of E3 on val-B, so E3 was kept.

Split, model, encoding	IC	95% CI	t	Events	Rank
val-A, d1-ultra, E0	0.076	[0.003, 0.141]	2.29	1,500	4
val-A, d1-ultra, E1	0.081	[0.024, 0.131]	3.11	1,500	1
val-A, d1-ultra, E2	0.066	[−0.001, 0.127]	2.14	1,500	5
val-A, d1-ultra, E3	0.080	[0.014, 0.138]	2.68	1,500	2
val-A, d1-ultra, E4	0.079	[0.012, 0.139]	2.64	1,500	3
val-A, span-02[‡], E0	0.065	[0.013, 0.113]	2.56	1,500	4
val-A, span-02[‡], E1	0.069	[0.017, 0.116]	2.72	1,500	2
val-A, span-02[‡], E2	0.069	[0.016, 0.120]	2.71	1,500	1
val-A, span-02[‡], E3	0.063	[0.017, 0.108]	2.79	1,500	5
val-A, span-02[‡], E4	0.068	[0.013, 0.118]	2.61	1,500	3
val-A, drex-v2.0, E0	0.085	[0.037, 0.131]	3.65	1,500	1
val-A, drex-v2.0, E1	0.074	[0.024, 0.122]	3.07	1,500	2
val-A, drex-v2.0, E2	0.070	[0.023, 0.115]	3.07	1,500	5
val-A, drex-v2.0, E3	0.071	[0.017, 0.123]	2.66	1,500	4
val-A, drex-v2.0, E4	0.073	[0.016, 0.127]	2.46	1,500	3
val-B, d1-ultra, E1	0.058	[0.013, 0.102]	2.53	1,500	2
val-B, d1-ultra, E3	0.063	[0.023, 0.102]	3.14	1,500	1
val-B, span-02[‡], E1	0.063	[0.012, 0.111]	2.58	1,500	2
val-B, span-02[‡], E3	0.077	[0.029, 0.124]	3.25	1,500	1
val-B, drex-v2.0, E1	0.030	[−0.030, 0.084]	1.01	1,500	2
val-B, drex-v2.0, E3	0.044	[−0.013, 0.096]	1.55	1,500	1

[‡]span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A5: Spend and requests of the whole project, from the ledgers.

Item	Value	Cap
OpenRouter spend, all chunks (USD)	15.73	30
OpenRouter spend, NA-05 (USD)	0.10	—
OpenRouter spend, NA-06 (USD)	3.53	—
OpenRouter spend, NA-07 (USD)	1.24	—
OpenRouter spend, NA-09 (USD)	0.04	—
OpenRouter spend, NA-11 (USD)	10.82	—
spend lines in the ledger	310,587	—
FMP requests (NA-10)	15,858	17,000
Kaggle GPU-hours, NA-08 (training)	23.38	—
Kaggle GPU-hours, NA-11 (test inference)	2.00	—
Kaggle GPU-hours, NA-08 + NA-11	25.38	—
rows of the trials ledger	1,620	—
lines of the test-access ledger	29	—

—: no cap.

Spend in USD and GPU-hours to two decimals; counts are whole numbers. Kaggle GPU-hours from the NA-08 and NA-11 reports (Kaggle keeps no ledger here). NA-12, NA-13, NA-13b and NA-14 made no API call, Kaggle run or FMP request. NA-13's deflated Sharpe ratios used the 545 rows the trials ledger held when NA-13 ran.

Table A6: Pre-registration deviations (PREREG §15), one line per row of the log.

Row	Date	Chunk	Change	Reason	Freeze
1	10-24	NA-00	Freeze hash stops at §15; §15 checked as append-only	The log must grow after the freeze	before
2	10-24	NA-00	Test-year adjusted closes sit outside the sealed folder	Data/ is read-only	before
3	10-24	operator	Amendment A1: convergence to V, H2-C, S-CONV	Operator, before any data	before
4	10-24	operator	Amendments A2, A3: battery v2, explaining moves	Operator, before any data	before
5	10-24	NA-01	Share classes grouped by issuer name	ATOM rarely has a CIK	before
6	10-24	NA-01	Company cap from one class; shares on the split basis	CIQ shares are company-level	before
7	10-24	NA-01	Test-month ranking closes rebuilt from adjusted ones	Tiers needed before the freeze	before
8	10-24	NA-01	Ids without Data/prices use ATOM's total return	Else about 7% of members drop	before
9	10-24	NA-01	Sector pool: one class per company, own company out	No company in its own factor	before
10	10-24	NA-01	Target details: volume, σ floor, multi-day scaling	PREREG leaves them open	before
11	10-24	NA-01 verif.	Share counts checked against Yahoo, used when CIQ is off	Bad CIQ shares for large names	before
12	10-24	NA-01 verif.	Price-level check at dividend dates; HON rescaled	ATOM's HON closes were halved	before
13	10-24	NA-01 verif.	ATOM raw volume put on the traded share basis	Pre-split volume was adjusted	before
14	10-24	NA-02	News tickers mapped through dated aliases	ATOM holds no renames	before
15	10-24	NA-02	Near-duplicates grouped in time order; novelty fixed	Keeps grouping point-in-time	before
16	10-24	NA-02	Rules of the after-the-fact price-talk detector	PREREG names categories only	before
17	10-24	NA-02	Leakage audit of vendor sentiment (z_{id} vs z_{on})	PREREG states it in words	before
18	10-25	NA-03	Rule filters reach the budget by seeded thinning	A yes-or-no rule has no threshold	before
19	10-25	NA-03	Filter comparison sample purged and embargoed	Purge and embargo left open	before
20	10-25	NA-03	F5, F6 trained on articles; tuning metric fixed	Training unit left open	before
21	10-25	NA-03	D2 events, third-party exclusion, data end defined	PREREG does not cover them	before
22	10-25	NA-04	The company block carries the GICS sector only	ATOM has no industry level	before
23	10-25	NA-04	Codebook scales for multi-day moves, market series	PREREG gives them no scale	before
24	10-25	NA-04	Publication lags of market, FRED, earnings fields	Availability left to NA-04	before
25	10-25	NA-04	Masking widened: periods, dates, tickers, names	Residue above the target	before
26	10-25	NA-04	No pre-market talk in Call B; question text written	Numbers stay in the codebook	before
27	10-25	NA-04	Move explainers and TradingView items left out	They carry the session's move	before
28	10-25	NA-04	Residual names (about 1% of states) accepted	Content alone identifies half	before
29	10-26	NA-04	Non-English articles and English price talk removed	Price talk in 2.5% of states	before
30	10-26	NA-04	More date forms; every price move and level removed	Date residue failed AC2	before
31	10-26	NA-04	Running market stories left out of pre-open states	Live pages held the outcome	before
32	10-26	NA-04	Residues below 0.5% of states accepted, disclosed	The seventh verification passed	before
33	10-26	NA-02	Articles of undecided parsing status allowed	Operator decision A	before
34	10-26	NA-06	Tournament samples, IC, confidence set, screens fixed	PREREG leaves details open	before
35	10-27	op. (NA-06)	No API model passes §8.3; top three kept, model-specific	Within-model tests stay valid	before
36	10-27	NA-07	Encoding search and anchoring test operationalized	PREREG gives rules only	before
37	10-27	NA-07	Numeracy gate for encodings other than E3	The fallback was undefined	before
38	10-27	op. (NA-10)	FMP request cap raised from 3,000 to 17,000	FMP serves 30 days a request	before
39	10-27	NA-10	Decision point D1: the 10:00 price, returns, scales	PREREG leaves them open	before
40	10-27	NA-10	Per-session basis of 30-minute bars; 5 bp bound failing	Vendor data need it	before
41	10-27	NA-10	The D1 Call B state and its half-hour wording	No phrase for the 10:00 move	before
42	10-27	NA-10 verif.	Request counts and the 5 bp gap corrected	Recomputed from the ledger	before
43	10-27	operator	Two attention cuts; timing and text robustness	Limited attention; leak checks	before
44	10-26	NA-08	Trained models read 1,036 tokens instead of 1,024	No state or question is cut	before
45	10-27	NA-08	One seed and one epoch per trained model for now	One week's Kaggle quota	before
46	10-27	NA-09	Baselines operationalized: events, features, LLM	PREREG names them only	before
47	10-28	NA-11	Analysis spec: samples, inference, bootstrap, families	Frozen code must fix them	before
48	10-28	NA-11	Freeze hash also covers inference and metrics code	The frozen spec runs them	before
49	10-28	op. (NA-11)	Trained instruments pinned to the seed-13 runs	Later seeds must not change them	before
50	10-28	NA-11	Final selection: val-B IC on common events	Event set left open	before
51	10-28	op. (NA-11)	Company identification with matched distractors	Removes baseline bias	before
52	10-28	NA-11	Live-format residue counted with the frozen pattern	First count was narrow	before
53	10-28	op. (NA-11)	H1b, the IC with z_{id}, joins the Holm family	Tests what S-D0 trades	before
54	10-28	op. (NA-11)	Production plan: d1-ultra as the API comparator	The PRD names no rule	before
55	10-28	op. (NA-11)	Three robustness variants dropped; 08:00 approximated	They need new states	before
56	10-28	op. (NA-11)	Canary baselines and drift check for API models	NA-05 had none for them	before
57	10-28	op. (NA-11)	span-02's canary STOP judged noise; proceed	Inside its own repeat range	after
58	10-28	op. (NA-11)	Price-only D2 test predictions moved to NA-12	A feature is a test outcome	after

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Table A6, continued

Row	Date	Chunk	Change	Reason	Freeze
59	10-28	operator	Anchor robustness runs fixed before unblinding	Near-constant anchor outputs	after
60	10-28	NA-12	NA-12's calls of the frozen spec fixed in advance	Caller fixes joins, families	after
61	10-28	NA-12	Figure data definitions and code fixes	No pre-registered number moves	after
62	10-28	op. (NA-12)	Exploratory analyses after unblinding, dated	Paper trail for post hoc work	after
63	10-28	NA-13	Strategy engine fixed before any test-period return	PREREG gives books, not joins	after
64	10-29	operator	Forward test (paper 2) pre-registered: F1 to F5	Confirmatory forward test	after
65	10-29	NA-13b	Exploratory selective study: rules fixed before its test	The prompt leaves joins open	after

Dates are in the sample's last calendar year (MM-DD); Freeze: written before or after the freeze. Summaries by NA-14; the full text is in PREREG §15.

C Priced at the open

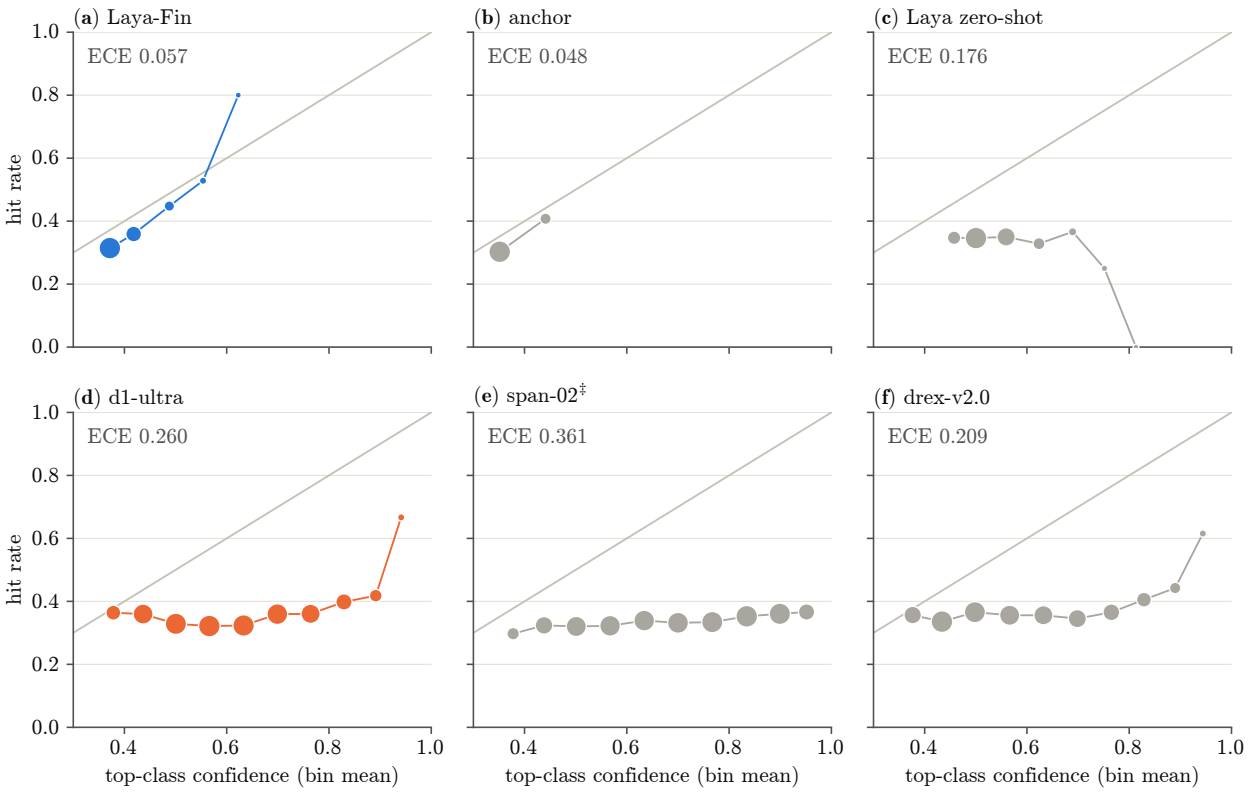


Figure A4: Reliability of the three-class forecasts. Reliability of Q1's three classes (fall, no move or rise at ±0.5 typical moves) against e_5 on the test period: the hit rate of the top class by bin of its confidence (15 bins; dot area grows with the events in the bin; the diagonal is perfect calibration; NA-12). The trained models are calibrated but barely informative: ECE 0.057 for Laya-Fin and 0.048 for the anchor (uninformative, near-constant outputs), with log-loss skill 0.001 for Laya-Fin. The zero-shot and API models are overconfident: ECE 0.176 for Laya zero-shot, 0.260 for d1-ultra, 0.361 for span-02 and 0.209 for drex-v2.0. ‡ span-02: the S5 canary fired a STOP after the freeze and the operator chose to proceed (PREREG §15).

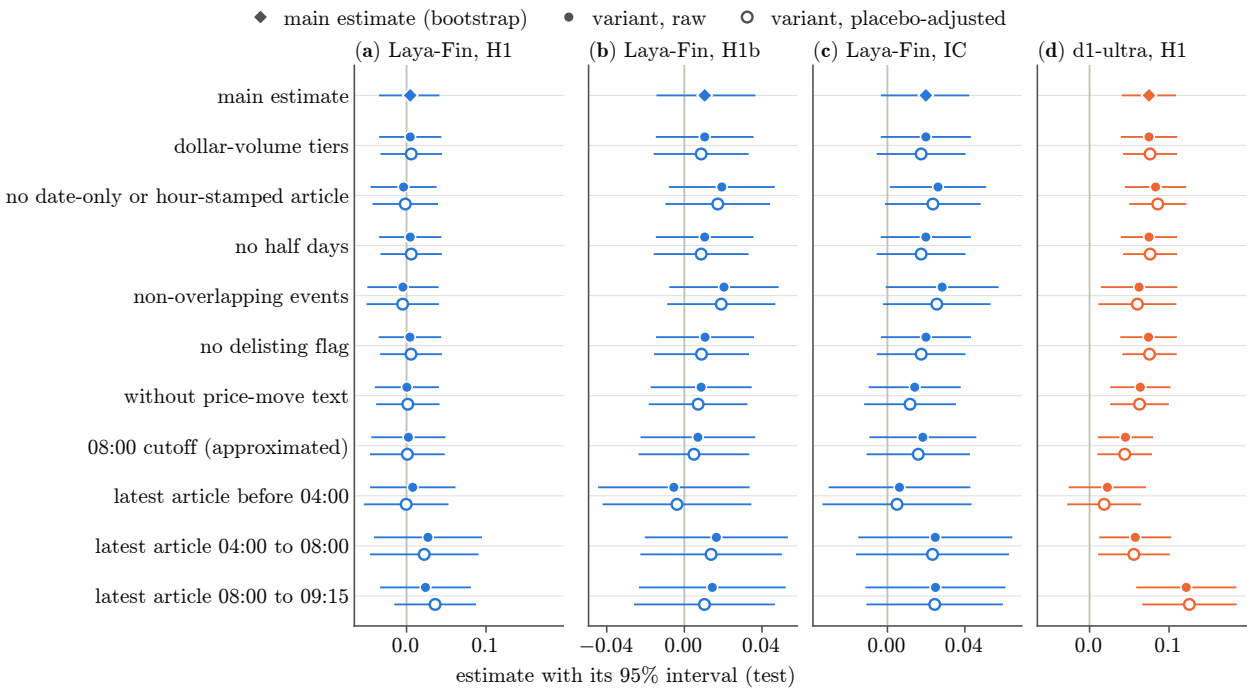


Figure A5: No robustness variant changes a conclusion. Robustness of the D0 results (§11.6): Laya-Fin's H1, H1b and IC of $\hat{N}$ with z_{cc}, and d1-ultra's H1, in each variant raw (filled) and placebo-adjusted (hollow), below the main estimate (diamond; stationary block bootstrap) with 95% intervals (± 1.96 Newey–West SE for the variants). (a) and (d), both H1, share a scale; (b) and (c) have their own. Laya-Fin's H1 and H1b stay within their intervals of zero in every variant. d1-ultra's H1 stays positive in every variant, from 0.022 when the latest article precedes 04:00 to 0.122 when it falls between 08:00 and 09:15; it is model-specific (§1.3).

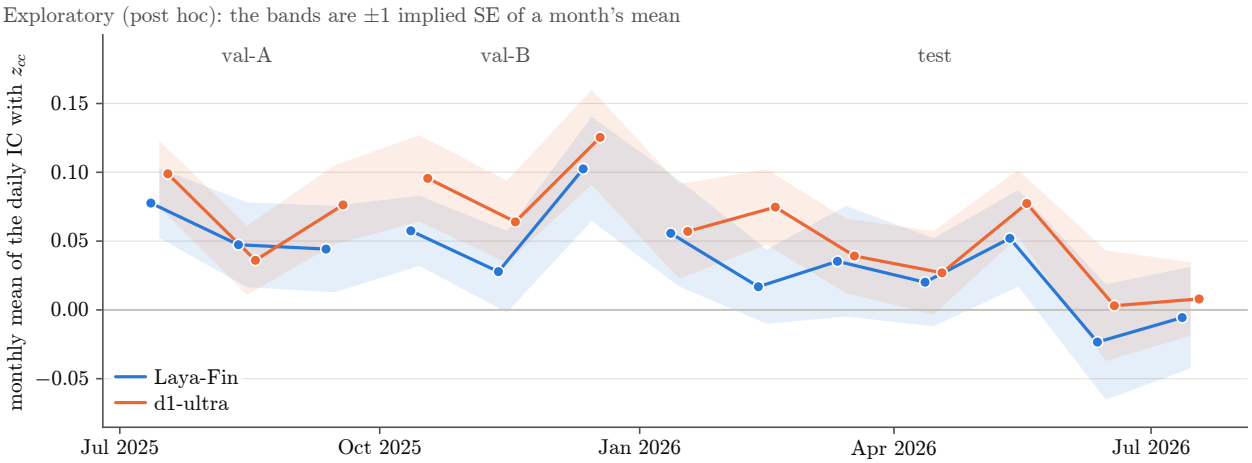


Figure A6: Exploratory: the monthly IC, with the implied standard error of a month's mean. Exploratory (post hoc): the bands come from NA-12 §13, requested after unblinding; no hypothesis, no claim. The monthly mean of the daily IC of $\hat{N}$ with z_{cc} over val-A, val-B and the test period for Laya-Fin and d1-ultra (model-specific), with ±1 implied standard error of a month's mean (iid; on test 0.037 for Laya-Fin). In the last two test months the ICs are about zero (Laya-Fin −0.02 and −0.01, d1-ultra +0.00 and +0.01), within about one standard error of a month's mean. Every release lies after the sample, where the time axis stops, which ends on 2026-07-31: Laya on 2026-09-18, the API models from 2026-10-12; a contamination cliff at a release would fall outside the data. The anchor (uninformative, near-constant outputs) has no implied SE; its monthly ICs are in the appendix's monthly table. ‡ span-02, whose release closes the API models' range: the S5 canary fired a STOP after the freeze and the operator chose to proceed (PREREG §15).

Table A7: The primary tests for every model.

Test	Estimate	95% CI	t	p	Holm p	Events
Laya-Fin						
H1	0.0046	[−0.0346, 0.0414]	0.23	0.409	0.853	13,748
H1b	0.0105	[−0.0144, 0.0366]	0.82	0.213	0.853	13,748
H2	−0.0440	—	—	0.616	0.853	—
H3	0.0038	[−0.0008, 0.0085]	1.52	0.054	0.269	17,044
H4	0.0012	[−0.0017, 0.0042]	—	0.216	0.853	13,732
anchor						
H1	0.0036	[−0.0271, 0.0342]	0.22	0.406	1.000	13,748
H1b	0.0020	[−0.0192, 0.0224]	0.18	0.427	1.000	13,748
H2	0.0150	—	—	0.294	1.000	—
H3	0.0013	[0.0002, 0.0024]	2.10	0.017	0.083	17,044
H4	−0.0004	[−0.0016, 0.0007]	—	0.773	1.000	13,732
Laya zero-shot						
H1	0.0540	[0.0231, 0.0867]	3.17	0.001	0.004	13,748
H1b	−0.0173	[−0.0438, 0.0086]	−1.25	0.900	1.000	13,748
H2	—	—	—	—	—	—
H3	—	—	—	—	—	—
H4	−0.0013	[−0.0023, −0.0002]	—	0.992	1.000	13,732
d1-ultra						
H1	0.0748	[0.0405, 0.1088]	4.13	<0.001	<0.001	13,748
H1b	−0.0007	[−0.0207, 0.0198]	−0.07	0.531	1.000	13,748
H2	−0.0756	—	—	0.991	1.000	—
H3	0.0007	[−0.0017, 0.0032]	0.52	0.285	1.000	17,044
H4	0.0004	[−0.0019, 0.0028]	—	0.354	1.000	13,732
span-02[‡]						
H1	0.0748	[0.0389, 0.1102]	4.05	<0.001	<0.001	13,748
H1b	−0.0005	[−0.0216, 0.0205]	−0.05	0.521	0.521	13,748
H2	—	—	—	—	—	—
H3	—	—	—	—	—	—
H4	—	—	—	—	—	—
drex-v2.0						
H1	0.0753	[0.0436, 0.1062]	4.66	<0.001	<0.001	13,748
H1b	0.0019	[−0.0178, 0.0224]	0.18	0.426	0.426	13,748
H2	—	—	—	—	—	—
H3	—	—	—	—	—	—
H4	—	—	—	—	—	—

—: the source table has no value for the row; the statistic does not apply to it or was not computed.

Holm within each model; the verdict is Laya-Fin's (Table 1).

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

[‡]span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

Table A8: Daily IC by target on the test period, with the quintile spread.

					Quintile spread	
Model, target	IC	95% CI	t	Events	Top − bottom	p
Laya-Fin, z_{cc}	0.020	[−0.003, 0.042]	1.67	13,748	0.127	0.024
Laya-Fin, z_{on}	0.015	[−0.011, 0.041]	1.08	13,439	0.174	0.044
Laya-Fin, z_{id}	0.010	[−0.015, 0.036]	0.82	13,439	0.050	0.201
anchor, z_{cc}	0.008	[−0.011, 0.026]	0.83	13,748	−0.001	0.500
anchor, z_{on}	0.006	[−0.016, 0.027]	0.50	13,439	−0.011	0.544
anchor, z_{id}	0.002	[−0.020, 0.022]	0.18	13,439	0.002	0.495
Laya zero-shot, z_{cc}	0.005	[−0.023, 0.032]	0.37	13,748	0.056	0.236
Laya zero-shot, z_{on}	0.037	[0.015, 0.058]	3.22	13,439	0.281	<0.001
Laya zero-shot, z_{id}	−0.017	[−0.043, 0.008]	−1.25	13,439	−0.102	0.953
d1-ultra, z_{cc}	0.039	[0.015, 0.062]	3.23	13,748	0.410	<0.001
d1-ultra, z_{on}	0.074	[0.043, 0.104]	4.74	13,439	0.812	<0.001
d1-ultra, z_{id}	−0.001	[−0.021, 0.020]	−0.07	13,439	−0.006	0.547
span-02‡, z_{cc}	0.041	[0.015, 0.064]	3.10	13,748	0.406	<0.001
span-02‡, z_{on}	0.074	[0.040, 0.106]	4.49	13,439	0.803	<0.001
span-02‡, z_{id}	−0.001	[−0.021, 0.020]	−0.05	13,439	−0.008	0.566
drex-v2.0, z_{cc}	0.044	[0.021, 0.067]	3.70	13,748	0.384	<0.001
drex-v2.0, z_{on}	0.077	[0.048, 0.105]	5.34	13,439	0.760	<0.001
drex-v2.0, z_{id}	0.002	[−0.018, 0.023]	0.18	13,439	−0.005	0.541

Quintile spread: the mean target in the top quintile of $\hat{N}$ minus the bottom quintile, in typical daily moves, and its p; the IC's own test is its t.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

‡span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

Table A9: Val-B daily ICs by target of the trained and zero-shot Laya models (NA-11, before the freeze).

Model	IC z_{cc}	t	IC z_{on}	t	IC z_{id}	t	Events
Laya-Fin	0.065	3.64	0.045	2.93	0.054	2.89	5,506
anchor	−0.020	−1.52	−0.002	−0.16	−0.011	−0.75	5,506
Laya zero-shot	0.029	1.94	0.047	2.55	0.002	0.12	5,506

From NA-11's run of the frozen spec on val-B, before the freeze.
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.

Table A10: Calibration of the three-class forecasts: the trained models are calibrated but barely informative.

Model, target	ECE	Log-loss skill	Brier skill	AUC	T	ECE before	ECE after
Laya-Fin, e_5	0.057	0.001	0.000	—	—	—	—
Laya-Fin, z_{cc}	0.033	0.010	0.010	—	—	—	—
Laya-Fin, $\lvert z_{cc}\rvert \geq 1$	—	—	—	0.60	—	—	—
Laya-Fin, e_5, temperature scaling	—	—	—	—	1.00	0.058	0.057
anchor, e_5	0.048	0.002	0.001	—	—	—	—
anchor, z_{cc}	0.016	0.015	0.015	—	—	—	—
anchor, $\lvert z_{cc}\rvert \geq 1$	—	—	—	0.58	—	—	—
anchor, e_5, temperature scaling	—	—	—	—	0.97	0.011	0.048
Laya zero-shot, e_5	0.176	−0.156	−0.110	—	—	—	—
Laya zero-shot, z_{cc}	0.179	−0.177	−0.120	—	—	—	—
Laya zero-shot, $\lvert z_{cc}\rvert \geq 1$	—	—	—	0.52	—	—	—
Laya zero-shot, e_5, temperature scaling	—	—	—	—	6.33	0.176	0.110
d1-ultra, e_5	0.260	−0.194	−0.194	—	—	—	—
d1-ultra, z_{cc}	0.249	−0.166	−0.165	—	—	—	—
d1-ultra, $\lvert z_{cc}\rvert \geq 1$	—	—	—	0.58	—	—	—
d1-ultra, e_5, temperature scaling	—	—	—	—	1.95	0.260	0.155
span-02[‡], e_5	0.361	−0.460	−0.361	—	—	—	—
span-02[‡], z_{cc}	0.342	−0.410	−0.315	—	—	—	—
span-02[‡], $\lvert z_{cc}\rvert \geq 1$	—	—	—	0.56	—	—	—
drex-v2.0, e_5	0.209	−0.135	−0.138	—	—	—	—
drex-v2.0, z_{cc}	0.204	−0.119	−0.120	—	—	—	—
drex-v2.0, $\lvert z_{cc}\rvert \geq 1$	—	—	—	0.57	—	—	—

—: the source table has no value for the row; the statistic does not apply to it or was not computed.
Q1's three classes against the targets; skill against val-A climatology; T the temperature fit on val-A.
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
[‡]span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

Table A11: Robustness of the primary results: no variant changes a conclusion.

Model, point, variant	H1	p	H1b	p	IC z_{cc}	b	p_b	c	p_c	IC y_2
Laya-Fin, D0, dollar-volume tiers, raw	0.005	0.409	0.010	0.213	0.020	—	—	—	—	—
Laya-Fin, D0, no date-only or hour-stamped article, raw	−0.004	0.577	0.019	0.091	0.026	—	—	—	—	—
Laya-Fin, D0, no half days, raw	0.005	0.409	0.010	0.213	0.020	—	—	—	—	—
Laya-Fin, D0, non-overlapping events, raw	−0.005	0.567	0.020	0.084	0.028	—	—	—	—	—
Laya-Fin, D0, no delisting flag, raw	0.004	0.418	0.011	0.209	0.020	—	—	—	—	—
Laya-Fin, D0, without price-move text, raw	0.000	0.494	0.009	0.262	0.014	—	—	—	—	—
Laya-Fin, D0, 08:00 cutoff (approximated), raw	0.002	0.461	0.007	0.315	0.018	—	—	—	—	—
Laya-Fin, D0, latest article 04:00 to 08:00, raw	0.027	0.199	0.016	0.185	0.025	—	—	—	—	—
Laya-Fin, D0, latest article 08:00 to 09:15, raw	0.024	0.204	0.014	0.232	0.025	—	—	—	—	—
Laya-Fin, D0, latest article before 04:00, raw	0.008	0.402	−0.005	0.591	0.006	—	—	—	—	—
Laya-Fin, D2, dollar-volume tiers, raw	—	—	—	—	—	−0.038	0.601	−0.014	0.114	—
Laya-Fin, D2, no date-only or hour-stamped article, raw	—	—	—	—	—	−0.011	0.528	−0.018	0.074	—
Laya-Fin, D2, no half days, raw	—	—	—	—	—	−0.044	0.616	−0.015	0.110	—
Laya-Fin, D2, non-overlapping events, raw	—	—	—	—	—	−0.012	0.526	0.005	0.639	—
Laya-Fin, D2, no delisting flag, raw	—	—	—	—	—	−0.050	0.630	−0.015	0.114	—
Laya-Fin, D0, dollar-volume tiers, placebo-adjusted	0.006	0.384	0.009	0.249	0.017	—	—	—	—	—
Laya-Fin, D0, no date-only or hour-stamped article, placebo-adjusted	−0.002	0.538	0.017	0.120	0.024	—	—	—	—	—
Laya-Fin, D0, no half days, placebo-adjusted	0.006	0.384	0.009	0.249	0.017	—	—	—	—	—
Laya-Fin, D0, non-overlapping events, placebo-adjusted	−0.005	0.573	0.019	0.096	0.026	—	—	—	—	—
Laya-Fin, D0, no delisting flag, placebo-adjusted	0.006	0.391	0.009	0.245	0.017	—	—	—	—	—
Laya-Fin, D0, without price-move text, placebo-adjusted	0.002	0.473	0.007	0.294	0.012	—	—	—	—	—
Laya-Fin, D0, 08:00 cutoff (approximated), placebo-adjusted	0.001	0.486	0.005	0.361	0.016	—	—	—	—	—
Laya-Fin, D0, latest article 04:00 to 08:00, placebo-adjusted	0.022	0.245	0.014	0.225	0.023	—	—	—	—	—
Laya-Fin, D0, latest article 08:00 to 09:15, placebo-adjusted	0.036	0.088	0.010	0.295	0.024	—	—	—	—	—
Laya-Fin, D0, latest article before 04:00, placebo-adjusted	0.000	0.523	−0.004	0.565	0.005	—	—	—	—	—
Laya-Fin, D2, dollar-volume tiers, placebo-adjusted	—	—	—	—	—	−0.036	0.597	−0.014	0.122	—
Laya-Fin, D2, no date-only or hour-stamped article, placebo-adjusted	—	—	—	—	—	−0.011	0.528	−0.018	0.074	—
Laya-Fin, D2, no half days, placebo-adjusted	—	—	—	—	—	−0.044	0.616	−0.015	0.110	—
Laya-Fin, D2, non-overlapping events, placebo-adjusted	—	—	—	—	—	−0.012	0.526	0.005	0.639	—
Laya-Fin, D2, no delisting flag, placebo-adjusted	—	—	—	—	—	−0.050	0.630	−0.015	0.114	—
Laya-Fin, D0+D2, placebo-adjusted	0.006	0.384	0.009	0.249	—	−0.044	0.616	−0.015	0.110	—
Laya-Fin, D0+D2, σ = 60-session SD, raw	0.007	0.367	0.009	0.238	—	−0.032	0.607	−0.013	0.109	—
Laya-Fin, D0, 2024-H1, raw	0.028	0.235	0.024	0.105	0.041	—	—	—	—	—
Laya-Fin, D0, 2024-H1, placebo-adjusted	0.022	0.291	0.025	0.103	—	—	—	—	—	—
Laya-Fin, D2, 2024-H1, raw	—	—	—	—	—	0.079	0.344	0.004	0.607	—
Laya-Fin, D2, 2024-H1, placebo-adjusted	—	—	—	—	—	0.079	0.344	0.004	0.607	—
anchor, D0, dollar-volume tiers, raw	0.004	0.406	0.002	0.427	0.008	—	—	—	—	—
anchor, D0, no date-only or hour-stamped article, raw	0.002	0.467	0.012	0.143	0.021	—	—	—	—	—

continued on the next page

Table A11, continued

Model, point, variant	H1	p	H1b	p	IC z_{cc}	b	p_b	c	p_c	IC y_2
anchor, D0, no half days, raw	0.004	0.406	0.002	0.427	0.008	—	—	—	—	—
anchor, D0, non-overlapping events, raw	−0.005	0.593	0.004	0.393	0.008	—	—	—	—	—
anchor, D0, no delisting flag, raw	0.004	0.404	0.002	0.426	0.008	—	—	—	—	—
anchor, D0, without price-move text, raw	0.008	0.294	−0.006	0.700	−0.002	—	—	—	—	—
anchor, D0, 08:00 cutoff (approximated), raw	−0.008	0.710	0.006	0.300	0.002	—	—	—	—	—
anchor, D0, latest article 04:00 to 08:00, raw	−0.030	0.840	0.041	0.023	0.034	—	—	—	—	—
anchor, D0, latest article 08:00 to 09:15, raw	0.047	0.070	−0.008	0.634	0.025	—	—	—	—	—
anchor, D0, latest article before 04:00, raw	−0.013	0.751	0.001	0.487	−0.006	—	—	—	—	—
anchor, D2, dollar-volume tiers, raw	—	—	—	—	—	0.775	0.292	−0.015	0.116	—
anchor, D2, no date-only or hour-stamped article, raw	—	—	—	—	—	0.306	0.408	−0.018	0.076	—
anchor, D2, no half days, raw	—	—	—	—	—	0.748	0.294	−0.015	0.111	—
anchor, D2, non-overlapping events, raw	—	—	—	—	—	3.898	0.034	0.006	0.655	—
anchor, D2, no delisting flag, raw	—	—	—	—	—	0.752	0.293	−0.015	0.114	—
anchor, D0, dollar-volume tiers, placebo-adjusted	0.006	0.345	0.002	0.429	0.008	—	—	—	—	—
anchor, D0, no date-only or hour-stamped article, placebo-adjusted	0.003	0.427	0.013	0.128	0.022	—	—	—	—	—
anchor, D0, no half days, placebo-adjusted	0.006	0.345	0.002	0.429	0.008	—	—	—	—	—
anchor, D0, non-overlapping events, placebo-adjusted	−0.002	0.533	0.004	0.395	0.007	—	—	—	—	—
anchor, D0, no delisting flag, placebo-adjusted	0.006	0.351	0.002	0.423	0.008	—	—	—	—	—
anchor, D0, without price-move text, placebo-adjusted	0.009	0.262	−0.006	0.701	−0.003	—	—	—	—	—
anchor, D0, 08:00 cutoff (approximated), placebo-adjusted	−0.005	0.627	0.005	0.323	0.002	—	—	—	—	—
anchor, D0, latest article 04:00 to 08:00, placebo-adjusted	−0.027	0.814	0.040	0.025	0.030	—	—	—	—	—
anchor, D0, latest article 08:00 to 09:15, placebo-adjusted	0.044	0.074	−0.005	0.585	0.022	—	—	—	—	—
anchor, D0, latest article before 04:00, placebo-adjusted	−0.009	0.689	0.000	0.501	−0.005	—	—	—	—	—
anchor, D2, dollar-volume tiers, placebo-adjusted	—	—	—	—	—	0.693	0.308	−0.014	0.124	—
anchor, D2, no date-only or hour-stamped article, placebo-adjusted	—	—	—	—	—	0.306	0.408	−0.018	0.076	—
anchor, D2, no half days, placebo-adjusted	—	—	—	—	—	0.748	0.294	−0.015	0.111	—
anchor, D2, non-overlapping events, placebo-adjusted	—	—	—	—	—	3.898	0.034	0.006	0.655	—
anchor, D2, no delisting flag, placebo-adjusted	—	—	—	—	—	0.752	0.293	−0.015	0.114	—
anchor, D0+D2, placebo-adjusted	0.006	0.345	0.002	0.429	—	0.748	0.294	−0.015	0.111	—
anchor, D0+D2, σ = 60-session SD, raw	0.005	0.372	−0.001	0.528	—	0.673	0.254	−0.013	0.110	—
anchor, D0, 2024-H1, raw	−0.043	0.895	0.026	0.127	0.011	—	—	—	—	—
anchor, D0, 2024-H1, placebo-adjusted	−0.042	0.889	0.024	0.151	—	—	—	—	—	—
anchor, D2, 2024-H1, raw	—	—	—	—	—	1.845	0.156	0.004	0.613	—
anchor, D2, 2024-H1, placebo-adjusted	—	—	—	—	—	1.845	0.156	0.004	0.613	—
d1-ultra, D0, dollar-volume tiers, raw	0.075	<0.001	−0.001	0.531	0.039	—	—	—	—	—
d1-ultra, D0, no date-only or hour-stamped article, raw	0.083	<0.001	−0.001	0.542	0.042	—	—	—	—	—
d1-ultra, D0, no half days, raw	0.075	<0.001	−0.001	0.531	0.039	—	—	—	—	—
d1-ultra, D0, non-overlapping events, raw	0.062	0.005	0.015	0.141	0.060	—	—	—	—	—

continued on the next page

Table A11, continued

Model, point, variant	H1	p	H1b	p	IC z_{cc}	b	p_b	c	p_c	IC y_2
d1-ultra, D0, no delisting flag, raw	0.074	<0.001	0.000	0.522	0.039	—	—	—	—	—
d1-ultra, D0, without price-move text, raw	0.064	<0.001	0.001	0.462	0.036	—	—	—	—	—
d1-ultra, D0, 08:00 cutoff (approximated), raw	0.045	0.003	0.000	0.509	0.023	—	—	—	—	—
d1-ultra, D0, latest article 04:00 to 08:00, raw	0.057	0.004	0.004	0.438	0.030	—	—	—	—	—
d1-ultra, D0, latest article 08:00 to 09:15, raw	0.122	<0.001	−0.006	0.621	0.057	—	—	—	—	—
d1-ultra, D0, latest article before 04:00, raw	0.022	0.170	0.001	0.465	0.019	—	—	—	—	—
d1-ultra, D2, dollar-volume tiers, raw	—	—	—	—	—	−0.074	0.990	−0.011	0.177	—
d1-ultra, D2, no date-only or hour-stamped article, raw	—	—	—	—	—	−0.074	0.984	−0.014	0.123	—
d1-ultra, D2, no half days, raw	—	—	—	—	—	−0.076	0.991	−0.011	0.171	—
d1-ultra, D2, non-overlapping events, raw	—	—	—	—	—	−0.062	0.950	0.008	0.711	—
d1-ultra, D2, no delisting flag, raw	—	—	—	—	—	−0.076	0.991	−0.011	0.176	—
d1-ultra, D0, dollar-volume tiers, placebo-adjusted	0.076	<0.001	−0.002	0.586	0.038	—	—	—	—	—
d1-ultra, D0, no date-only or hour-stamped article, placebo-adjusted	0.086	<0.001	−0.004	0.636	0.041	—	—	—	—	—
d1-ultra, D0, no half days, placebo-adjusted	0.076	<0.001	−0.002	0.586	0.038	—	—	—	—	—
d1-ultra, D0, non-overlapping events, placebo-adjusted	0.060	0.007	0.016	0.137	0.064	—	—	—	—	—
d1-ultra, D0, no delisting flag, placebo-adjusted	0.075	<0.001	−0.002	0.572	0.038	—	—	—	—	—
d1-ultra, D0, without price-move text, placebo-adjusted	0.063	<0.001	0.002	0.459	0.036	—	—	—	—	—
d1-ultra, D0, 08:00 cutoff (approximated), placebo-adjusted	0.044	0.003	0.001	0.474	0.027	—	—	—	—	—
d1-ultra, D0, latest article 04:00 to 08:00, placebo-adjusted	0.056	0.005	0.004	0.435	0.029	—	—	—	—	—
d1-ultra, D0, latest article 08:00 to 09:15, placebo-adjusted	0.126	<0.001	−0.006	0.630	0.050	—	—	—	—	—
d1-ultra, D0, latest article before 04:00, placebo-adjusted	0.018	0.212	0.005	0.372	0.024	—	—	—	—	—
d1-ultra, D2, dollar-volume tiers, placebo-adjusted	—	—	—	—	—	−0.075	0.991	−0.011	0.189	—
d1-ultra, D2, no date-only or hour-stamped article, placebo-adjusted	—	—	—	—	—	−0.074	0.984	−0.014	0.123	—
d1-ultra, D2, no half days, placebo-adjusted	—	—	—	—	—	−0.076	0.991	−0.011	0.171	—
d1-ultra, D2, non-overlapping events, placebo-adjusted	—	—	—	—	—	−0.062	0.950	0.008	0.711	—
d1-ultra, D2, no delisting flag, placebo-adjusted	—	—	—	—	—	−0.076	0.991	−0.011	0.176	—
d1-ultra, D0+D2, placebo-adjusted	0.076	<0.001	−0.002	0.586	—	−0.076	0.991	−0.011	0.171	—
d1-ultra, D0+D2, σ = 60-session SD, raw	0.075	<0.001	−0.001	0.555	—	−0.056	0.991	−0.009	0.179	—
d1-ultra, D0, 2024-H1, raw	0.153	0.001	0.020	0.255	0.109	—	—	—	—	—
d1-ultra, D0, 2024-H1, placebo-adjusted	0.146	<0.001	0.021	0.234	—	—	—	—	—	—
d1-ultra, D2, 2024-H1, raw	—	—	—	—	—	0.015	0.382	0.004	0.601	—
d1-ultra, D2, 2024-H1, placebo-adjusted	—	—	—	—	—	0.015	0.382	0.004	0.601	—
d1-ultra, D0, unmasked twins: masked reading, raw	0.063	0.003	−0.009	0.736	0.026	—	—	—	—	—
d1-ultra, D0, unmasked twins: unmasked reading, raw	0.074	<0.001	−0.009	0.735	0.029	—	—	—	—	—
d1-ultra, D2, Call B, news removed: J (real), raw	—	—	—	—	—	—	—	—	—	−0.020
d1-ultra, D2, Call B, news removed: J (ablated), raw	—	—	—	—	—	—	—	—	—	−0.012
d1-ultra, D2, Call B, news removed: C (real), raw	—	—	—	—	—	—	—	—	—	−0.019
d1-ultra, D2, Call B, news removed: C (ablated), raw	—	—	—	—	—	—	—	—	—	−0.014

continued on the next page

Table A11, continued

Model, point, variant	H1	p	H1b	p	IC z_{cc}	b	p_b	c	p_c	IC y_2
d1-ultra, D2, Call B, news removed: $\hat{N}$ (real), raw	—	—	—	—	—	—	—	—	—	−0.015
d1-ultra, D2, Call B, news removed: $\hat{N}$ (ablated), raw	—	—	—	—	—	—	—	—	—	0.018
d1-ultra, D2, Call B, shuffled news: J (real), raw	—	—	—	—	—	—	—	—	—	−0.021
d1-ultra, D2, Call B, shuffled news: J (ablated), raw	—	—	—	—	—	—	—	—	—	−0.001
d1-ultra, D2, Call B, shuffled news: C (real), raw	—	—	—	—	—	—	—	—	—	−0.022
d1-ultra, D2, Call B, shuffled news: C (ablated), raw	—	—	—	—	—	—	—	—	—	−0.005
d1-ultra, D2, Call B, shuffled news: $\hat{N}$ (real), raw	—	—	—	—	—	—	—	—	—	−0.013
d1-ultra, D2, Call B, shuffled news: $\hat{N}$ (ablated), raw	—	—	—	—	—	—	—	—	—	0.008
all tiers, summary text, not run	—	—	—	—	—	—	—	—	—	—
all tiers, vendor tags in the text, not run	—	—	—	—	—	—	—	—	—	—
all tiers, 09:30 cutoff, not run	—	—	—	—	—	—	—	—	—	—
all tiers, gated S-D1, not run	—	—	—	—	—	—	—	—	—	—
all tiers, unmasked, trained models, not run	—	—	—	—	—	—	—	—	—	—

—: the source table has no value for the row; the statistic does not apply to it or was not computed.
Each variant raw and placebo-adjusted (§5.7). Call B ablations: IC of J, C and $\hat{N}$ with y_2, real against ablated.
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A12: Monthly mean of the daily IC with z_{cc}.

Model, split, month	IC	Days
Laya-Fin, val-A, 2025-07	0.078	17
Laya-Fin, val-A, 2025-08	0.047	21
Laya-Fin, val-A, 2025-09	0.044	21
Laya-Fin, val-B, 2025-10	0.057	18
Laya-Fin, val-B, 2025-11	0.028	19
Laya-Fin, val-B, 2025-12	0.102	22
Laya-Fin, test, 2026-01	0.056	15
Laya-Fin, test, 2026-02	0.017	19
Laya-Fin, test, 2026-03	0.035	22
Laya-Fin, test, 2026-04	0.020	21
Laya-Fin, test, 2026-05	0.052	20
Laya-Fin, test, 2026-06	−0.023	21
Laya-Fin, test, 2026-07	−0.006	22
anchor, val-A, 2025-07	0.022	17
anchor, val-A, 2025-08	0.003	21
anchor, val-A, 2025-09	0.046	21
anchor, val-B, 2025-10	−0.003	18
anchor, val-B, 2025-11	−0.007	19
anchor, val-B, 2025-12	−0.045	22
anchor, test, 2026-01	0.039	15
anchor, test, 2026-02	−0.028	19
anchor, test, 2026-03	−0.003	22
anchor, test, 2026-04	0.016	21
anchor, test, 2026-05	0.009	20
anchor, test, 2026-06	0.028	21
anchor, test, 2026-07	0.000	22
Laya zero-shot, val-A, 2025-07	0.059	17
Laya zero-shot, val-A, 2025-08	0.013	21
Laya zero-shot, val-A, 2025-09	0.018	21
Laya zero-shot, val-B, 2025-10	0.044	18
Laya zero-shot, val-B, 2025-11	0.009	19
Laya zero-shot, val-B, 2025-12	0.035	22
Laya zero-shot, test, 2026-01	0.036	15
Laya zero-shot, test, 2026-02	0.016	19
Laya zero-shot, test, 2026-03	0.062	22
Laya zero-shot, test, 2026-04	−0.020	21
Laya zero-shot, test, 2026-05	0.029	20
Laya zero-shot, test, 2026-06	−0.055	21
Laya zero-shot, test, 2026-07	−0.021	22
d1-ultra, val-A, 2025-07	0.099	17
d1-ultra, val-A, 2025-08	0.036	21
d1-ultra, val-A, 2025-09	0.076	21
d1-ultra, val-B, 2025-10	0.096	18
d1-ultra, val-B, 2025-11	0.064	19
d1-ultra, val-B, 2025-12	0.125	22
d1-ultra, test, 2026-01	0.057	15
d1-ultra, test, 2026-02	0.075	19
d1-ultra, test, 2026-03	0.039	22
d1-ultra, test, 2026-04	0.027	21
d1-ultra, test, 2026-05	0.077	20
d1-ultra, test, 2026-06	0.003	21
d1-ultra, test, 2026-07	0.008	22
span-02[‡], val-B, 2025-10	0.085	18
span-02[‡], val-B, 2025-11	0.082	19

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Table A12, continued

Model, split, month	IC	Days
span-02‡, val-B, 2025-12	0.123	22
span-02‡, test, 2026-01	0.057	15
span-02‡, test, 2026-02	0.065	19
span-02‡, test, 2026-03	0.043	22
span-02‡, test, 2026-04	0.040	21
span-02‡, test, 2026-05	0.099	20
span-02‡, test, 2026-06	−0.001	21
span-02‡, test, 2026-07	−0.007	22
drex-v2.0, val-B, 2025-10	0.073	18
drex-v2.0, val-B, 2025-11	0.043	19
drex-v2.0, val-B, 2025-12	0.105	22
drex-v2.0, test, 2026-01	0.055	15
drex-v2.0, test, 2026-02	0.076	19
drex-v2.0, test, 2026-03	0.053	22
drex-v2.0, test, 2026-04	0.013	21
drex-v2.0, test, 2026-05	0.095	20
drex-v2.0, test, 2026-06	0.007	21
drex-v2.0, test, 2026-07	0.019	22

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).
‡span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

D Over- and under-reaction, and convergence

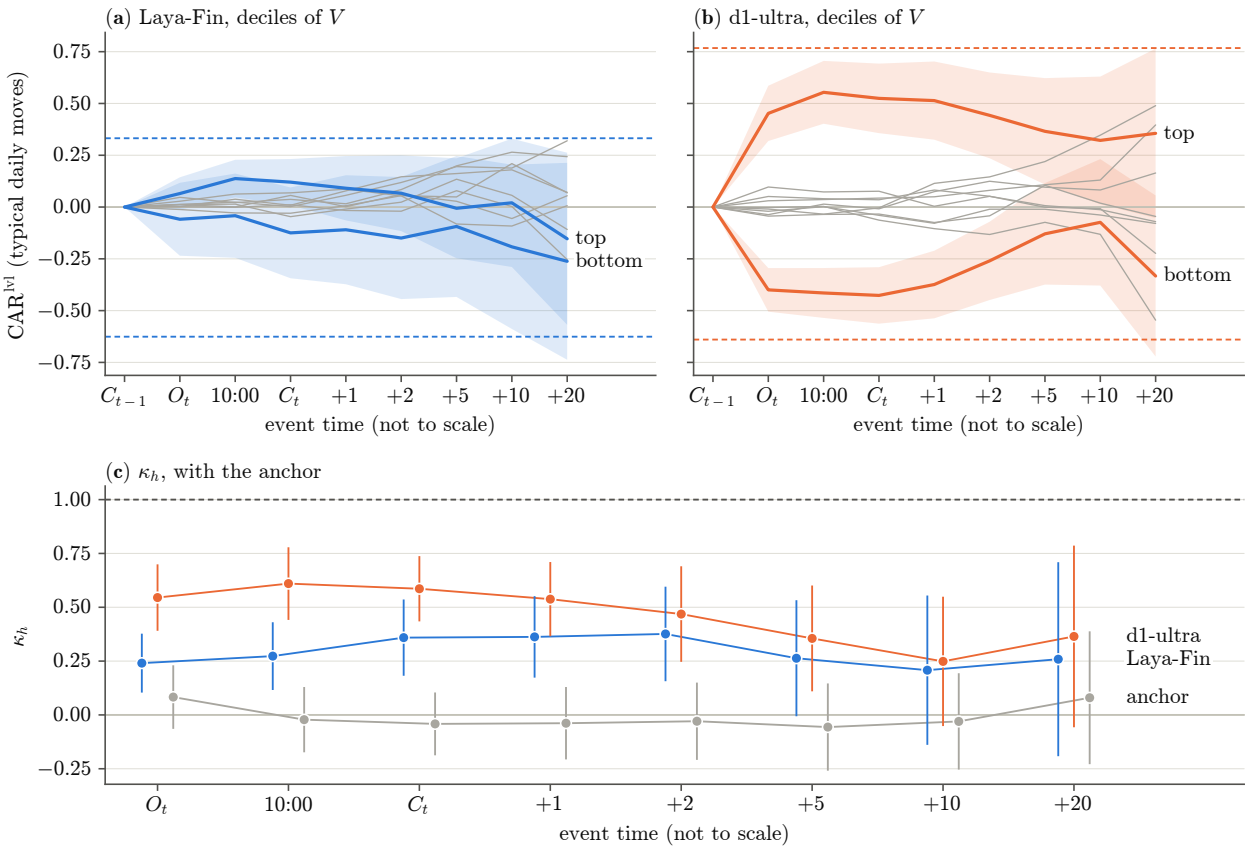


Figure A7: Convergence in detail, every decile and the anchor. (a, b) $\mathrm{CAR}^{\mathrm{lvl}}$ of every decile of V for Laya-Fin and d1-ultra (gray: deciles 2 to 9; the extreme deciles with SE bands and their mean V dashed), on one scale. Laya-Fin's deciles are ordered like d1-ultra's but smaller and noisier: its top decile (mean V 0.33) stands at +0.12 at the close. (c) The ladder κ_h with the anchor, whose readings are uninformative (near-constant outputs): its κ_h is about zero throughout (−0.04 at the close). Event time is not to scale; d1-ultra is model-specific (§1.3).

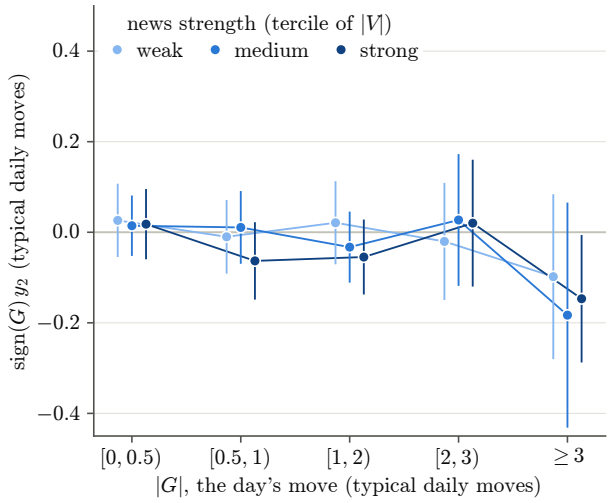


Figure A8: Laya-Fin's drift after the close by the size of the move. Figure 4 (b) for Laya-Fin: $\text{sign}(G)\, y_2$ by bin of $|G|$, one line per tercile of news strength $|V|$, with 95% intervals. Moves with $|G| \geq 3$ reverse slightly in every tercile (strong -0.15, medium -0.18, weak -0.10); smaller moves do not drift.

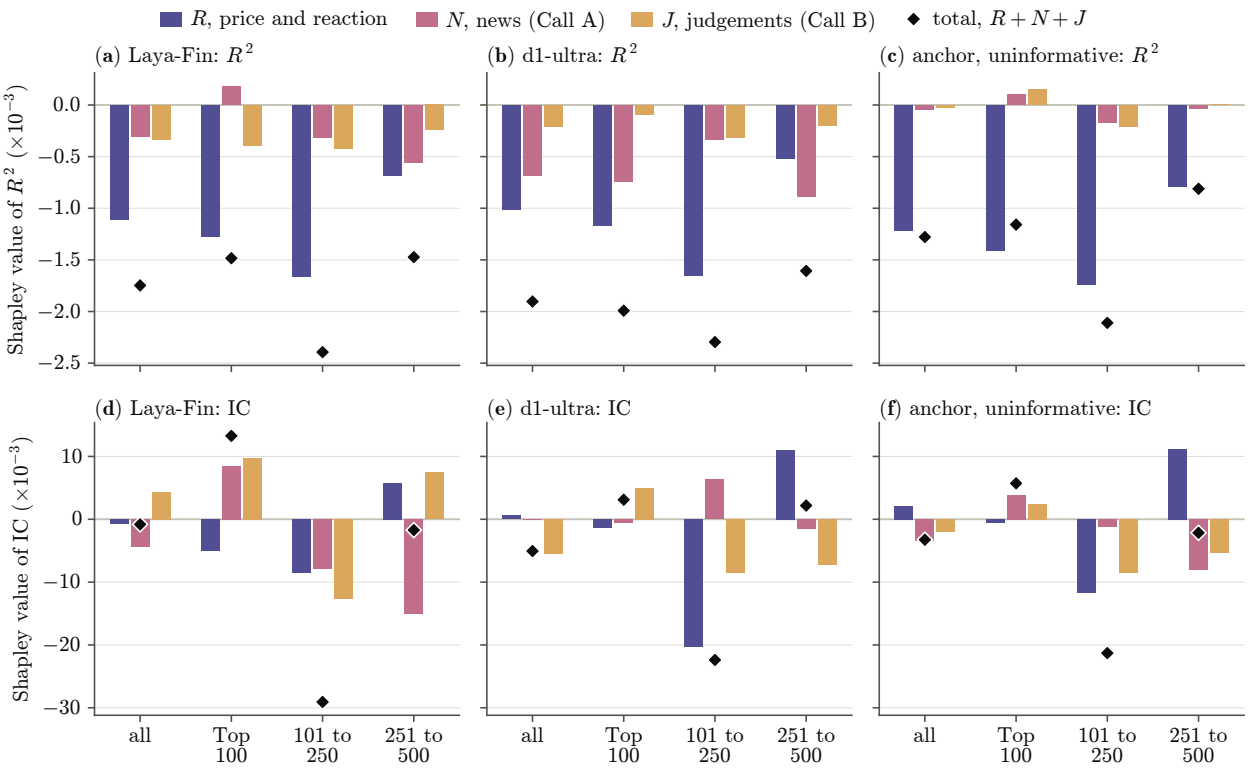


Figure A9: The value of information at the close is about zero for every feature group. Exact Shapley values of the test-period out-of-sample R^2 (a to c) and mean daily IC (d to f) of y_2 at D2 over the feature groups R (price and reaction), N (news, Call A) and J (judgements, Call B), for all member-days and by size tier, with the full model's total $R + N + J$ (diamond), from the H3 ridge fits (§11.5); values in units of 10^{-3}, and each row on one scale. One column per instrument: Laya-Fin, d1-ultra (model-specific) and the anchor (uninformative, near-constant outputs). Every R^2 total is negative, in every tier; for all member-days it is -0.0017 for Laya-Fin, -0.0019 for d1-ultra and -0.0013 for the anchor, so the Shapley shares are shares of nothing and are not drawn. The IC totals are small and of both signs, from -0.029 (Laya-Fin, 101 to 250) to 0.013 (Laya-Fin, Top-100). H3 is not supported for Laya-Fin (Holm p 0.269).

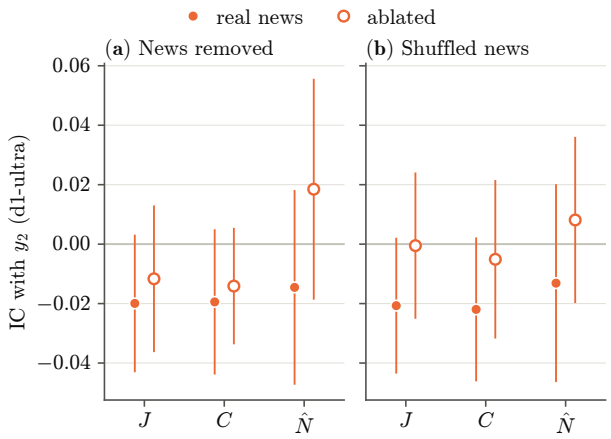


Figure A10: Call B's judgments carry no information about the drift, with or without the news. Call B ablations for d1-ultra (S8, on the same test events): the IC of J, C and $\hat{N}$ with y_2, real news (filled) against (a) the news removed and (b) shuffled news (hollow), with 95% intervals (± 1.96 Newey–West SE). J's IC is -0.02 with the news and -0.01 without it, and -0.02 against -0.00 with shuffled news. d1-ultra is model-specific (§1.3).

Table A13: The over/under regression for every model, decision point and $|G|$ bin.

Model, regression	b	SE	p_b	c	SE	p_c	p_b DK	p_c DK	Events
Laya-Fin, D2	−0.044	(0.149)	0.616	−0.015	(0.012)	0.110	0.615	0.117	17,188
Laya-Fin, D1	0.229	(0.148)	0.062	0.013	(0.012)	0.861	0.105	0.824	13,203
Laya-Fin, D0	0.278	(0.212)	0.096	—	—	—	0.121	—	13,439
Laya-Fin, D2, $\|G\| \in [0,1)$	−0.187	(0.242)	0.779	−0.016	(0.037)	0.338	0.758	0.330	9,795
Laya-Fin, D2, $\|G\| \in [1,2)$	−0.172	(0.273)	0.735	−0.025	(0.026)	0.171	0.736	0.179	4,647
Laya-Fin, D2, $\|G\| \in [2,\infty)$	0.092	(0.254)	0.358	−0.013	(0.012)	0.138	0.315	0.157	2,746
anchor, D2	0.748	(1.377)	0.294	−0.015	(0.012)	0.111	0.341	0.121	17,188
anchor, D1	1.964	(2.482)	0.215	0.014	(0.012)	0.874	0.269	0.838	13,203
anchor, D0	−4.510	(3.236)	0.917	—	—	—	0.898	—	13,439
anchor, D2, $\|G\| \in [0,1)$	−0.086	(3.523)	0.510	−0.016	(0.037)	0.331	0.513	0.323	9,795
anchor, D2, $\|G\| \in [1,2)$	1.336	(3.596)	0.355	−0.026	(0.026)	0.167	0.371	0.174	4,647
anchor, D2, $\|G\| \in [2,\infty)$	0.273	(2.604)	0.458	−0.013	(0.012)	0.150	0.456	0.168	2,746
Laya zero-shot, D2	—	—	—	—	—	—	—	—	—
Laya zero-shot, D1	−0.001	(0.063)	0.504	0.014	(0.012)	0.871	0.504	0.833	13,203
Laya zero-shot, D0	−0.021	(0.069)	0.619	—	—	—	0.608	—	13,439
d1-ultra, D2	−0.076	(0.031)	0.991	−0.011	(0.012)	0.171	0.985	0.175	17,188
d1-ultra, D1	−0.018	(0.032)	0.711	0.015	(0.012)	0.884	0.707	0.853	13,203
d1-ultra, D0	0.022	(0.037)	0.277	—	—	—	0.280	—	13,439
d1-ultra, D2, $\|G\| \in [0,1)$	−0.067	(0.043)	0.939	−0.015	(0.037)	0.348	0.898	0.340	9,795
d1-ultra, D2, $\|G\| \in [1,2)$	−0.086	(0.053)	0.947	−0.023	(0.026)	0.189	0.972	0.196	4,647
d1-ultra, D2, $\|G\| \in [2,\infty)$	−0.080	(0.056)	0.920	−0.009	(0.013)	0.237	0.948	0.249	2,746
span-02‡, D2	—	—	—	—	—	—	—	—	—
span-02‡, D1	−0.010	(0.023)	0.666	0.014	(0.012)	0.882	0.660	0.850	13,203
span-02‡, D0	0.013	(0.023)	0.282	—	—	—	0.293	—	13,439
drex-v2.0, D2	—	—	—	—	—	—	—	—	—
drex-v2.0, D1	−0.032	(0.026)	0.889	0.015	(0.012)	0.890	0.859	0.857	13,203
drex-v2.0, D0	0.011	(0.030)	0.356	—	—	—	0.354	—	13,439

—: not estimated for that row (at D0 there is no G, so no c).

DK: Driscoll–Kraay standard errors (5 lags) as robustness.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

‡span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

Table A14: Secondary hypotheses: only H5 survives Benjamini–Hochberg for Laya-Fin.

Model, test	Estimate	95% CI	SE	p	BH q	Pooled q
Laya-Fin, H5	0.0025	—	—	0.010	0.058	0.078
Laya-Fin, H6	—	—	—	—	—	—
Laya-Fin, H7	0.0146	[−0.0068, 0.0363]	—	0.097	0.238	0.317
Laya-Fin, H2-C, κ at h^*	0.2588	—	(0.2299)	0.131	0.238	0.317
Laya-Fin, H2-C, κ increase	0.0182	[−0.3614, 0.4903]	—	0.467	0.561	0.748
Laya-Fin, H2-C, λ_{t+5}	0.0265	—	(0.0264)	0.159	0.238	0.317
Laya-Fin, H2-C, naive tracking	1.0007	[0.9995, 1.0018]	—	0.867	0.867	0.946
Laya-Fin, A3.4	—	—	—	—	—	—
anchor, H5	0.0003	—	—	0.072	0.358	—
anchor, H6	—	—	—	—	—	—
anchor, H7	0.0024	[−0.0294, 0.0355]	—	0.442	—	—
anchor, H2-C, κ at h^*	0.0798	—	(0.1574)	0.307	0.511	—
anchor, H2-C, κ increase	−0.0031	[−0.2669, 0.4215]	—	0.514	0.643	—
anchor, H2-C, λ_{t+5}	0.0270	—	(0.0266)	0.156	0.390	—
anchor, H2-C, naive tracking	1.0020	[1.0002, 1.0036]	—	0.992	0.992	—
anchor, A3.4	—	—	—	—	—	—
d1-ultra, H5	0.0082	—	—	<0.001	<0.001	—
d1-ultra, H6	0.0104	—	(0.0192)	0.705	0.946	0.940
d1-ultra, H7	—	—	—	—	—	—
d1-ultra, H2-C, κ at h^*	0.3645	—	(0.2152)	0.046	0.163	—
d1-ultra, H2-C, κ increase	−0.1805	[−0.5343, 0.2158]	—	0.834	0.946	—
d1-ultra, H2-C, λ_{t+5}	0.0216	—	(0.0262)	0.206	0.480	—
d1-ultra, H2-C, naive tracking	1.0005	[0.9993, 1.0017]	—	0.781	0.946	—
d1-ultra, A3.4	−0.0619	—	(0.0382)	0.946	0.946	0.946

—: the source table has no value for the row; the statistic does not apply to it or was not computed.

BH at $q = 0.10$ within each instrument; the pooled family adds d1-ultra's H6 and A3.4 to Laya-Fin's. The anchor's H7 is reported outside its family, so it has no q. Each test has either an SE or a 95% CI, as NA-12 states it.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A15: Heterogeneity of H2: no cut differs.

Model, cut	Events	Wald F	p	BH q	Δb	SE	Δc	SE
Laya-Fin, Top 100	4,516	1.83	0.164	0.741	0.490	(0.281)	−0.025	(0.021)
Laya-Fin, 101 to 250	5,121	0.10	0.904	0.904	−0.008	(0.258)	−0.007	(0.016)
Laya-Fin, 251 to 500	7,551	1.71	0.185	0.741	−0.326	(0.253)	0.023	(0.016)
Laya-Fin, quantified	15,030	0.66	0.521	0.833	0.470	(0.609)	0.022	(0.026)
Laya-Fin, small move	9,795	0.29	0.751	0.859	−0.193	(0.255)	0.005	(0.035)
Laya-Fin, press wire	9,655	0.75	0.477	0.833	−0.058	(0.330)	−0.024	(0.020)
Laya-Fin, low usual coverage	8,387	0.44	0.645	0.859	0.040	(0.247)	0.014	(0.016)
Laya-Fin, busy day	7,805	0.93	0.398	0.833	−0.194	(0.263)	−0.019	(0.019)
anchor, Top 100	4,516	2.18	0.117	0.469	6.973	(4.120)	−0.023	(0.020)
anchor, 101 to 250	5,121	0.52	0.594	0.679	−2.559	(2.719)	−0.007	(0.016)
anchor, 251 to 500	7,551	1.18	0.312	0.679	−2.500	(3.035)	0.021	(0.015)
anchor, quantified	15,030	2.84	0.062	0.469	7.109	(3.308)	0.024	(0.026)
anchor, small move	9,795	0.01	0.992	0.992	0.018	(2.650)	0.004	(0.035)
anchor, press wire	9,655	0.79	0.457	0.679	−1.650	(3.084)	−0.024	(0.020)
anchor, low usual coverage	8,387	0.75	0.477	0.679	2.272	(2.652)	0.014	(0.016)
anchor, busy day	7,805	0.58	0.561	0.679	1.140	(2.570)	−0.020	(0.019)
d1-ultra, Top 100	4,516	1.10	0.337	0.719	0.059	(0.060)	−0.024	(0.020)
d1-ultra, 101 to 250	5,121	0.88	0.418	0.719	0.063	(0.055)	−0.010	(0.016)
d1-ultra, 251 to 500	7,551	3.36	0.038	0.301	−0.095	(0.050)	0.026	(0.015)
d1-ultra, quantified	15,030	0.53	0.591	0.719	0.014	(0.062)	0.023	(0.027)
d1-ultra, small move	9,795	0.15	0.857	0.857	0.023	(0.044)	0.002	(0.035)
d1-ultra, press wire	9,655	0.67	0.515	0.719	−0.021	(0.049)	−0.022	(0.020)
d1-ultra, low usual coverage	8,387	2.52	0.084	0.336	0.096	(0.046)	0.010	(0.016)
d1-ultra, busy day	7,805	0.47	0.629	0.719	0.002	(0.043)	−0.019	(0.019)

Wald test of $\hat{N} \cdot 1[\text{cut}]$ and $G \cdot 1[\text{cut}]$ added to H2 at D2; BH over each model's eight cuts. Events: the D2 events in the cut.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A16: The convergence ladder κ_h in every variant.

Model, variant, rung	κ_h	SE	p	p DK	Events
Laya-Fin, ladder, O_t	0.241	(0.070)	<0.001	<0.001	13,439
Laya-Fin, ladder, 10:00	0.273	(0.080)	<0.001	<0.001	13,287
Laya-Fin, ladder, C_t	0.359	(0.090)	<0.001	<0.001	13,748
Laya-Fin, ladder, t+1	0.362	(0.097)	<0.001	<0.001	13,615
Laya-Fin, ladder, t+2	0.376	(0.112)	<0.001	<0.001	13,439
Laya-Fin, ladder, t+5	0.263	(0.137)	0.029	0.026	12,850
Laya-Fin, ladder, t+10	0.208	(0.177)	0.121	0.124	11,993
Laya-Fin, ladder, t+20	0.259	(0.230)	0.131	0.136	11,201
Laya-Fin, ladder, O_t to t+20	0.018	—	0.467	—	—
Laya-Fin, ladder, clean windows, O_t	0.241	(0.070)	<0.001	<0.001	13,439
Laya-Fin, ladder, clean windows, 10:00	0.273	(0.080)	<0.001	<0.001	13,287
Laya-Fin, ladder, clean windows, C_t	0.359	(0.090)	<0.001	<0.001	13,748
Laya-Fin, ladder, clean windows, t+1	0.325	(0.166)	0.026	0.024	7,705
Laya-Fin, ladder, clean windows, t+2	0.538	(0.279)	0.028	0.033	5,130
Laya-Fin, ladder, clean windows, t+5	0.114	(0.346)	0.371	0.383	2,091
Laya-Fin, ladder, clean windows, t+10	0.338	(1.005)	0.369	0.358	853
Laya-Fin, ladder, clean windows, t+20	−0.715	(4.653)	0.561	0.555	205
Laya-Fin, ladder, clean windows, O_t to t+20	−0.955	—	0.550	—	—
Laya-Fin, ladder, $h^* = t$+10, O_t	0.241	(0.070)	<0.001	<0.001	13,439
Laya-Fin, ladder, $h^* = t$+10, 10:00	0.273	(0.080)	<0.001	<0.001	13,287
Laya-Fin, ladder, $h^* = t$+10, C_t	0.359	(0.090)	<0.001	<0.001	13,748
Laya-Fin, ladder, $h^* = t$+10, t+1	0.362	(0.097)	<0.001	<0.001	13,615
Laya-Fin, ladder, $h^* = t$+10, t+2	0.376	(0.112)	<0.001	<0.001	13,439
Laya-Fin, ladder, $h^* = t$+10, t+5	0.263	(0.137)	0.029	0.026	12,850
Laya-Fin, ladder, $h^* = t$+10, t+10	0.208	(0.177)	0.121	0.124	11,993
Laya-Fin, ladder, $h^* = t$+10, t+20	0.259	(0.230)	0.131	0.136	11,201
Laya-Fin, ladder, $h^* = t$+10, O_t to t+10	−0.033	—	0.581	—	—
Laya-Fin, ladder, plateau, t+16 to t+20	0.273	(0.218)	0.106	0.112	11,201
anchor, ladder, O_t	0.083	(0.075)	0.136	0.150	13,439
anchor, ladder, 10:00	−0.022	(0.077)	0.612	0.608	13,287
anchor, ladder, C_t	−0.042	(0.074)	0.712	0.699	13,748
anchor, ladder, t+1	−0.039	(0.086)	0.673	0.668	13,615
anchor, ladder, t+2	−0.029	(0.092)	0.625	0.636	13,439
anchor, ladder, t+5	−0.056	(0.103)	0.707	0.699	12,850
anchor, ladder, t+10	−0.030	(0.114)	0.604	0.602	11,993
anchor, ladder, t+20	0.080	(0.157)	0.307	0.340	11,201
anchor, ladder, O_t to t+20	−0.003	—	0.514	—	—
anchor, ladder, clean windows, O_t	0.083	(0.075)	0.136	0.150	13,439
anchor, ladder, clean windows, 10:00	−0.022	(0.077)	0.612	0.608	13,287
anchor, ladder, clean windows, C_t	−0.042	(0.074)	0.712	0.699	13,748
anchor, ladder, clean windows, t+1	0.316	(0.179)	0.039	0.033	7,705
anchor, ladder, clean windows, t+2	0.806	(0.167)	<0.001	<0.001	5,130
anchor, ladder, clean windows, t+5	0.567	(0.298)	0.030	0.021	2,091
anchor, ladder, clean windows, t+10	0.234	(1.573)	0.441	0.432	853
anchor, ladder, clean windows, t+20	7.068	(13.111)	0.296	0.308	205
anchor, ladder, clean windows, O_t to t+20	6.985	—	0.212	—	—
anchor, ladder, $h^* = t$+10, O_t	0.083	(0.075)	0.136	0.150	13,439
anchor, ladder, $h^* = t$+10, 10:00	−0.022	(0.077)	0.612	0.608	13,287
anchor, ladder, $h^* = t$+10, C_t	−0.042	(0.074)	0.712	0.699	13,748
anchor, ladder, $h^* = t$+10, t+1	−0.039	(0.086)	0.673	0.668	13,615
anchor, ladder, $h^* = t$+10, t+2	−0.029	(0.092)	0.625	0.636	13,439
anchor, ladder, $h^* = t$+10, t+5	−0.056	(0.103)	0.707	0.699	12,850
anchor, ladder, $h^* = t$+10, t+10	−0.030	(0.114)	0.604	0.602	11,993
anchor, ladder, $h^* = t$+10, t+20	0.080	(0.157)	0.307	0.340	11,201

continued on the next page

Table A16, continued

Model, variant, rung	κ_h	SE	p	p DK	Events
anchor, ladder, $h^* = t{+}10$, O_t to $t{+}10$	−0.113	—	0.838	—	—
anchor, ladder, plateau, $t{+}16$ to $t{+}20$	0.077	(0.149)	0.302	0.326	11,201
d1-ultra, ladder, O_t	0.545	(0.079)	<0.001	<0.001	13,439
d1-ultra, ladder, 10:00	0.610	(0.086)	<0.001	<0.001	13,287
d1-ultra, ladder, C_t	0.586	(0.077)	<0.001	<0.001	13,748
d1-ultra, ladder, $t{+}1$	0.538	(0.088)	<0.001	<0.001	13,615
d1-ultra, ladder, $t{+}2$	0.468	(0.113)	<0.001	<0.001	13,439
d1-ultra, ladder, $t{+}5$	0.355	(0.125)	0.003	0.011	12,850
d1-ultra, ladder, $t{+}10$	0.249	(0.153)	0.054	0.077	11,993
d1-ultra, ladder, $t{+}20$	0.365	(0.215)	0.046	0.044	11,201
d1-ultra, ladder, O_t to $t{+}20$	−0.181	—	0.834	—	—
d1-ultra, ladder, clean windows, O_t	0.545	(0.079)	<0.001	<0.001	13,439
d1-ultra, ladder, clean windows, 10:00	0.610	(0.086)	<0.001	<0.001	13,287
d1-ultra, ladder, clean windows, C_t	0.586	(0.077)	<0.001	<0.001	13,748
d1-ultra, ladder, clean windows, $t{+}1$	0.175	(0.094)	0.032	0.064	7,705
d1-ultra, ladder, clean windows, $t{+}2$	0.164	(0.123)	0.093	0.131	5,130
d1-ultra, ladder, clean windows, $t{+}5$	−0.107	(0.280)	0.649	0.641	2,091
d1-ultra, ladder, clean windows, $t{+}10$	0.360	(0.751)	0.317	0.312	853
d1-ultra, ladder, clean windows, $t{+}20$	−1.225	(2.251)	0.706	0.697	205
d1-ultra, ladder, clean windows, O_t to $t{+}20$	−1.770	—	0.875	—	—
d1-ultra, ladder, $h^* = t{+}10$, O_t	0.545	(0.079)	<0.001	<0.001	13,439
d1-ultra, ladder, $h^* = t{+}10$, 10:00	0.610	(0.086)	<0.001	<0.001	13,287
d1-ultra, ladder, $h^* = t{+}10$, C_t	0.586	(0.077)	<0.001	<0.001	13,748
d1-ultra, ladder, $h^* = t{+}10$, $t{+}1$	0.538	(0.088)	<0.001	<0.001	13,615
d1-ultra, ladder, $h^* = t{+}10$, $t{+}2$	0.468	(0.113)	<0.001	<0.001	13,439
d1-ultra, ladder, $h^* = t{+}10$, $t{+}5$	0.355	(0.125)	0.003	0.011	12,850
d1-ultra, ladder, $h^* = t{+}10$, $t{+}10$	0.249	(0.153)	0.054	0.077	11,993
d1-ultra, ladder, $h^* = t{+}10$, $t{+}20$	0.365	(0.215)	0.046	0.044	11,201
d1-ultra, ladder, $h^* = t{+}10$, O_t to $t{+}10$	−0.296	—	0.983	—	—
d1-ultra, ladder, plateau, $t{+}16$ to $t{+}20$	0.367	(0.206)	0.039	0.043	11,201

—: the source table has no value for the row; the statistic does not apply to it or was not computed.
Variants: the ladder; clean windows (no other selected event in the window); $h^* = t{+}10$; the plateau.
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A17: Gap closure λ_h and ψ_h at D1 and D2.

Model, point, variant, rung	λ_h	SE	p	ψ_h	p (two-sided)	Events
Laya-Fin, D2, all tiers, t+1	−0.007	(0.012)	0.728	−0.002	0.962	17,695
Laya-Fin, D2, all tiers, t+5	0.026	(0.026)	0.159	−0.067	0.460	17,188
Laya-Fin, D2, all tiers, t+20	0.050	(0.051)	0.164	−0.243	0.209	14,985
Laya-Fin, D2, clean, t+1	0.006	(0.016)	0.350	0.085	0.141	10,266
Laya-Fin, D2, clean, t+5	0.004	(0.063)	0.473	−0.201	0.487	2,877
Laya-Fin, D2, clean, t+20	0.358	(0.711)	0.308	0.639	0.822	314
Laya-Fin, D2, all tiers, half-life (sessions)	∞	—	—	—	—	—
Laya-Fin, D1, all tiers, C_t	−0.011	(0.013)	0.805	0.064	0.095	13,287
Laya-Fin, D1, all tiers, t+1	−0.024	(0.020)	0.885	0.137	0.058	13,287
Laya-Fin, D1, all tiers, t+5	−0.002	(0.034)	0.527	0.048	0.718	12,711
Laya-Fin, D1, all tiers, t+20	−0.013	(0.062)	0.585	−0.003	0.989	11,087
Laya-Fin, D1, clean, C_t	−0.011	(0.013)	0.805	0.064	0.095	13,287
Laya-Fin, D1, clean, t+1	0.022	(0.030)	0.232	0.189	0.084	7,554
Laya-Fin, D1, clean, t+5	−0.014	(0.093)	0.558	0.046	0.885	2,073
Laya-Fin, D1, clean, t+20	−0.225	(1.025)	0.587	−0.191	0.964	203
Laya-Fin, D1, all tiers, half-life (sessions)	∞	—	—	—	—	—
anchor, D2, all tiers, t+1	−0.007	(0.012)	0.730	0.049	0.428	17,695
anchor, D2, all tiers, t+5	0.027	(0.027)	0.156	−0.034	0.813	17,188
anchor, D2, all tiers, t+20	0.052	(0.051)	0.158	0.157	0.481	14,985
anchor, D2, clean, t+1	0.005	(0.016)	0.377	0.292	0.039	10,266
anchor, D2, clean, t+5	0.002	(0.063)	0.485	−0.621	0.186	2,877
anchor, D2, clean, t+20	0.310	(0.682)	0.326	3.333	0.487	314
anchor, D2, all tiers, half-life (sessions)	∞	—	—	—	—	—
anchor, D1, all tiers, C_t	−0.012	(0.013)	0.819	0.010	0.780	13,287
anchor, D1, all tiers, t+1	−0.025	(0.020)	0.899	0.039	0.486	13,287
anchor, D1, all tiers, t+5	−0.003	(0.034)	0.533	−0.043	0.624	12,711
anchor, D1, all tiers, t+20	−0.013	(0.062)	0.582	0.053	0.711	11,087
anchor, D1, clean, C_t	−0.012	(0.013)	0.819	0.010	0.780	13,287
anchor, D1, clean, t+1	0.020	(0.030)	0.255	0.280	0.011	7,554
anchor, D1, clean, t+5	−0.010	(0.093)	0.542	0.319	0.354	2,073
anchor, D1, clean, t+20	−0.302	(0.949)	0.625	9.321	0.410	203
anchor, D1, all tiers, half-life (sessions)	∞	—	—	—	—	—
d1-ultra, D2, all tiers, t+1	−0.009	(0.012)	0.773	−0.041	0.306	17,695
d1-ultra, D2, all tiers, t+5	0.022	(0.026)	0.206	−0.160	0.094	17,188
d1-ultra, D2, all tiers, t+20	0.049	(0.051)	0.170	−0.134	0.423	14,985
d1-ultra, D2, clean, t+1	0.003	(0.016)	0.418	−0.065	0.177	10,266
d1-ultra, D2, clean, t+5	−0.003	(0.063)	0.521	−0.416	0.019	2,877
d1-ultra, D2, clean, t+20	0.325	(0.693)	0.320	−3.506	0.166	314
d1-ultra, D2, all tiers, half-life (sessions)	∞	—	—	—	—	—
d1-ultra, D1, all tiers, C_t	−0.013	(0.013)	0.829	−0.003	0.940	13,287
d1-ultra, D1, all tiers, t+1	−0.028	(0.020)	0.921	−0.051	0.316	13,287
d1-ultra, D1, all tiers, t+5	−0.012	(0.034)	0.636	−0.234	0.036	12,711
d1-ultra, D1, all tiers, t+20	−0.020	(0.062)	0.624	−0.183	0.372	11,087
d1-ultra, D1, clean, C_t	−0.013	(0.013)	0.829	−0.003	0.940	13,287
d1-ultra, D1, clean, t+1	0.014	(0.030)	0.317	−0.182	0.008	7,554
d1-ultra, D1, clean, t+5	−0.029	(0.094)	0.622	−0.396	0.149	2,073
d1-ultra, D1, clean, t+20	−0.232	(1.008)	0.591	−1.009	0.661	203
d1-ultra, D1, all tiers, half-life (sessions)	∞	—	—	—	—	—

—: the half-life is a rung, not an estimate, so it has no standard error, p, ψ_h or event count.
The half-life rows hold the first rung with $\lambda_h \geq 0.5$ (∞: not reached within t+20).
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A18: Tracking error of V: no better than a naive zero.

Model, rung	RMSE	Ratio to naive	p	Ratio to price-only	p	Events
Laya-Fin, O_t	1.381	1.019	1.000	1.014	0.996	13,439
Laya-Fin, 10:00	1.661	1.012	0.999	1.007	0.978	13,287
Laya-Fin, C_t	2.019	1.005	0.949	1.003	0.800	13,748
Laya-Fin, t+1	2.708	1.003	0.952	1.002	0.848	13,615
Laya-Fin, t+2	3.218	1.002	0.896	1.001	0.699	13,439
Laya-Fin, t+5	4.279	1.002	0.967	1.001	0.839	12,850
Laya-Fin, t+10	5.454	1.001	0.950	1.001	0.788	11,993
Laya-Fin, t+20	7.047	1.001	0.867	1.000	0.625	11,201
anchor, O_t	1.427	1.053	1.000	1.047	1.000	13,439
anchor, 10:00	1.715	1.044	1.000	1.040	1.000	13,287
anchor, C_t	2.073	1.033	1.000	1.030	1.000	13,748
anchor, t+1	2.747	1.018	1.000	1.016	1.000	13,615
anchor, t+2	3.252	1.012	1.000	1.011	1.000	13,439
anchor, t+5	4.302	1.007	1.000	1.007	1.000	12,850
anchor, t+10	5.471	1.004	1.000	1.004	1.000	11,993
anchor, t+20	7.056	1.002	0.992	1.002	0.969	11,201
d1-ultra, O_t	1.350	0.997	0.290	0.991	0.032	13,439
d1-ultra, 10:00	1.633	0.995	0.094	0.990	0.011	13,287
d1-ultra, C_t	2.002	0.997	0.164	0.994	0.034	13,748
d1-ultra, t+1	2.699	1.000	0.402	0.998	0.232	13,615
d1-ultra, t+2	3.216	1.001	0.657	1.000	0.483	13,439
d1-ultra, t+5	4.278	1.002	0.910	1.001	0.772	12,850
d1-ultra, t+10	5.455	1.001	0.927	1.001	0.794	11,993
d1-ultra, t+20	7.045	1.000	0.781	1.000	0.508	11,201

RMSE of $\mathrm{CAR}_h^{\mathrm{lvl}} - V$ relative to a naive 0 and to the price-only V.
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A19: Convergence by cut.

Model, cut, level	κ_{t+20}	p	κ_{C_t}	p	λ_{t+5}	p	Events (κ)	Events (λ)
Laya-Fin, tier, Top 100	0.711	0.088	0.749	<0.001	0.065	0.074	2,872	4,516
Laya-Fin, tier, 101 to 250	−0.182	0.657	0.228	0.103	0.037	0.170	3,294	5,121
Laya-Fin, tier, 251 to 500	0.325	0.050	0.252	0.016	0.003	0.465	5,035	7,551
Laya-Fin, vendor quantified flag, no	0.044	0.479	0.508	0.143	0.077	0.119	1,310	2,158
Laya-Fin, vendor quantified flag, yes	0.248	0.151	0.354	<0.001	0.023	0.203	9,891	15,030
Laya-Fin, $\|G\|$, D2 gap, $[0, 1)$	—	—	—	—	0.037	0.329	—	9,795
Laya-Fin, $\|G\|$, D2 gap, $[1, 2)$	—	—	—	—	0.055	0.179	—	4,647
Laya-Fin, $\|G\|$, D2 gap, $[2, \infty)$	—	—	—	—	0.020	0.236	—	2,746
Laya-Fin, $\|G\|$, D1 gap, $[0, 1)$	—	—	—	—	0.225	0.053	—	7,006
Laya-Fin, $\|G\|$, D1 gap, $[1, 2)$	—	—	—	—	0.020	0.412	—	3,439
Laya-Fin, $\|G\|$, D1 gap, $[2, \infty)$	—	—	—	—	−0.025	0.704	—	2,184
anchor, tier, Top 100	0.589	0.046	0.090	0.262	0.065	0.067	2,872	4,516
anchor, tier, 101 to 250	0.047	0.440	−0.147	0.825	0.038	0.164	3,294	5,121
anchor, tier, 251 to 500	−0.216	0.846	−0.060	0.680	0.004	0.448	5,035	7,551
anchor, vendor quantified flag, no	−0.726	0.698	0.516	0.013	0.073	0.134	1,310	2,158
anchor, vendor quantified flag, yes	0.077	0.322	−0.062	0.785	0.023	0.200	9,891	15,030
anchor, $\|G\|$, D2 gap, $[0, 1)$	—	—	—	—	0.036	0.333	—	9,795
anchor, $\|G\|$, D2 gap, $[1, 2)$	—	—	—	—	0.055	0.176	—	4,647
anchor, $\|G\|$, D2 gap, $[2, \infty)$	—	—	—	—	0.020	0.233	—	2,746
anchor, $\|G\|$, D1 gap, $[0, 1)$	—	—	—	—	0.225	0.053	—	7,006
anchor, $\|G\|$, D1 gap, $[1, 2)$	—	—	—	—	0.020	0.414	—	3,439
anchor, $\|G\|$, D1 gap, $[2, \infty)$	—	—	—	—	−0.027	0.720	—	2,184
d1-ultra, tier, Top 100	0.734	0.041	0.648	<0.001	0.060	0.092	2,872	4,516
d1-ultra, tier, 101 to 250	0.046	0.459	0.500	0.002	0.040	0.119	3,294	5,121
d1-ultra, tier, 251 to 500	0.346	0.111	0.607	<0.001	−0.006	0.576	5,035	7,551
d1-ultra, vendor quantified flag, no	0.264	0.358	0.378	0.081	0.076	0.126	1,310	2,158
d1-ultra, vendor quantified flag, yes	0.390	0.039	0.593	<0.001	0.017	0.262	9,891	15,030
d1-ultra, model quantified, narrative	−0.039	0.549	0.176	0.025	0.020	0.310	7,450	11,665
d1-ultra, model quantified, numbers	0.784	0.002	0.860	<0.001	0.017	0.278	3,751	5,523
d1-ultra, $\|G\|$, D2 gap, $[0, 1)$	—	—	—	—	0.036	0.333	—	9,795
d1-ultra, $\|G\|$, D2 gap, $[1, 2)$	—	—	—	—	0.055	0.179	—	4,647
d1-ultra, $\|G\|$, D2 gap, $[2, \infty)$	—	—	—	—	0.016	0.270	—	2,746
d1-ultra, $\|G\|$, D1 gap, $[0, 1)$	—	—	—	—	0.226	0.053	—	7,006
d1-ultra, $\|G\|$, D1 gap, $[1, 2)$	—	—	—	—	0.018	0.420	—	3,439
d1-ultra, $\|G\|$, D1 gap, $[2, \infty)$	—	—	—	—	−0.033	0.765	—	2,184

—: the source table has no value for the row; the statistic does not apply to it or was not computed.
Events (κ): the D0 events behind κ; events (λ): the gap events behind λ_{t+5}.
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A20: Shapley values of the out-of-sample R^2 and IC of y_2 at D2: every total is about zero.

Model, tier, metric, group	Shapley value	Share	Events
Laya-Fin, all tiers, R^2, R	−0.0011	0.636	17,044
Laya-Fin, all tiers, R^2, N	−0.0003	0.175	17,044
Laya-Fin, all tiers, R^2, J	−0.0003	0.189	17,044
Laya-Fin, all tiers, R^2, R+N+J	−0.0017	—	17,044
Laya-Fin, all tiers, IC, R	−0.0008	0.969	17,044
Laya-Fin, all tiers, IC, N	−0.0043	5.365	17,044
Laya-Fin, all tiers, IC, J	0.0043	−5.334	17,044
Laya-Fin, all tiers, IC, R+N+J	−0.0008	—	17,044
Laya-Fin, Top 100, R^2, R	−0.0013	0.860	4,516
Laya-Fin, Top 100, R^2, N	0.0002	−0.124	4,516
Laya-Fin, Top 100, R^2, J	−0.0004	0.264	4,516
Laya-Fin, Top 100, R^2, R+N+J	−0.0015	—	4,516
Laya-Fin, Top 100, IC, R	−0.0050	−0.376	4,516
Laya-Fin, Top 100, IC, N	0.0085	0.642	4,516
Laya-Fin, Top 100, IC, J	0.0097	0.734	4,516
Laya-Fin, Top 100, IC, R+N+J	0.0133	—	4,516
Laya-Fin, 101 to 250, R^2, R	−0.0017	0.692	5,056
Laya-Fin, 101 to 250, R^2, N	−0.0003	0.133	5,056
Laya-Fin, 101 to 250, R^2, J	−0.0004	0.175	5,056
Laya-Fin, 101 to 250, R^2, R+N+J	−0.0024	—	5,056
Laya-Fin, 101 to 250, IC, R	−0.0086	0.295	5,056
Laya-Fin, 101 to 250, IC, N	−0.0078	0.269	5,056
Laya-Fin, 101 to 250, IC, J	−0.0127	0.435	5,056
Laya-Fin, 101 to 250, IC, R+N+J	−0.0291	—	5,056
Laya-Fin, 251 to 500, R^2, R	−0.0007	0.461	7,472
Laya-Fin, 251 to 500, R^2, N	−0.0006	0.375	7,472
Laya-Fin, 251 to 500, R^2, J	−0.0002	0.164	7,472
Laya-Fin, 251 to 500, R^2, R+N+J	−0.0015	—	7,472
Laya-Fin, 251 to 500, IC, R	0.0058	−3.378	7,472
Laya-Fin, 251 to 500, IC, N	−0.0150	8.722	7,472
Laya-Fin, 251 to 500, IC, J	0.0075	−4.344	7,472
Laya-Fin, 251 to 500, IC, R+N+J	−0.0017	—	7,472
anchor, all tiers, R^2, R	−0.0012	0.951	17,044
anchor, all tiers, R^2, N	0.0000	0.031	17,044
anchor, all tiers, R^2, J	0.0000	0.018	17,044
anchor, all tiers, R^2, R+N+J	−0.0013	—	17,044
anchor, all tiers, IC, R	0.0021	−0.661	17,044
anchor, all tiers, IC, N	−0.0035	1.062	17,044
anchor, all tiers, IC, J	−0.0019	0.599	17,044
anchor, all tiers, IC, R+N+J	−0.0033	—	17,044
anchor, Top 100, R^2, R	−0.0014	1.214	4,516
anchor, Top 100, R^2, N	0.0001	−0.087	4,516
anchor, Top 100, R^2, J	0.0001	−0.127	4,516
anchor, Top 100, R^2, R+N+J	−0.0012	—	4,516
anchor, Top 100, IC, R	−0.0006	−0.100	4,516
anchor, Top 100, IC, N	0.0038	0.670	4,516
anchor, Top 100, IC, J	0.0025	0.430	4,516
anchor, Top 100, IC, R+N+J	0.0057	—	4,516
anchor, 101 to 250, R^2, R	−0.0017	0.823	5,056
anchor, 101 to 250, R^2, N	−0.0002	0.079	5,056
anchor, 101 to 250, R^2, J	−0.0002	0.099	5,056
anchor, 101 to 250, R^2, R+N+J	−0.0021	—	5,056
anchor, 101 to 250, IC, R	−0.0117	0.548	5,056
anchor, 101 to 250, IC, N	−0.0012	0.056	5,056

continued on the next page

Table A20, continued

Model, tier, metric, group	Shapley value	Share	Events
anchor, 101 to 250, IC, J	−0.0084	0.397	5,056
anchor, 101 to 250, IC, R+N+J	−0.0213	—	5,056
anchor, 251 to 500, R^2, R	−0.0008	0.968	7,472
anchor, 251 to 500, R^2, N	0.0000	0.039	7,472
anchor, 251 to 500, R^2, J	0.0000	−0.007	7,472
anchor, 251 to 500, R^2, R+N+J	−0.0008	—	7,472
anchor, 251 to 500, IC, R	0.0112	−5.216	7,472
anchor, 251 to 500, IC, N	−0.0080	3.704	7,472
anchor, 251 to 500, IC, J	−0.0054	2.512	7,472
anchor, 251 to 500, IC, R+N+J	−0.0021	—	7,472
d1-ultra, all tiers, R^2, R	−0.0010	0.532	17,044
d1-ultra, all tiers, R^2, N	−0.0007	0.359	17,044
d1-ultra, all tiers, R^2, J	−0.0002	0.109	17,044
d1-ultra, all tiers, R^2, R+N+J	−0.0019	—	17,044
d1-ultra, all tiers, IC, R	0.0006	−0.119	17,044
d1-ultra, all tiers, IC, N	−0.0001	0.020	17,044
d1-ultra, all tiers, IC, J	−0.0056	1.099	17,044
d1-ultra, all tiers, IC, R+N+J	−0.0051	—	17,044
d1-ultra, Top 100, R^2, R	−0.0012	0.586	4,516
d1-ultra, Top 100, R^2, N	−0.0007	0.370	4,516
d1-ultra, Top 100, R^2, J	−0.0001	0.044	4,516
d1-ultra, Top 100, R^2, R+N+J	−0.0020	—	4,516
d1-ultra, Top 100, IC, R	−0.0014	−0.454	4,516
d1-ultra, Top 100, IC, N	−0.0005	−0.158	4,516
d1-ultra, Top 100, IC, J	0.0050	1.612	4,516
d1-ultra, Top 100, IC, R+N+J	0.0031	—	4,516
d1-ultra, 101 to 250, R^2, R	−0.0017	0.720	5,056
d1-ultra, 101 to 250, R^2, N	−0.0003	0.144	5,056
d1-ultra, 101 to 250, R^2, J	−0.0003	0.136	5,056
d1-ultra, 101 to 250, R^2, R+N+J	−0.0023	—	5,056
d1-ultra, 101 to 250, IC, R	−0.0202	0.903	5,056
d1-ultra, 101 to 250, IC, N	0.0063	−0.283	5,056
d1-ultra, 101 to 250, IC, J	−0.0085	0.380	5,056
d1-ultra, 101 to 250, IC, R+N+J	−0.0224	—	5,056
d1-ultra, 251 to 500, R^2, R	−0.0005	0.325	7,472
d1-ultra, 251 to 500, R^2, N	−0.0009	0.549	7,472
d1-ultra, 251 to 500, R^2, J	−0.0002	0.126	7,472
d1-ultra, 251 to 500, R^2, R+N+J	−0.0016	—	7,472
d1-ultra, 251 to 500, IC, R	0.0110	5.038	7,472
d1-ultra, 251 to 500, IC, N	−0.0016	−0.714	7,472
d1-ultra, 251 to 500, IC, J	−0.0073	−3.324	7,472
d1-ultra, 251 to 500, IC, R+N+J	0.0022	—	7,472

—: a total has no share.
Shares are of a total that is about zero, so they carry no information.
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A21: Sums of figure-data cells cited in captions.

Quantity	Value
Laya-Fin's events with $\lvert\hat{N}\rvert \geq 0.5$ (its three RG classes)	146

E Explaining realized moves

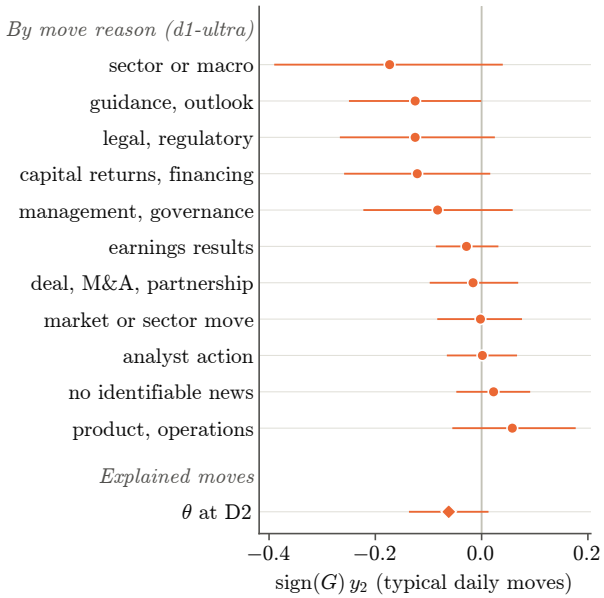


Figure A11: Explained moves do not persist more. Persistence after D2, the mean of $\text{sign}(G)\, y_2$, by d1-ultra's reason for the move (95% intervals), and θ, the D2 regression of persistence on being explained: −0.06 (SE 0.038, p 0.946). No reason persists significantly. Days that d1-ultra judges explained are 16% of event days and carry 41% of their squared moves. d1-ultra is model-specific (§1.3).

Table A22: The explanation ladder for every model and cut.

Model, cut, step	R^2	95% CI	p	ΔR^2	p	Member-days
all tiers, market and sector (Roll)	0.2525	—	—	—	—	68,761
Laya-Fin, all tiers, ex-ante news	0.0013	[−0.0002, 0.0028]	0.045	—	—	68,761
Laya-Fin, all tiers, same-day news	0.0025	[0.0004, 0.0045]	0.010	0.0012	0.010	68,761
Laya-Fin, all tiers, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.5317	—	—	0.0012	—	—
Laya-Fin, event days, ex-ante news	0.0037	[−0.0008, 0.0078]	0.037	—	—	18,291
Laya-Fin, event days, same-day news	0.0063	[0.0001, 0.0120]	0.015	0.0026	0.040	18,291
Laya-Fin, event days, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.5875	—	—	0.0026	—	—
Laya-Fin, non-event days, ex-ante news	0.0000	[−0.0001, 0.0002]	0.389	—	—	50,470
Laya-Fin, Top 100, ex-ante news	0.0060	[0.0000, 0.0132]	0.045	—	—	14,007
Laya-Fin, Top 100, same-day news	0.0050	[0.0014, 0.0091]	0.009	−0.0010	0.737	14,007
Laya-Fin, Top 100, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	1.2030	—	—	−0.0010	—	—
Laya-Fin, 101 to 250, ex-ante news	−0.0007	[−0.0029, 0.0014]	0.724	—	—	20,612
Laya-Fin, 101 to 250, same-day news	0.0018	[−0.0010, 0.0044]	0.080	0.0025	0.008	20,612
Laya-Fin, 101 to 250, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	−0.3820	—	—	0.0025	—	—
Laya-Fin, 251 to 500, ex-ante news	0.0010	[−0.0011, 0.0029]	0.161	—	—	34,142
Laya-Fin, 251 to 500, same-day news	0.0018	[−0.0011, 0.0046]	0.107	0.0008	0.173	34,142
Laya-Fin, 251 to 500, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.5685	—	—	0.0008	—	—
anchor, all tiers, ex-ante news	0.0001	[−0.0001, 0.0004]	0.190	—	—	68,761
anchor, all tiers, same-day news	0.0003	[−0.0001, 0.0008]	0.072	0.0002	0.094	68,761
anchor, all tiers, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.3563	—	—	0.0002	—	—
anchor, event days, ex-ante news	0.0003	[−0.0003, 0.0008]	0.141	—	—	18,291
anchor, event days, same-day news	0.0006	[−0.0005, 0.0017]	0.111	0.0004	0.140	18,291
anchor, event days, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.4187	—	—	0.0004	—	—
anchor, non-event days, ex-ante news	0.0000	[−0.0001, 0.0002]	0.389	—	—	50,470
anchor, Top 100, ex-ante news	−0.0005	[−0.0013, 0.0003]	0.914	—	—	14,007
anchor, Top 100, same-day news	−0.0008	[−0.0019, 0.0003]	0.934	−0.0003	0.880	14,007
anchor, Top 100, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.6497	—	—	−0.0003	—	—
anchor, 101 to 250, ex-ante news	−0.0013	[−0.0022, −0.0004]	0.996	—	—	20,612
anchor, 101 to 250, same-day news	−0.0007	[−0.0021, 0.0005]	0.861	0.0006	0.155	20,612
anchor, 101 to 250, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	1.7733	—	—	0.0006	—	—
anchor, 251 to 500, ex-ante news	0.0000	[−0.0007, 0.0006]	0.558	—	—	34,142
anchor, 251 to 500, same-day news	0.0003	[−0.0014, 0.0019]	0.366	0.0003	0.314	34,142
anchor, 251 to 500, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	−0.1546	—	—	0.0003	—	—
d1-ultra, all tiers, ex-ante news	0.0028	[−0.0005, 0.0059]	0.033	—	—	68,761
d1-ultra, all tiers, same-day news	0.0082	[0.0040, 0.0120]	<0.001	0.0053	<0.001	68,761
d1-ultra, all tiers, explanation (reads the move)	0.1576	[0.1271, 0.1852]	<0.001	0.1494	<0.001	68,761
d1-ultra, all tiers, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.3478	—	—	0.0053	—	—
d1-ultra, event days, ex-ante news	0.0078	[0.0011, 0.0131]	<0.001	—	—	18,291
d1-ultra, event days, same-day news	0.0192	[0.0110, 0.0261]	<0.001	0.0114	<0.001	18,291
d1-ultra, event days, explanation (reads the move)	0.3753	[0.3564, 0.3924]	<0.001	0.3561	<0.001	18,291

continued on the next page

Table A22, continued

Model, cut, step	R^2	95% CI	p	ΔR^2	p	Member-days
d1-ultra, event days, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.4085	—	—	0.0114	—	—
d1-ultra, non-event days, ex-ante news	0.0000	[−0.0001, 0.0002]	0.389	—	—	50,470
d1-ultra, Top 100, ex-ante news	0.0053	[−0.0001, 0.0107]	0.026	—	—	14,007
d1-ultra, Top 100, same-day news	0.0088	[0.0055, 0.0121]	<0.001	0.0035	0.005	14,007
d1-ultra, Top 100, explanation (reads the move)	0.1985	[0.1653, 0.2314]	<0.001	0.1897	<0.001	14,007
d1-ultra, Top 100, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.6017	—	—	0.0035	—	—
d1-ultra, 101 to 250, ex-ante news	0.0017	[−0.0006, 0.0040]	0.065	—	—	20,612
d1-ultra, 101 to 250, same-day news	0.0047	[−0.0010, 0.0102]	0.048	0.0030	0.061	20,612
d1-ultra, 101 to 250, explanation (reads the move)	0.1564	[0.1215, 0.1898]	<0.001	0.1516	<0.001	20,612
d1-ultra, 101 to 250, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.3693	—	—	0.0030	—	—
d1-ultra, 251 to 500, ex-ante news	0.0032	[0.0000, 0.0063]	0.020	—	—	34,142
d1-ultra, 251 to 500, same-day news	0.0066	[0.0024, 0.0105]	<0.001	0.0034	0.001	34,142
d1-ultra, 251 to 500, explanation (reads the move)	0.1401	[0.1088, 0.1662]	<0.001	0.1335	<0.001	34,142
d1-ultra, 251 to 500, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.4906	—	—	0.0034	—	—
d1-ultra, analyst action, ex-ante news	0.0031	[−0.0024, 0.0087]	0.138	—	—	3,629
d1-ultra, analyst action, same-day news	0.0117	[0.0040, 0.0208]	0.007	0.0087	<0.001	3,629
d1-ultra, analyst action, explanation (reads the move)	0.4335	[0.4048, 0.4624]	<0.001	0.4218	<0.001	3,629
d1-ultra, analyst action, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.2617	—	—	0.0087	—	—
d1-ultra, capital returns, financing, ex-ante news	−0.0340	[−0.0856, 0.0097]	0.903	—	—	643
d1-ultra, capital returns, financing, same-day news	−0.0374	[−0.0887, 0.0043]	0.928	−0.0035	0.641	643
d1-ultra, capital returns, financing, explanation (reads the move)	0.3168	[0.2605, 0.3675]	<0.001	0.3542	<0.001	643
d1-ultra, capital returns, financing, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.9075	—	—	−0.0035	—	—
d1-ultra, deal, M&A, partnership, ex-ante news	−0.0049	[−0.0175, 0.0080]	0.773	—	—	1,266
d1-ultra, deal, M&A, partnership, same-day news	0.0144	[0.0004, 0.0288]	0.026	0.0192	<0.001	1,266
d1-ultra, deal, M&A, partnership, explanation (reads the move)	0.3804	[0.3414, 0.4247]	<0.001	0.3660	<0.001	1,266
d1-ultra, deal, M&A, partnership, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	−0.3382	—	—	0.0192	—	—
d1-ultra, earnings results, ex-ante news	0.0141	[0.0046, 0.0216]	<0.001	—	—	7,028
d1-ultra, earnings results, same-day news	0.0265	[0.0122, 0.0393]	<0.001	0.0124	<0.001	7,028
d1-ultra, earnings results, explanation (reads the move)	0.4003	[0.3768, 0.4213]	<0.001	0.3739	<0.001	7,028
d1-ultra, earnings results, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.5327	—	—	0.0124	—	—
d1-ultra, guidance, outlook, ex-ante news	−0.0353	[−0.0716, −0.0059]	0.971	—	—	506

continued on the next page

Table A22, continued

Model, cut, step	R^2	95% CI	p	ΔR^2	p	Member-days
d1-ultra, guidance, outlook, same-day news	−0.0073	[−0.0307, 0.0142]	0.717	0.0279	0.001	506
d1-ultra, guidance, outlook, explanation (reads the move)	−0.0518	[−0.2231, 0.0300]	0.770	−0.0444	0.753	506
d1-ultra, guidance, outlook, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	4.7994	—	—	0.0279	—	—
d1-ultra, legal, regulatory, ex-ante news	−0.0185	[−0.0616, 0.0165]	0.807	—	—	467
d1-ultra, legal, regulatory, same-day news	−0.0147	[−0.0607, 0.0235]	0.732	0.0037	0.080	467
d1-ultra, legal, regulatory, explanation (reads the move)	−1.1605	[−4.6786, 0.4163]	0.769	−1.1458	0.766	467
d1-ultra, legal, regulatory, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	1.2524	—	—	0.0037	—	—
d1-ultra, management, governance, ex-ante news	0.0056	[−0.0076, 0.0165]	0.164	—	—	319
d1-ultra, management, governance, same-day news	0.0022	[−0.0020, 0.0064]	0.153	−0.0034	0.737	319
d1-ultra, management, governance, explanation (reads the move)	0.1698	[0.0991, 0.2492]	<0.001	0.1676	<0.001	319
d1-ultra, management, governance, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	2.5507	—	—	−0.0034	—	—
d1-ultra, other, ex-ante news	−0.0041	[−0.0079, −0.0004]	0.984	—	—	3,010
d1-ultra, other, same-day news	−0.0042	[−0.0116, 0.0038]	0.856	0.0000	0.498	3,010
d1-ultra, other, explanation (reads the move)	0.1943	[0.1657, 0.2299]	<0.001	0.1984	<0.001	3,010
d1-ultra, other, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	0.9883	—	—	0.0000	—	—
d1-ultra, product, operations, ex-ante news	−0.0034	[−0.0101, 0.0031]	0.843	—	—	1,072
d1-ultra, product, operations, same-day news	−0.0031	[−0.0098, 0.0034]	0.823	0.0003	0.013	1,072
d1-ultra, product, operations, explanation (reads the move)	0.3555	[0.3082, 0.3956]	<0.001	0.3586	<0.001	1,072
d1-ultra, product, operations, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	1.0901	—	—	0.0003	—	—
d1-ultra, sector or macro, ex-ante news	−0.0027	[−0.0084, 0.0018]	0.851	—	—	351
d1-ultra, sector or macro, same-day news	0.0307	[−0.0086, 0.0579]	0.009	0.0334	0.003	351
d1-ultra, sector or macro, explanation (reads the move)	0.3492	[0.1449, 0.5054]	<0.001	0.3185	<0.001	351
d1-ultra, sector or macro, ex-ante ÷ ex-post (increment: ex-post − ex-ante)	−0.0880	—	—	0.0334	—	—

—: the source table has no value for the row; the statistic does not apply to it or was not computed.
Out-of-sample R^2 of z_{cc}, cumulative by step; bootstrap CIs.
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A23: Explained share, θ and persistence by reason (d1-ultra): explained moves do not persist more.

Item	Value	Share of events	SE	p	Events
explained share (all)	0.410	0.156	—	—	18,291
explained share (101 to 250)	0.421	—	—	—	—
explained share (251 to 500)	0.401	—	—	—	—
explained share (Top 100)	0.413	—	—	—	—
explained share (analyst action)	0.342	—	—	—	3,269
explained share (capital returns, financing)	0.278	—	—	—	410
explained share (deal, M&A, partnership)	0.336	—	—	—	1,192
explained share (earnings results)	0.536	—	—	—	4,592
explained share (guidance, outlook)	0.500	—	—	—	622
explained share (legal, regulatory)	0.313	—	—	—	423
explained share (management, governance)	0.071	—	—	—	263
explained share (market or sector move)	0.036	—	—	—	2,113
explained share (no identifiable news)	0.021	—	—	—	4,387
explained share (product, operations)	0.338	—	—	—	782
explained share (sector or macro)	0.622	—	—	—	238
θ at D2	−0.062	—	(0.038)	0.946	17,188
θ at D1	−0.040	—	(0.036)	0.868	13,203
persistence (analyst action)	0.001	—	—	0.491	3,159
persistence (capital returns, financing)	−0.121	—	—	0.960	392
persistence (deal, M&A, partnership)	−0.016	—	—	0.646	1,136
persistence (earnings results)	−0.028	—	—	0.842	4,204
persistence (guidance, outlook)	−0.125	—	—	0.976	572
persistence (legal, regulatory)	−0.125	—	—	0.956	408
persistence (management, governance)	−0.083	—	—	0.877	254
persistence (market or sector move)	−0.002	—	—	0.540	1,986
persistence (no identifiable news)	0.023	—	—	0.260	4,091
persistence (product, operations)	0.058	—	—	0.168	758
persistence (sector or macro)	−0.173	—	—	0.945	228

—: the source table has no value for the row; the statistic does not apply to it or was not computed.
The exploratory A3.5 market-level replication is in Table A35.
The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A24: Question battery v2: mechanism cuts, horizon validity and width.

Model, analysis, label, level	b	p_b	c	p_c	κ_{t+20}	ρ	p	ΔR^2	Agreement	Events
Laya-Fin, H2 at D2, vendor event type, analyst action	0.365	0.336	0.041	0.895	—	—	—	—	—	1,734
Laya-Fin, H2 at D2, vendor event type, capital returns	−4.884	0.814	0.100	0.692	—	—	—	—	—	343
Laya-Fin, H2 at D2, vendor event type, competition	−3.675	0.680	−0.196	0.192	—	—	—	—	—	215
Laya-Fin, H2 at D2, vendor event type, earnings results	−0.028	0.570	−0.021	0.041	—	—	—	—	—	10,828
Laya-Fin, H2 at D2, vendor event type, financing dilution	—	—	—	—	—	—	—	—	—	—
Laya-Fin, H2 at D2, vendor event type, guidance outlook	0.050	0.475	0.035	0.779	—	—	—	—	—	1,549
Laya-Fin, H2 at D2, vendor event type, insider ownership	−3.853	0.825	0.121	0.796	—	—	—	—	—	254
Laya-Fin, H2 at D2, vendor event type, leadership governance	−0.371	0.518	−0.101	0.392	—	—	—	—	—	160
Laya-Fin, H2 at D2, vendor event type, legal regulatory	−1.938	0.619	0.270	0.798	—	—	—	—	—	117
Laya-Fin, H2 at D2, vendor event type, M&A deal	−12.871	0.664	−0.224	0.363	—	—	—	—	—	141
Laya-Fin, H2 at D2, vendor event type, macro sector	−1.326	0.676	0.057	0.680	—	—	—	—	—	516
Laya-Fin, H2 at D2, vendor event type, operations supply	2.450	0.234	−0.132	0.276	—	—	—	—	—	300
Laya-Fin, H2 at D2, vendor event type, other corporate	−8.550	0.913	−0.160	0.091	—	—	—	—	—	233
Laya-Fin, H2 at D2, vendor event type, partnership customer	0.218	0.460	−0.061	0.413	—	—	—	—	—	342
Laya-Fin, H2 at D2, vendor event type, product launch	2.262	0.284	−0.154	0.111	—	—	—	—	—	420
Laya-Fin, H2 at D2, vendor quantified flag, no	−0.156	0.601	−0.037	0.115	—	—	—	—	—	2,158
Laya-Fin, H2 at D2, vendor quantified flag, yes	−0.015	0.538	−0.013	0.145	—	—	—	—	—	15,030
Laya-Fin, ladder, vendor event type, analyst action	—	—	—	—	2.622	—	—	—	—	1,049
Laya-Fin, ladder, vendor event type, capital returns	—	—	—	—	−2.533	—	—	—	—	233
Laya-Fin, ladder, vendor event type, competition	—	—	—	—	6.758	—	—	—	—	91
Laya-Fin, ladder, vendor event type, earnings results	—	—	—	—	0.259	—	—	—	—	7,453
Laya-Fin, ladder, vendor event type, guidance outlook	—	—	—	—	0.980	—	—	—	—	852
Laya-Fin, ladder, vendor event type, insider ownership	—	—	—	—	1.171	—	—	—	—	176
Laya-Fin, ladder, vendor event type, leadership governance	—	—	—	—	−1.746	—	—	—	—	112
Laya-Fin, ladder, vendor event type, legal regulatory	—	—	—	—	27.043	—	—	—	—	75
Laya-Fin, ladder, vendor event type, M&A deal	—	—	—	—	2.751	—	—	—	—	88
Laya-Fin, ladder, vendor event type, macro sector	—	—	—	—	0.430	—	—	—	—	245
Laya-Fin, ladder, vendor event type, operations supply	—	—	—	—	9.130	—	—	—	—	191
Laya-Fin, ladder, vendor event type, other corporate	—	—	—	—	−10.376	—	—	—	—	153
Laya-Fin, ladder, vendor event type, partnership customer	—	—	—	—	2.230	—	—	—	—	216
Laya-Fin, ladder, vendor event type, product launch	—	—	—	—	6.641	—	—	—	—	244
Laya-Fin, ladder, vendor quantified flag, no	—	—	—	—	0.044	—	—	—	—	1,310
Laya-Fin, ladder, vendor quantified flag, yes	—	—	—	—	0.248	—	—	—	—	9,891
anchor, H2 at D2, vendor event type, analyst action	17.803	0.129	0.043	0.901	—	—	—	—	—	1,734
anchor, H2 at D2, vendor event type, capital returns	−66.994	0.553	0.105	0.711	—	—	—	—	—	343
anchor, H2 at D2, vendor event type, competition	577.271	0.324	−0.175	0.246	—	—	—	—	—	215
anchor, H2 at D2, vendor event type, earnings results	0.253	0.433	−0.021	0.043	—	—	—	—	—	10,828
anchor, H2 at D2, vendor event type, financing dilution	—	—	—	—	—	—	—	—	—	—

continued on the next page

Table A24, continued

Model, analysis, label, level	b	p_b	c	p_c	κ_{t+20}	ρ	p	ΔR^2	Agreement	Events
anchor, H2 at D2, vendor event type, guidance outlook	1.477	0.443	0.035	0.780	—	—	—	—	—	1,549
anchor, H2 at D2, vendor event type, insider ownership	111.904	0.445	0.152	0.830	—	—	—	—	—	254
anchor, H2 at D2, vendor event type, leadership governance	196.719	0.419	−0.093	0.407	—	—	—	—	—	160
anchor, H2 at D2, vendor event type, legal regulatory	539.007	0.326	0.227	0.745	—	—	—	—	—	117
anchor, H2 at D2, vendor event type, M&A deal	86.356	0.489	−0.218	0.357	—	—	—	—	—	141
anchor, H2 at D2, vendor event type, macro sector	−9.396	0.680	0.049	0.657	—	—	—	—	—	516
anchor, H2 at D2, vendor event type, operations supply	114.038	0.424	−0.118	0.295	—	—	—	—	—	300
anchor, H2 at D2, vendor event type, other corporate	594.767	0.260	−0.101	0.142	—	—	—	—	—	233
anchor, H2 at D2, vendor event type, partnership customer	452.483	0.266	−0.045	0.432	—	—	—	—	—	342
anchor, H2 at D2, vendor event type, product launch	−326.856	0.838	−0.146	0.124	—	—	—	—	—	420
anchor, H2 at D2, vendor quantified flag, no	−7.930	0.746	−0.036	0.122	—	—	—	—	—	2,158
anchor, H2 at D2, vendor quantified flag, yes	1.112	0.205	−0.013	0.149	—	—	—	—	—	15,030
anchor, ladder, vendor event type, analyst action	—	—	—	—	5.034	—	—	—	—	1,049
anchor, ladder, vendor event type, capital returns	—	—	—	—	−10.957	—	—	—	—	233
anchor, ladder, vendor event type, competition	—	—	—	—	14.767	—	—	—	—	91
anchor, ladder, vendor event type, earnings results	—	—	—	—	0.017	—	—	—	—	7,453
anchor, ladder, vendor event type, guidance outlook	—	—	—	—	0.448	—	—	—	—	852
anchor, ladder, vendor event type, insider ownership	—	—	—	—	6.976	—	—	—	—	176
anchor, ladder, vendor event type, leadership governance	—	—	—	—	13.068	—	—	—	—	112
anchor, ladder, vendor event type, legal regulatory	—	—	—	—	−3.293	—	—	—	—	75
anchor, ladder, vendor event type, M&A deal	—	—	—	—	63.554	—	—	—	—	88
anchor, ladder, vendor event type, macro sector	—	—	—	—	5.512	—	—	—	—	245
anchor, ladder, vendor event type, operations supply	—	—	—	—	24.972	—	—	—	—	191
anchor, ladder, vendor event type, other corporate	—	—	—	—	−50.410	—	—	—	—	153
anchor, ladder, vendor event type, partnership customer	—	—	—	—	−1.182	—	—	—	—	216
anchor, ladder, vendor event type, product launch	—	—	—	—	16.391	—	—	—	—	244
anchor, ladder, vendor quantified flag, no	—	—	—	—	−0.726	—	—	—	—	1,310
anchor, ladder, vendor quantified flag, yes	—	—	—	—	0.077	—	—	—	—	9,891
d1-ultra, H2 at D2, own event type (Call A, same-day news), analyst action	−0.132	0.987	0.006	0.593	—	—	—	—	—	3,474
d1-ultra, H2 at D2, own event type (Call A, same-day news), capital returns financing	−0.403	0.931	0.009	0.541	—	—	—	—	—	609
d1-ultra, H2 at D2, own event type (Call A, same-day news), deal, M&A, partnership	0.047	0.360	−0.049	0.177	—	—	—	—	—	1,212
d1-ultra, H2 at D2, own event type (Call A, same-day news), earnings results	−0.088	0.991	−0.020	0.061	—	—	—	—	—	6,501
d1-ultra, H2 at D2, own event type (Call A, same-day news), guidance outlook	0.032	0.437	0.029	0.633	—	—	—	—	—	463

continued on the next page

Table A24, continued

Model, analysis, label, level	b	p_b	c	p_c	κ_{t+20}	ρ	p	ΔR^2	Agreement	Events
d1-ultra, H2 at D2, own event type (Call A, same-day news), legal regulatory	−0.138	0.757	−0.014	0.468	—	—	—	—	—	443
d1-ultra, H2 at D2, own event type (Call A, same-day news), management governance	0.036	0.468	−0.055	0.383	—	—	—	—	—	306
d1-ultra, H2 at D2, own event type (Call A, same-day news), other	0.003	0.490	0.006	0.567	—	—	—	—	—	2,810
d1-ultra, H2 at D2, own event type (Call A, same-day news), product operations	0.133	0.243	0.003	0.517	—	—	—	—	—	1,039
d1-ultra, H2 at D2, own event type (Call A, same-day news), sector or macro	−0.230	0.662	0.171	0.867	—	—	—	—	—	331
d1-ultra, H2 at D2, own quantified flag (Call A, same-day news), narrative	−0.066	0.950	−0.008	0.322	—	—	—	—	—	11,665
d1-ultra, H2 at D2, own quantified flag (Call A, same-day news), numbers	−0.061	0.912	−0.010	0.219	—	—	—	—	—	5,523
d1-ultra, H2 at D2, vendor event type, analyst action	−0.095	0.885	0.048	0.923	—	—	—	—	—	1,734
d1-ultra, H2 at D2, vendor event type, capital returns	−0.650	0.939	0.100	0.697	—	—	—	—	—	343
d1-ultra, H2 at D2, vendor event type, competition	−0.164	0.598	−0.162	0.269	—	—	—	—	—	215
d1-ultra, H2 at D2, vendor event type, earnings results	−0.077	0.982	−0.018	0.072	—	—	—	—	—	10,828
d1-ultra, H2 at D2, vendor event type, financing dilution	—	—	—	—	—	—	—	—	—	—
d1-ultra, H2 at D2, vendor event type, guidance outlook	0.012	0.467	0.035	0.777	—	—	—	—	—	1,549
d1-ultra, H2 at D2, vendor event type, insider ownership	−0.201	0.656	0.149	0.844	—	—	—	—	—	254
d1-ultra, H2 at D2, vendor event type, leadership governance	−0.513	0.776	−0.080	0.409	—	—	—	—	—	160
d1-ultra, H2 at D2, vendor event type, legal regulatory	0.233	0.315	0.321	0.850	—	—	—	—	—	117
d1-ultra, H2 at D2, vendor event type, M&A deal	−0.100	0.525	−0.210	0.376	—	—	—	—	—	141
d1-ultra, H2 at D2, vendor event type, macro sector	−0.076	0.604	0.052	0.669	—	—	—	—	—	516
d1-ultra, H2 at D2, vendor event type, operations supply	0.147	0.326	−0.127	0.288	—	—	—	—	—	300
d1-ultra, H2 at D2, vendor event type, other corporate	−0.446	0.693	−0.110	0.146	—	—	—	—	—	233
d1-ultra, H2 at D2, vendor event type, partnership customer	−0.110	0.589	−0.050	0.432	—	—	—	—	—	342
d1-ultra, H2 at D2, vendor event type, product launch	0.291	0.183	−0.157	0.107	—	—	—	—	—	420
d1-ultra, H2 at D2, vendor quantified flag, no	−0.001	0.503	−0.038	0.116	—	—	—	—	—	2,158
d1-ultra, H2 at D2, vendor quantified flag, yes	−0.084	0.994	−0.009	0.227	—	—	—	—	—	15,030
d1-ultra, ladder, own event type (Call A, pre-open news), analyst action	—	—	—	—	0.254	—	—	—	—	1,804
d1-ultra, ladder, own event type (Call A, pre-open news), capital returns financing	—	—	—	—	−1.389	—	—	—	—	397
d1-ultra, ladder, own event type (Call A, pre-open news), deal, M&A, partnership	—	—	—	—	−1.086	—	—	—	—	694
d1-ultra, ladder, own event type (Call A, pre-open news), earnings results	—	—	—	—	0.596	—	—	—	—	4,966

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Table A24, continued

Model, analysis, label, level	b	p_b	c	p_c	κ_{t+20}	ρ	p	ΔR^2	Agreement	Events
d1-ultra, ladder, own event type (Call A, pre-open news), guidance outlook	—	—	—	—	2.117	—	—	—	—	315
d1-ultra, ladder, own event type (Call A, pre-open news), legal regulatory	—	—	—	—	−1.178	—	—	—	—	233
d1-ultra, ladder, own event type (Call A, pre-open news), management governance	—	—	—	—	0.206	—	—	—	—	200
d1-ultra, ladder, own event type (Call A, pre-open news), other	—	—	—	—	1.346	—	—	—	—	1,818
d1-ultra, ladder, own event type (Call A, pre-open news), product operations	—	—	—	—	−0.150	—	—	—	—	569
d1-ultra, ladder, own event type (Call A, pre-open news), sector or macro	—	—	—	—	16.931	—	—	—	—	205
d1-ultra, ladder, own quantified flag (Call A, pre-open news), narrative	—	—	—	—	−0.039	—	—	—	—	7,450
d1-ultra, ladder, own quantified flag (Call A, pre-open news), numbers	—	—	—	—	0.784	—	—	—	—	3,751
d1-ultra, ladder, vendor event type, analyst action	—	—	—	—	0.940	—	—	—	—	1,049
d1-ultra, ladder, vendor event type, capital returns	—	—	—	—	0.118	—	—	—	—	233
d1-ultra, ladder, vendor event type, competition	—	—	—	—	0.536	—	—	—	—	91
d1-ultra, ladder, vendor event type, earnings results	—	—	—	—	0.446	—	—	—	—	7,453
d1-ultra, ladder, vendor event type, guidance outlook	—	—	—	—	0.329	—	—	—	—	852
d1-ultra, ladder, vendor event type, insider ownership	—	—	—	—	2.799	—	—	—	—	176
d1-ultra, ladder, vendor event type, leadership governance	—	—	—	—	1.937	—	—	—	—	112
d1-ultra, ladder, vendor event type, legal regulatory	—	—	—	—	−7.286	—	—	—	—	75
d1-ultra, ladder, vendor event type, M&A deal	—	—	—	—	−4.585	—	—	—	—	88
d1-ultra, ladder, vendor event type, macro sector	—	—	—	—	3.215	—	—	—	—	245
d1-ultra, ladder, vendor event type, operations supply	—	—	—	—	2.668	—	—	—	—	191
d1-ultra, ladder, vendor event type, other corporate	—	—	—	—	−4.239	—	—	—	—	153
d1-ultra, ladder, vendor event type, partnership customer	—	—	—	—	−1.230	—	—	—	—	216
d1-ultra, ladder, vendor event type, product launch	—	—	—	—	0.196	—	—	—	—	244
d1-ultra, ladder, vendor quantified flag, no	—	—	—	—	0.264	—	—	—	—	1,310
d1-ultra, ladder, vendor quantified flag, yes	—	—	—	—	0.390	—	—	—	—	9,891
d1-ultra, horizon validity: Spearman(H, share of the t+20 move by C_t)	—	—	—	—	—	−0.068	<0.001	—	—	9,683
d1-ultra, H4 width: ΔR^2 of the uncertainty score and P(no material effect)	—	—	—	—	—	—	0.245	0.0001	—	—
d1-ultra, reading validity (NA-06, val-A): credibility	—	—	—	—	—	—	—	—	0.990	500
d1-ultra, reading validity (NA-06, val-A): event type	—	—	—	—	—	—	—	—	0.572	500

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Table A24, continued

Model, analysis, label, level	b	p_b	c	p_c	κ_{t+20}	ρ	p	ΔR^2	Agreement	Events
d1-ultra, reading validity (NA-06, val-A): event type vs raw tag	—	—	—	—	—	—	—	—	0.454	500
d1-ultra, reading validity (NA-06, val-A): quantified	—	—	—	—	—	—	—	—	0.494	500
d1-ultra, reading validity (NA-06, val-A): scheduled	—	—	—	—	—	—	—	—	0.552	500

—: the source table has no value for the row; the statistic does not apply to it or was not computed.
Reading validity: the share of the 500 val-A events on which d1-ultra's answer agrees with the vendor's tag (event type, the raw tag, quantified; the rumor tag for credibility) or with the earnings calendar (scheduled).
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

F Economic significance

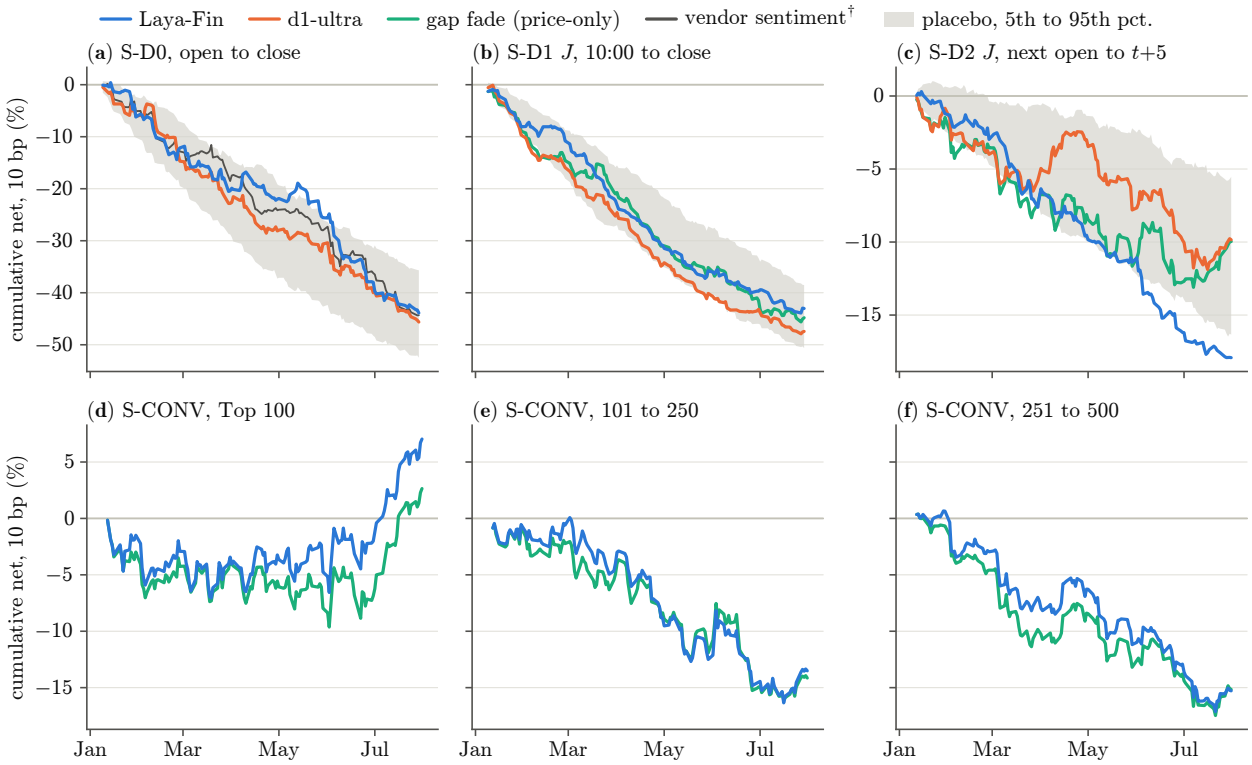


Figure A12: The other books, and S-CONV by tier. Cumulative net returns at 10 bp per side over the test period (NA-13). (a) S-D0 from the open to the close, with vendor sentiment (†: a possibly look-ahead-contaminated upper bound, not the benchmark); (b) S-D1 J from 10:00 to the close; (c) S-D2 J from the next open to t+5, each against the gap fade where it applies and the placebo's 5th to 95th percentile band of Laya-Fin's universe. The steep declines of the one-day books are mostly costs (net Sharpe -6.03 and -9.70, gross $+0.35$ and $+0.76$). (d to f) Laya-Fin's S-CONV and the gap fade by tier: descriptive sub-books of a null book, not separate tests. (a) and (b) share a scale, as do (d) to (f); (c) has its own, since S-D2 loses far less. d1-ultra is model-specific (§1.3).

Table A25: Every pre-registered book and the benchmarks, test and val-B.

Book	Gross	5 bp	10 bp	20 bp	Turnover	Placebo	Placebo net	DSR	Capacity (USD)
test									
S-CONV, Laya-Fin, D2 U, to t+1	−3.15	−6.89	−10.64	−18.10	4.65	0	51	<0.001	27,612,849
S-CONV, Laya-Fin, D2 U, to t+5	0.61	−0.69	−1.98	−4.58	0.90	68	95	<0.001	137,902,233
S-CONV, Laya-Fin, D2 U, to t+20	0.22	−0.36	−0.94	−2.11	0.20	58	76	<0.001	530,743,321
S-CONV, Laya-Fin, D1 U, to C_t	1.61	−2.56	−6.74	−15.08	4.68	83	100	<0.001	20,000,923
S-CONV, Laya-Fin, D1 U, to t+1	−0.97	−3.48	−5.99	−10.99	2.32	20	55	<0.001	40,001,846
S-CONV, Laya-Fin, D1 U, to t+5	1.23	0.02	−1.19	−3.62	0.75	84	89	<0.001	119,937,047
S-CONV, Laya-Fin, D1 U, to t+20	−0.67	−1.22	−1.77	−2.87	0.19	36	45	<0.001	417,986,337
S-CONV, d1-ultra, D2 U, to t+1	−3.48	−7.22	−10.97	−18.47	4.62	0	44	<0.001	27,146,457
S-CONV, d1-ultra, D2 U, to t+5	−0.09	−1.40	−2.72	−5.35	0.90	48	88	<0.001	134,260,626
S-CONV, d1-ultra, D2 U, to t+20	0.02	−0.62	−1.26	−2.54	0.20	48	67	<0.001	519,885,636
S-CONV, d1-ultra, D1 U, to C_t	0.03	−4.19	−8.42	−16.80	4.71	49	100	<0.001	20,125,210
S-CONV, d1-ultra, D1 U, to t+1	−2.26	−4.81	−7.35	−12.41	2.34	6	19	<0.001	40,250,419
S-CONV, d1-ultra, D1 U, to t+5	0.32	−0.98	−2.27	−4.87	0.75	65	76	<0.001	120,074,026
S-CONV, d1-ultra, D1 U, to t+20	−0.88	−1.43	−1.98	−3.08	0.19	31	41	<0.001	418,848,150
S-CONV, anchor, D2 U, to t+1	−2.91	−6.47	−10.04	−17.14	4.64	0	64	<0.001	27,580,447
S-CONV, anchor, D2 U, to t+5	0.71	−0.50	−1.72	−4.15	0.90	68	96	<0.001	135,732,285
S-CONV, anchor, D2 U, to t+20	0.13	−0.39	−0.92	−1.98	0.20	56	77	<0.001	527,355,164
S-CONV, anchor, D1 U, to C_t	1.10	−3.18	−7.45	−15.96	4.65	76	100	<0.001	20,192,885
S-CONV, anchor, D1 U, to t+1	−1.38	−3.96	−6.54	−11.68	2.31	15	39	<0.001	40,385,770
S-CONV, anchor, D1 U, to t+5	0.90	−0.37	−1.64	−4.18	0.75	81	88	<0.001	120,074,026
S-CONV, anchor, D1 U, to t+20	−0.68	−1.23	−1.79	−2.89	0.19	36	45	<0.001	417,986,337
S-D0, Laya-Fin, D0 $\tilde{N}$, to C_t	0.35	−2.85	−6.03	−12.35	4.36	64	98	<0.001	19,992,020
S-D0, Laya-Fin, D0 $\tilde{N}$, to t+1	−0.95	−2.77	−4.58	−8.19	2.16	22	76	<0.001	39,984,041
S-D0, Laya-Fin, D0 $\tilde{N}$, to C_t, gated	1.18	−0.43	−2.03	−5.23	4.64	90	94	<0.001	6,876,824
S-D0, Laya-Fin, D0 $\tilde{N}$, to t+1, gated	1.88	0.75	−0.38	−2.65	2.31	93	96	<0.001	13,753,648
S-D0, d1-ultra, D0 $\tilde{N}$, to C_t	0.10	−3.16	−6.41	−12.90	4.41	58	96	<0.001	19,911,776
S-D0, d1-ultra, D0 $\tilde{N}$, to t+1	−1.05	−2.99	−4.93	−8.80	2.19	20	66	<0.001	39,823,551
S-D0, d1-ultra, D0 $\tilde{N}$, to C_t, gated	1.15	−0.69	−2.52	−6.15	4.73	74	93	<0.001	8,410,532
S-D0, d1-ultra, D0 $\tilde{N}$, to t+1, gated	0.20	−1.05	−2.30	−4.79	2.35	62	73	<0.001	16,821,064
S-D0, anchor, D0 $\tilde{N}$, to C_t	−0.72	−4.58	−8.44	−16.12	4.33	36	56	<0.001	20,326,238
S-D0, anchor, D0 $\tilde{N}$, to t+1	−0.79	−3.50	−6.21	−11.62	2.15	26	29	<0.001	40,652,476
S-D0, anchor, D0 $\tilde{N}$, to C_t, gated	−0.51	−1.80	−3.09	−5.70	4.79	34	55	<0.001	8,192,136
S-D0, anchor, D0 $\tilde{N}$, to t+1, gated	0.11	−0.82	−1.74	−3.59	2.38	51	67	<0.001	16,384,273
S-D1, Laya-Fin, D1 J, to C_t	0.76	−4.48	−9.70	−20.05	4.36	71	85	<0.001	18,999,288
S-D1, Laya-Fin, D1 J, to t+1	−0.15	−3.10	−6.03	−11.87	2.16	49	59	<0.001	37,998,576
S-D1, Laya-Fin, D1 J, to t+5	−2.35	−3.99	−5.63	−8.88	0.70	6	3	<0.001	112,478,752

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Table A25, continued (test)

Book	Gross	5 bp	10 bp	20 bp	Turnover	Placebo	Placebo net	DSR	Capacity (USD)
S-D1, Laya-Fin, D1 RG, to C_t	0.02	−0.15	−0.32	−0.63	0.12	46	44	<0.001	3,012,124
S-D1, Laya-Fin, D1 RG, to t+1	−0.04	−0.19	−0.33	−0.62	0.06	47	46	<0.001	6,024,248
S-D1, Laya-Fin, D1 RG, to t+5	1.12	1.07	1.02	0.92	0.02	74	74	0.008	18,072,745
S-D1, d1-ultra, D1 J, to C_t	−0.28	−5.80	−11.35	−22.43	4.51	40	57	<0.001	19,068,165
S-D1, d1-ultra, D1 J, to t+1	−0.70	−3.70	−6.69	−12.64	2.24	36	36	<0.001	38,136,331
S-D1, d1-ultra, D1 J, to t+5	−0.24	−1.63	−3.02	−5.77	0.72	45	52	<0.001	114,346,804
S-D1, d1-ultra, D1 RG, to C_t	−1.21	−4.04	−6.87	−12.51	4.79	22	54	<0.001	11,224,824
S-D1, d1-ultra, D1 RG, to t+1	−3.20	−4.83	−6.46	−9.71	2.38	2	9	<0.001	22,449,648
S-D1, d1-ultra, D1 RG, to t+5	−0.88	−1.70	−2.52	−4.17	0.77	26	41	<0.001	66,287,585
S-D1, anchor, D1 J, to C_t	0.24	−4.75	−9.73	−19.66	4.44	51	85	<0.001	20,058,238
S-D1, anchor, D1 J, to t+1	−1.17	−4.04	−6.91	−12.63	2.20	27	34	<0.001	40,116,476
S-D1, anchor, D1 J, to t+5	0.85	−0.58	−2.00	−4.86	0.71	76	80	<0.001	118,417,650
S-D2, Laya-Fin, D2 J, to t+1	−1.14	−5.88	−10.61	−20.05	4.26	20	51	<0.001	25,269,525
S-D2, Laya-Fin, D2 J, to t+5	−2.69	−4.57	−6.45	−10.18	0.83	3	4	<0.001	125,469,463
S-D2, Laya-Fin, D2 J, to t+20	−1.77	−2.66	−3.55	−5.32	0.18	11	12	<0.001	471,903,675
S-D2, Laya-Fin, D2 J, to t+1, gated	−1.52	−3.83	−6.14	−10.75	4.67	12	19	<0.001	11,191,386
S-D2, Laya-Fin, D2 J, to t+5, gated	−0.90	−1.91	−2.91	−4.93	0.91	21	21	<0.001	55,124,870
S-D2, Laya-Fin, D2 J, to t+20, gated	−1.01	−1.51	−2.01	−3.00	0.20	31	29	<0.001	209,010,082
S-D2, Laya-Fin, D2 RG, to t+1	1.87	1.75	1.61	1.31	0.24	90	92	0.005	3,829,744
S-D2, Laya-Fin, D2 RG, to t+5	1.69	1.60	1.50	1.31	0.05	92	92	0.024	19,148,722
S-D2, Laya-Fin, D2 RG, to t+20	−0.55	−0.60	−0.64	−0.73	0.01	40	40	<0.001	70,615,845
S-D2, Laya-Fin, D2 RG, to t+1, gated	0.24	0.07	−0.11	−0.44	0.10	59	66	<0.001	3,244,676
S-D2, Laya-Fin, D2 RG, to t+5, gated	0.29	0.24	0.19	0.09	0.02	68	68	0.002	16,223,380
S-D2, Laya-Fin, D2 RG, to t+20, gated	−0.83	−0.86	−0.88	−0.94	0.01	39	42	<0.001	64,893,519
S-D2, d1-ultra, D2 J, to t+1	−1.84	−5.97	−10.08	−18.23	4.40	6	64	<0.001	26,520,852
S-D2, d1-ultra, D2 J, to t+5	0.39	−0.99	−2.37	−5.12	0.85	64	91	<0.001	132,436,588
S-D2, d1-ultra, D2 J, to t+20	0.33	−0.30	−0.93	−2.18	0.19	59	77	<0.001	520,649,966
S-D2, d1-ultra, D2 J, to t+1, gated	−1.54	−4.48	−7.41	−13.21	4.50	12	14	<0.001	11,897,252
S-D2, d1-ultra, D2 J, to t+5, gated	−0.42	−1.40	−2.39	−4.36	0.87	44	46	<0.001	59,486,258
S-D2, d1-ultra, D2 J, to t+20, gated	0.42	−0.06	−0.54	−1.49	0.19	62	65	<0.001	237,117,063
S-D2, d1-ultra, D2 RG, to t+1	−2.55	−5.53	−8.52	−14.49	4.72	5	22	<0.001	15,897,165
S-D2, d1-ultra, D2 RG, to t+5	0.15	−0.92	−1.98	−4.11	0.92	59	78	<0.001	79,485,824
S-D2, d1-ultra, D2 RG, to t+20	0.42	−0.05	−0.52	−1.45	0.20	61	69	<0.001	314,288,110
S-D2, d1-ultra, D2 RG, to t+1, gated	−0.74	−3.07	−5.41	−10.10	4.76	30	44	<0.001	9,408,673
S-D2, d1-ultra, D2 RG, to t+5, gated	0.39	−0.41	−1.21	−2.80	0.93	64	74	<0.001	47,043,366
S-D2, d1-ultra, D2 RG, to t+20, gated	0.32	0.00	−0.32	−0.96	0.21	53	62	<0.001	183,614,525
S-D2, anchor, D2 J, to t+1	−3.43	−8.16	−12.89	−22.34	4.38	0	12	<0.001	25,739,243
S-D2, anchor, D2 J, to t+5	0.13	−1.44	−3.01	−6.15	0.85	57	80	<0.001	128,837,992

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Table A25, continued (test)

Book	Gross	5 bp	10 bp	20 bp	Turnover	Placebo	Placebo net	DSR	Capacity (USD)
S-D2, anchor, D2 J, to t+20	−0.03	−0.83	−1.62	−3.21	0.19	46	58	<0.001	512,252,552
S-D2, anchor, D2 J, to t+1, gated	−2.93	−5.98	−9.01	−15.00	4.59	3	3	<0.001	11,143,207
S-D2, anchor, D2 J, to t+5, gated	0.44	−0.67	−1.78	−4.01	0.89	65	55	<0.001	57,903,787
S-D2, anchor, D2 J, to t+20, gated	1.48	1.00	0.51	−0.46	0.20	79	78	0.003	207,401,770
gap fade, D2 gap, to t+1	−3.52	−7.19	−10.87	−18.24	4.64	0	47	<0.001	27,580,447
gap fade, D2 gap, to t+5	0.44	−0.78	−2.01	−4.45	0.90	66	94	<0.001	135,732,285
gap fade, D2 gap, to t+20	−0.09	−0.61	−1.13	−2.16	0.20	46	71	<0.001	530,743,321
gap fade, D1 gap, to C_t	0.78	−3.44	−7.66	−16.06	4.69	70	100	<0.001	20,086,107
gap fade, D1 gap, to t+1	−1.57	−4.13	−6.69	−11.79	2.33	13	33	<0.001	40,172,214
gap fade, D1 gap, to t+5	0.40	−0.84	−2.08	−4.57	0.75	68	81	<0.001	120,408,453
gap fade, D1 gap, to t+20	−0.98	−1.54	−2.10	−3.22	0.19	27	36	<0.001	419,210,328
vendor sentiment†, D0, to C_t	0.27	−2.99	−6.24	−12.75	4.36	62	97	<0.001	20,113,795
vendor sentiment†, D0, to t+1	−1.36	−3.34	−5.32	−9.28	2.16	11	51	<0.001	40,227,590

Table A25, continued (val-B)

Book	Gross	5 bp	10 bp	20 bp	Turnover	Placebo	Placebo net	DSR	Capacity (USD)
val-B									
S-CONV, Laya-Fin, D2 U, to t+1	−4.53	−10.23	−15.89	−27.09	4.41	1	17	<0.001	19,513,622
S-CONV, Laya-Fin, D2 U, to t+5	−1.43	−3.51	−5.59	−9.76	0.82	27	39	<0.001	99,719,178
S-CONV, Laya-Fin, D2 U, to t+20	2.20	1.04	−0.11	−2.39	0.15	77	78	<0.001	501,912,710
S-CONV, Laya-Fin, D1 U, to C_t	2.41	−4.65	−11.75	−25.98	4.42	87	76	<0.001	18,439,211
S-CONV, Laya-Fin, D1 U, to t+1	0.19	−3.67	−7.58	−15.51	2.18	55	45	<0.001	37,034,751
S-CONV, Laya-Fin, D1 U, to t+5	−1.45	−3.09	−4.74	−8.04	0.67	23	34	<0.001	112,958,485
S-CONV, Laya-Fin, D1 U, to t+20	1.16	0.18	−0.80	−2.79	0.14	69	68	<0.001	451,328,844
S-CONV, d1-ultra, D2 U, to t+1	−4.11	−9.71	−15.30	−26.47	4.44	2	23	<0.001	20,283,922
S-CONV, d1-ultra, D2 U, to t+5	−1.03	−3.00	−4.97	−8.90	0.83	30	48	<0.001	104,448,766
S-CONV, d1-ultra, D2 U, to t+20	3.58	2.49	1.41	−0.71	0.15	91	93	0.007	530,494,663
S-CONV, d1-ultra, D1 U, to C_t	3.12	−3.69	−10.58	−24.55	4.40	93	91	<0.001	16,443,615
S-CONV, d1-ultra, D1 U, to t+1	2.25	−1.12	−4.52	−11.38	2.16	83	86	<0.001	33,213,081
S-CONV, d1-ultra, D1 U, to t+5	−0.36	−1.99	−3.63	−6.92	0.67	47	56	<0.001	102,961,769
S-CONV, d1-ultra, D1 U, to t+20	0.68	−0.33	−1.35	−3.42	0.14	64	61	<0.001	447,112,638
S-CONV, anchor, D2 U, to t+1	−4.46	−9.68	−14.86	−25.12	4.43	1	27	<0.001	19,329,748
S-CONV, anchor, D2 U, to t+5	−1.83	−3.84	−5.84	−9.81	0.82	24	35	<0.001	97,568,109
S-CONV, anchor, D2 U, to t+20	1.97	0.83	−0.31	−2.56	0.15	71	73	<0.001	501,912,710
S-CONV, anchor, D1 U, to C_t	1.69	−4.92	−11.59	−24.99	4.40	79	78	<0.001	16,443,615
S-CONV, anchor, D1 U, to t+1	1.07	−2.13	−5.35	−11.87	2.17	71	78	<0.001	33,213,081

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Table A25, continued (val-B)

Book	Gross	5 bp	10 bp	20 bp	Turnover	Placebo	Placebo net	DSR	Capacity (USD)
S-CONV, anchor, D1 U, to t+5	−1.16	−2.84	−4.52	−7.88	0.67	32	43	<0.001	102,961,769
S-CONV, anchor, D1 U, to t+20	0.52	−0.34	−1.20	−2.93	0.14	64	64	<0.001	459,302,380
S-D0, Laya-Fin, D0 $\tilde{N}$, to C_t	3.03	−1.02	−5.05	−13.07	4.29	93	99	<0.001	15,947,679
S-D0, Laya-Fin, D0 $\tilde{N}$, to t+1	1.96	−0.64	−3.25	−8.50	2.11	84	94	<0.001	31,912,853
S-D0, Laya-Fin, D0 $\tilde{N}$, to C_t, gated	3.96	1.40	−1.19	−6.42	4.44	94	88	<0.001	5,974,189
S-D0, Laya-Fin, D0 $\tilde{N}$, to t+1, gated	2.02	0.66	−0.70	−3.44	2.18	86	86	<0.001	11,977,152
S-D0, d1-ultra, D0 $\tilde{N}$, to C_t	1.48	−2.26	−5.99	−13.45	4.27	78	96	<0.001	16,190,065
S-D0, d1-ultra, D0 $\tilde{N}$, to t+1	3.83	1.65	−0.53	−4.87	2.10	97	100	<0.001	32,722,473
S-D0, d1-ultra, D0 $\tilde{N}$, to C_t, gated	4.51	2.19	−0.15	−4.90	4.46	98	98	<0.001	7,317,338
S-D0, d1-ultra, D0 $\tilde{N}$, to t+1, gated	3.97	2.35	0.73	−2.51	2.19	97	98	0.003	14,712,657
S-D0, anchor, D0 $\tilde{N}$, to C_t	1.26	−2.94	−7.16	−15.67	4.27	77	89	<0.001	16,953,052
S-D0, anchor, D0 $\tilde{N}$, to t+1	1.02	−1.81	−4.64	−10.28	2.09	75	83	<0.001	34,293,148
S-D0, anchor, D0 $\tilde{N}$, to C_t, gated	−0.96	−2.24	−3.52	−6.09	4.37	32	40	<0.001	8,997,206
S-D0, anchor, D0 $\tilde{N}$, to t+1, gated	−1.70	−2.92	−4.13	−6.58	2.19	21	24	<0.001	17,994,412
S-D1, Laya-Fin, D1 J, to C_t	−1.25	−6.20	−11.08	−20.63	4.25	31	84	<0.001	16,489,018
S-D1, Laya-Fin, D1 J, to t+1	−1.09	−3.97	−6.84	−12.49	2.09	33	65	<0.001	33,156,164
S-D1, Laya-Fin, D1 J, to t+5	−0.01	−1.54	−3.08	−6.17	0.64	49	59	<0.001	100,309,837
S-D1, Laya-Fin, D1 RG, to C_t	1.41	1.10	0.76	0.05	0.47	61	65	0.001	2,327,469
S-D1, Laya-Fin, D1 RG, to t+1	−0.87	−1.27	−1.66	−2.43	0.23	36	32	<0.001	4,654,938
S-D1, Laya-Fin, D1 RG, to t+5	0.58	0.36	0.15	−0.28	0.07	65	61	0.001	13,964,814
S-D1, d1-ultra, D1 J, to C_t	0.51	−5.91	−12.39	−25.48	4.30	58	69	<0.001	14,183,895
S-D1, d1-ultra, D1 J, to t+1	−3.23	−6.62	−10.02	−16.81	2.12	7	14	<0.001	28,937,869
S-D1, d1-ultra, D1 J, to t+5	−3.45	−4.96	−6.47	−9.46	0.66	7	14	<0.001	89,214,376
S-D1, d1-ultra, D1 RG, to C_t	2.94	−2.34	−7.65	−18.15	4.40	91	68	<0.001	10,073,699
S-D1, d1-ultra, D1 RG, to t+1	1.23	−1.35	−3.93	−9.13	2.16	80	69	<0.001	20,408,932
S-D1, d1-ultra, D1 RG, to t+5	0.69	−0.71	−2.12	−4.97	0.68	66	67	<0.001	63,947,316
S-D1, anchor, D1 J, to C_t	−1.51	−6.76	−12.03	−22.61	4.25	28	71	<0.001	14,183,895
S-D1, anchor, D1 J, to t+1	3.67	0.52	−2.64	−9.00	2.10	94	100	<0.001	30,010,198
S-D1, anchor, D1 J, to t+5	0.87	−0.81	−2.48	−5.81	0.65	62	71	<0.001	93,443,343
S-D2, Laya-Fin, D2 J, to t+1	0.63	−4.74	−10.10	−20.77	4.21	65	82	<0.001	19,410,645
S-D2, Laya-Fin, D2 J, to t+5	−2.55	−5.12	−7.69	−12.80	0.79	17	15	<0.001	99,991,310
S-D2, Laya-Fin, D2 J, to t+20	2.23	1.03	−0.16	−2.54	0.14	77	77	<0.001	484,278,372
S-D2, Laya-Fin, D2 J, to t+1, gated	−2.56	−5.25	−7.92	−13.09	4.15	6	9	<0.001	8,956,888
S-D2, Laya-Fin, D2 J, to t+5, gated	−2.59	−3.72	−4.85	−7.10	0.81	18	23	<0.001	45,451,969
S-D2, Laya-Fin, D2 J, to t+20, gated	0.80	0.17	−0.45	−1.68	0.15	68	68	<0.001	198,129,105
S-D2, Laya-Fin, D2 RG, to t+1	−4.84	−4.93	−5.01	−5.14	0.56	1	1	<0.001	1,800,823
S-D2, Laya-Fin, D2 RG, to t+5	−3.35	−3.56	−3.75	−4.09	0.10	6	6	<0.001	9,004,117
S-D2, Laya-Fin, D2 RG, to t+20	−2.35	−2.46	−2.58	−2.80	0.03	7	8	<0.001	36,016,467

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Table A25, continued (val-B)

Book	Gross	5 bp	10 bp	20 bp	Turnover	Placebo	Placebo net	DSR	Capacity (USD)
S-D2, Laya-Fin, D2 RG, to t+1, gated	−4.20	−4.30	−4.38	−4.51	0.39	1	1	<0.001	1,800,823
S-D2, Laya-Fin, D2 RG, to t+5, gated	1.58	1.44	1.30	1.00	0.07	86	87	0.004	9,004,117
S-D2, Laya-Fin, D2 RG, to t+20, gated	−0.62	−0.72	−0.83	−1.04	0.02	46	46	<0.001	36,016,467
S-D2, d1-ultra, D2 J, to t+1	−1.47	−6.75	−12.05	−22.71	4.29	27	61	<0.001	19,311,796
S-D2, d1-ultra, D2 J, to t+5	−0.42	−2.50	−4.57	−8.69	0.80	36	55	<0.001	97,167,123
S-D2, d1-ultra, D2 J, to t+20	3.75	2.47	1.20	−1.28	0.14	93	91	0.008	526,437,216
S-D2, d1-ultra, D2 J, to t+1, gated	−3.21	−6.31	−9.41	−15.63	4.38	3	12	<0.001	8,557,777
S-D2, d1-ultra, D2 J, to t+5, gated	−3.88	−5.30	−6.72	−9.55	0.81	3	2	<0.001	44,406,572
S-D2, d1-ultra, D2 J, to t+20, gated	1.72	1.10	0.48	−0.77	0.15	79	80	0.002	219,583,704
S-D2, d1-ultra, D2 RG, to t+1	−3.88	−7.44	−11.00	−18.06	4.39	5	18	<0.001	12,558,160
S-D2, d1-ultra, D2 RG, to t+5	−1.02	−2.45	−3.88	−6.74	0.81	29	36	<0.001	63,897,089
S-D2, d1-ultra, D2 RG, to t+20	4.66	3.92	3.19	1.74	0.15	98	99	0.046	278,933,314
S-D2, d1-ultra, D2 RG, to t+1, gated	−1.72	−4.23	−6.73	−11.68	4.46	31	43	<0.001	7,965,042
S-D2, d1-ultra, D2 RG, to t+5, gated	−4.34	−5.51	−6.67	−9.00	0.83	2	2	<0.001	42,687,403
S-D2, d1-ultra, D2 RG, to t+20, gated	−0.85	−1.40	−1.96	−3.04	0.15	37	36	<0.001	185,278,844
S-D2, anchor, D2 J, to t+1	0.78	−5.83	−12.41	−25.43	4.25	67	57	<0.001	19,100,245
S-D2, anchor, D2 J, to t+5	1.60	−1.14	−3.89	−9.34	0.80	80	72	<0.001	95,951,498
S-D2, anchor, D2 J, to t+20	1.38	−0.04	−1.47	−4.32	0.14	65	57	<0.001	497,257,456
S-D2, anchor, D2 J, to t+1, gated	4.10	0.74	−2.62	−9.30	4.51	98	95	<0.001	7,856,932
S-D2, anchor, D2 J, to t+5, gated	3.61	2.43	1.26	−1.10	0.83	95	94	0.006	38,364,620
S-D2, anchor, D2 J, to t+20, gated	−0.58	−1.11	−1.64	−2.70	0.15	41	40	<0.001	165,158,070
gap fade, D2 gap, to t+1	−4.79	−10.22	−15.63	−26.38	4.43	1	20	<0.001	19,513,622
gap fade, D2 gap, to t+5	−2.02	−3.97	−5.92	−9.78	0.82	22	35	<0.001	98,917,336
gap fade, D2 gap, to t+20	1.47	0.37	−0.73	−2.90	0.15	65	69	<0.001	511,866,048
gap fade, D1 gap, to C_t	1.41	−5.47	−12.39	−26.30	4.43	77	67	<0.001	15,965,175
gap fade, D1 gap, to t+1	0.89	−2.49	−5.89	−12.78	2.18	65	71	<0.001	32,408,789
gap fade, D1 gap, to t+5	−1.58	−3.21	−4.84	−8.09	0.67	22	30	<0.001	99,639,242
gap fade, D1 gap, to t+20	0.02	−0.76	−1.54	−3.12	0.14	53	60	<0.001	447,112,638
vendor sentiment†, D0, to C_t	1.19	−2.73	−6.69	−14.70	4.30	74	91	<0.001	16,532,408
vendor sentiment†, D0, to t+1	2.80	0.53	−1.74	−6.28	2.11	92	99	<0.001	33,301,874

Annualized Sharpe ratios gross and net of 5, 10 and 20 bp per side; placebo percentiles among 100 random-signal books; DSR at 10 bp with the 545 trials in the ledger when NA-13 computed it; capacity at 1% of the 20-session ADV (median over decisions). Empty RG books of the anchor are left out. The exploratory MA-cross fade (post hoc) is in Table A34 and Figure A15.

†Vendor sentiment: its labels were produced retrospectively, so its books are a possibly look-ahead-contaminated upper bound, not a benchmark; the clean benchmarks are the price-only gap fade and the random-signal placebo.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A27: S-CONV against the price-only gap fade: the same positions on the same side.

Book	Shared	Same side	Correlation	Days
Laya-Fin, D2 to t+5	0.922	0.998	0.96	139
Laya-Fin, D1 to C_t	0.891	0.998	0.93	138
d1-ultra, D2 to t+5	0.903	0.998	0.94	139
anchor, D2 to t+5	0.960	1.000	0.98	139

Shared: the share of the book's (session, symbol) positions also held by the gap fade; same side among those; correlation of the daily gross returns. Recomputed from NA-13's stored positions and books.

The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.

Table A28: The random-signal placebo: 5th, 50th and 95th percentile of the Sharpe ratio per universe.

	Gross			Net of 10 bp			
Decision point, exit: books	5th	50th	95th	5th	50th	95th	Books
D0, t+1: S-D0 gated (Laya-Fin)	−2.59	−0.12	2.12	−5.15	−2.63	−0.52	100
D0, t+1: S-D0, sentiment†	−1.78	0.12	2.74	−7.49	−5.34	−2.91	100
D0, t+1: S-D0 gated (anchor)	−2.22	0.05	2.50	−4.73	−2.33	0.09	100
D0, t+1: S-D0 gated (d1-ultra)	−2.03	−0.12	2.16	−5.43	−3.43	−1.33	100
D0, C_t: S-D0 gated (Laya-Fin)	−2.25	−0.25	1.60	−6.13	−4.06	−2.01	100
D0, C_t: S-D0, sentiment†	−1.99	−0.29	2.35	−11.01	−8.77	−6.46	100
D0, C_t: S-D0 gated (anchor)	−2.63	0.03	1.91	−5.79	−3.31	−1.48	100
D0, C_t: S-D0 gated (d1-ultra)	−2.04	−0.16	1.95	−6.99	−4.60	−2.42	100
D1, t+1: S-D1 RG (Laya-Fin)	−2.01	0.13	2.49	−2.12	−0.09	2.35	100
D1, t+1: S-D1 RG (d1-ultra)	−2.85	−0.36	2.24	−6.99	−4.43	−1.48	100
D1, t+1: S-CONV, gap fade	−2.27	−0.05	2.19	−8.51	−6.15	−3.60	100
D1, t+1: S-D1 J	−2.20	−0.10	2.05	−8.54	−6.45	−3.95	100
D1, t+20: S-CONV, gap fade	−2.06	−0.09	1.70	−3.78	−1.61	0.17	100
D1, t+5: S-D1 RG (Laya-Fin)	−2.20	−0.05	2.46	−2.30	−0.18	2.37	100
D1, t+5: S-D1 RG (d1-ultra)	−1.98	−0.05	1.87	−3.99	−2.07	0.00	100
D1, t+5: S-CONV, gap fade	−1.72	−0.09	2.36	−4.97	−3.13	−0.70	100
D1, t+5: S-D1 J	−2.38	−0.14	1.94	−5.57	−3.08	−1.21	100
D1, C_t: S-D1 RG (Laya-Fin)	−1.84	0.37	2.13	−1.95	−0.07	2.01	100
D1, C_t: S-D1 RG (d1-ultra)	−2.40	0.11	2.05	−9.80	−7.02	−4.99	100
D1, C_t: S-CONV, gap fade	−2.04	0.12	2.63	−14.11	−11.60	−9.03	100
D1, C_t: S-D1 J	−2.19	0.22	2.11	−14.04	−11.78	−8.91	100
D2, t+1: S-D2 RG (d1-ultra)	−2.45	0.16	2.48	−10.21	−7.37	−4.71	100
D2, t+1: S-CONV, S-D2 J, gap fade	−2.00	0.08	2.76	−13.32	−10.74	−7.86	100
D2, t+1: S-D2 J gated (Laya-Fin)	−2.07	0.29	2.10	−7.50	−4.70	−2.90	100
D2, t+1: S-D2 RG gated (Laya-Fin)	−2.29	−0.25	2.20	−2.33	−1.13	1.83	100
D2, t+1: S-D2 J gated (anchor)	−2.51	−0.15	2.15	−7.98	−5.63	−3.34	100
D2, t+1: S-D2 RG (Laya-Fin)	−2.27	−0.04	2.38	−2.62	−0.84	1.89	100
D2, t+1: S-D2 RG gated (d1-ultra)	−2.17	0.03	2.21	−7.69	−5.23	−2.74	100
D2, t+1: S-D2 J gated (d1-ultra)	−2.42	0.10	1.87	−8.33	−5.87	−3.55	100
D2, t+20: S-D2 RG (d1-ultra)	−2.11	0.04	2.49	−3.41	−1.24	1.18	100
D2, t+20: S-CONV, S-D2 J, gap fade	−2.39	0.06	2.55	−4.27	−1.81	0.58	100
D2, t+20: S-D2 J gated (Laya-Fin)	−2.10	−0.59	1.94	−2.99	−1.55	1.09	100
D2, t+20: S-D2 RG gated (Laya-Fin)	−3.05	−0.31	2.57	−3.10	−0.37	2.50	100
D2, t+20: S-D2 J gated (anchor)	−1.54	0.44	2.57	−2.48	−0.43	1.70	100
D2, t+20: S-D2 RG (Laya-Fin)	−2.70	−0.24	2.94	−2.80	−0.39	2.81	100
D2, t+20: S-D2 RG gated (d1-ultra)	−2.25	0.10	2.22	−2.99	−0.69	1.41	100
D2, t+20: S-D2 J gated (d1-ultra)	−1.84	0.11	2.16	−2.82	−0.88	1.14	100
D2, t+5: S-D2 RG (d1-ultra)	−2.53	0.00	1.98	−5.33	−2.72	−0.84	100
D2, t+5: S-CONV, S-D2 J, gap fade	−2.12	−0.04	2.23	−6.25	−4.05	−1.99	100
D2, t+5: S-D2 J gated (Laya-Fin)	−1.82	0.22	2.02	−3.83	−1.83	−0.04	100
D2, t+5: S-D2 RG gated (Laya-Fin)	−2.15	−0.57	1.82	−2.24	−0.85	1.67	100
D2, t+5: S-D2 J gated (anchor)	−2.10	−0.14	1.77	−4.16	−1.90	−0.15	100
D2, t+5: S-D2 RG (Laya-Fin)	−2.29	0.05	1.95	−2.48	−0.23	1.71	100
D2, t+5: S-D2 RG gated (d1-ultra)	−2.41	−0.09	1.69	−4.25	−1.94	−0.24	100
D2, t+5: S-D2 J gated (d1-ultra)	−2.03	−0.06	2.18	−4.37	−2.17	−0.04	100

Annualized Sharpe ratios of the 100 random-signal books of each test universe (a decision point, an exit and a set of events), numpy's linear quantiles; from NA-13's stored placebo books. A universe is named by the books that trade on it, with the model where its events are that model's alone.
†Vendor sentiment: its labels were produced retrospectively, so its books are a possibly look-ahead-contaminated upper bound, not a benchmark; the clean benchmarks are the price-only gap fade and the random-signal placebo.

Table A29: Break-even cost per side of every test book.

Book	Gross return (annual)	Turnover	Break-even (bp)
S-CONV, Laya-Fin, D2 U, to t+1	−0.494	4.65	−4.2
S-CONV, Laya-Fin, D2 U, to t+5	0.054	0.90	2.4
S-CONV, Laya-Fin, D2 U, to t+20	0.010	0.20	1.9
S-CONV, Laya-Fin, D1 U, to C_t	0.228	4.68	1.9
S-CONV, Laya-Fin, D1 U, to t+1	−0.113	2.32	−1.9
S-CONV, Laya-Fin, D1 U, to t+5	0.096	0.75	5.1
S-CONV, Laya-Fin, D1 U, to t+20	−0.029	0.19	−6.1
S-CONV, d1-ultra, D2 U, to t+1	−0.545	4.62	−4.7
S-CONV, d1-ultra, D2 U, to t+5	−0.008	0.90	−0.3
S-CONV, d1-ultra, D2 U, to t+20	0.001	0.20	0.2
S-CONV, d1-ultra, D1 U, to C_t	0.005	4.71	0.0
S-CONV, d1-ultra, D1 U, to t+1	−0.262	2.34	−4.4
S-CONV, d1-ultra, D1 U, to t+5	0.023	0.75	1.2
S-CONV, d1-ultra, D1 U, to t+20	−0.038	0.19	−8.0
S-CONV, anchor, D2 U, to t+1	−0.479	4.64	−4.1
S-CONV, anchor, D2 U, to t+5	0.067	0.90	2.9
S-CONV, anchor, D2 U, to t+20	0.006	0.20	1.3
S-CONV, anchor, D1 U, to C_t	0.151	4.65	1.3
S-CONV, anchor, D1 U, to t+1	−0.155	2.31	−2.7
S-CONV, anchor, D1 U, to t+5	0.067	0.75	3.6
S-CONV, anchor, D1 U, to t+20	−0.029	0.19	−6.2
S-D0, Laya-Fin, D0 $\tilde{N}$, to C_t	0.060	4.36	0.5
S-D0, Laya-Fin, D0 $\tilde{N}$, to t+1	−0.143	2.16	−2.6
S-D0, Laya-Fin, D0 $\tilde{N}$, to C_t, gated	0.429	4.64	3.7
S-D0, Laya-Fin, D0 $\tilde{N}$, to t+1, gated	0.483	2.31	8.3
S-D0, d1-ultra, D0 $\tilde{N}$, to C_t	0.016	4.41	0.1
S-D0, d1-ultra, D0 $\tilde{N}$, to t+1	−0.149	2.19	−2.7
S-D0, d1-ultra, D0 $\tilde{N}$, to C_t, gated	0.372	4.73	3.1
S-D0, d1-ultra, D0 $\tilde{N}$, to t+1, gated	0.048	2.35	0.8
S-D0, anchor, D0 $\tilde{N}$, to C_t	−0.102	4.33	−0.9
S-D0, anchor, D0 $\tilde{N}$, to t+1	−0.079	2.15	−1.5
S-D0, anchor, D0 $\tilde{N}$, to C_t, gated	−0.241	4.79	−2.0
S-D0, anchor, D0 $\tilde{N}$, to t+1, gated	0.035	2.38	0.6
S-D1, Laya-Fin, D1 J, to C_t	0.079	4.36	0.7
S-D1, Laya-Fin, D1 J, to t+1	−0.014	2.16	−0.3
S-D1, Laya-Fin, D1 J, to t+5	−0.125	0.70	−7.1
S-D1, Laya-Fin, D1 RG, to C_t	0.002	0.12	0.6
S-D1, Laya-Fin, D1 RG, to t+1	−0.002	0.06	−1.4
S-D1, Laya-Fin, D1 RG, to t+5	0.054	0.02	109.0
S-D1, d1-ultra, D1 J, to C_t	−0.029	4.51	−0.3
S-D1, d1-ultra, D1 J, to t+1	−0.066	2.24	−1.2
S-D1, d1-ultra, D1 J, to t+5	−0.016	0.72	−0.9
S-D1, d1-ultra, D1 RG, to C_t	−0.257	4.79	−2.1
S-D1, d1-ultra, D1 RG, to t+1	−0.586	2.38	−9.8
S-D1, d1-ultra, D1 RG, to t+5	−0.104	0.77	−5.4
S-D1, anchor, D1 J, to C_t	0.027	4.44	0.2
S-D1, anchor, D1 J, to t+1	−0.113	2.20	−2.0
S-D1, anchor, D1 J, to t+5	0.053	0.71	3.0
S-D2, Laya-Fin, D2 J, to t+1	−0.129	4.26	−1.2
S-D2, Laya-Fin, D2 J, to t+5	−0.148	0.83	−7.1
S-D2, Laya-Fin, D2 J, to t+20	−0.046	0.18	−9.9
S-D2, Laya-Fin, D2 J, to t+1, gated	−0.388	4.67	−3.3
S-D2, Laya-Fin, D2 J, to t+5, gated	−0.102	0.91	−4.5
S-D2, Laya-Fin, D2 J, to t+20, gated	−0.052	0.20	−10.2

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Table A29, continued

Book	Gross return (annual)	Turnover	Break-even (bp)
S-D2, Laya-Fin, D2 RG, to t+1	0.316	0.24	52.6
S-D2, Laya-Fin, D2 RG, to t+5	0.101	0.05	85.1
S-D2, Laya-Fin, D2 RG, to t+20	−0.016	0.01	−61.9
S-D2, Laya-Fin, D2 RG, to t+1, gated	0.017	0.10	6.9
S-D2, Laya-Fin, D2 RG, to t+5, gated	0.015	0.02	29.3
S-D2, Laya-Fin, D2 RG, to t+20, gated	−0.021	0.01	−156.7
S-D2, d1-ultra, D2 J, to t+1	−0.246	4.40	−2.2
S-D2, d1-ultra, D2 J, to t+5	0.030	0.85	1.4
S-D2, d1-ultra, D2 J, to t+20	0.012	0.19	2.6
S-D2, d1-ultra, D2 J, to t+1, gated	−0.294	4.50	−2.6
S-D2, d1-ultra, D2 J, to t+5, gated	−0.046	0.87	−2.1
S-D2, d1-ultra, D2 J, to t+20, gated	0.021	0.19	4.4
S-D2, d1-ultra, D2 RG, to t+1	−0.507	4.72	−4.3
S-D2, d1-ultra, D2 RG, to t+5	0.016	0.92	0.7
S-D2, d1-ultra, D2 RG, to t+20	0.023	0.20	4.5
S-D2, d1-ultra, D2 RG, to t+1, gated	−0.190	4.76	−1.6
S-D2, d1-ultra, D2 RG, to t+5, gated	0.057	0.93	2.4
S-D2, d1-ultra, D2 RG, to t+20, gated	0.026	0.21	5.0
S-D2, anchor, D2 J, to t+1	−0.400	4.38	−3.6
S-D2, anchor, D2 J, to t+5	0.009	0.85	0.4
S-D2, anchor, D2 J, to t+20	−0.001	0.19	−0.2
S-D2, anchor, D2 J, to t+1, gated	−0.556	4.59	−4.8
S-D2, anchor, D2 J, to t+5, gated	0.045	0.89	2.0
S-D2, anchor, D2 J, to t+20, gated	0.076	0.20	15.3
gap fade, D2 gap, to t+1	−0.563	4.64	−4.8
gap fade, D2 gap, to t+5	0.041	0.90	1.8
gap fade, D2 gap, to t+20	−0.004	0.20	−0.9
gap fade, D1 gap, to C_t	0.109	4.69	0.9
gap fade, D1 gap, to t+1	−0.180	2.33	−3.1
gap fade, D1 gap, to t+5	0.030	0.75	1.6
gap fade, D1 gap, to t+20	−0.042	0.19	−8.8
vendor sentiment†, D0, to C_t	0.046	4.36	0.4
vendor sentiment†, D0, to t+1	−0.187	2.16	−3.4

Gross mean ÷ turnover, from NA-13's strategy table. The exploratory MA-cross fade (post hoc) is in Table A34 and Figure A15.

†Vendor sentiment: its labels were produced retrospectively, so its books are a possibly look-ahead-contaminated upper bound, not a benchmark; the clean benchmarks are the price-only gap fade and the random-signal placebo. The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

G Memorisation

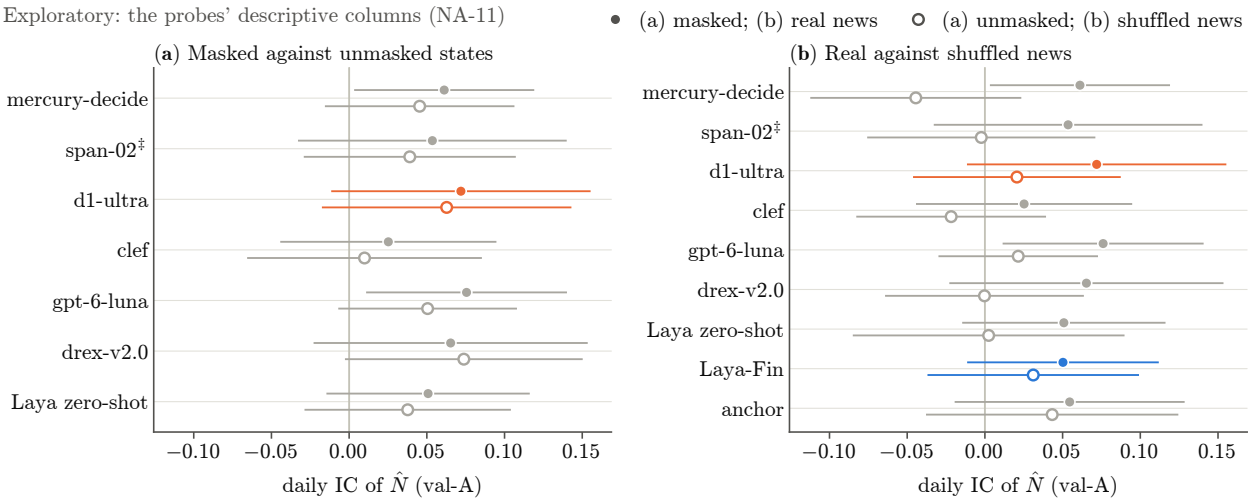


Figure A13: Exploratory: the probes' descriptive columns. Exploratory (post hoc): the probes' descriptive columns (NA-11, val-A), not hypotheses. (a) ICs on masked and unmasked states: unmasking does not raise them materially (d1-ultra 0.072 masked and 0.063 unmasked; the largest rise, drex-v2.0's, is +0.01, t 0.48). Both panels share the IC scale. (b) ICs on real against shuffled news on the same events (Laya-Fin 0.050 and 0.031, inconclusive); the anchor is uninformative (near-constant outputs). ‡ span-02: the S5 canary fired a STOP after the freeze and the operator chose to proceed (PREREG §15).

Table A30: Release dates: every decision model came out after the sample; only the anchor's checkpoint predates it.

Model	Days after the sample
end of the test period, 2026-07-31	0
Laya (open weights; Laya-Fin's base), 2026-09-18	49
drex-v2.0, 2026-10-12	73
d1-ultra, 2026-10-15	76
span-02‡, 2026-10-16	77
ModernBERT-large (the anchor's checkpoint), 2024-12	—

—: the anchor's checkpoint is dated by month only, before the sample, so it has no day count.

Release dates from the NA-05 catalog; Laya's from the operator (Convai's release, weights on Hugging Face); the anchor's checkpoint is known to the month only.

‡span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

Table A31: Company identification on masked states: the API models name the company far above the matched baseline.

Model	Identified	States	Matched baseline	Screen bar
clef	0.660	400	0.282	0.40
mercury-decide	0.620	400	0.282	0.40
d1-ultra	0.743	400	0.282	0.40
Laya zero-shot	0.320	400	0.282	0.40
drex-v2.0	0.680	400	0.282	0.40
gpt-6-luna	0.682	399	0.282	0.40
span-02‡	0.662	400	0.282	0.40

NA-06's val-A events with activity-matched distractors (NA-11); States: the valid answers.

‡span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

Table A32: The disclosed val-A probes (descriptive columns exploratory).

Model	Flip rate	Right way	Shift (SD)	IC masked	IC unmasked	IC real	IC shuffled
mercury-decide	0.56	0.99	1.40	0.061	0.045	0.061	−0.044
span-02[‡]	0.64	0.99	1.61	0.054	0.039	0.054	−0.002
d1-ultra	0.68	0.98	1.83	0.072	0.063	0.072	0.021
clef	0.53	0.98	1.24	0.025	0.010	0.025	−0.022
gpt-6-luna	0.57	0.95	1.47	0.076	0.051	0.076	0.021
drex-v2.0	0.71	0.98	1.59	0.065	0.074	0.065	0.000
Laya zero-shot	0.04	0.94	0.41	0.051	0.038	0.051	0.002
Laya-Fin	0.00	0.80	0.20	—	—	0.050	0.031
anchor	—	0.43	0.00	—	—	0.055	0.043

—: the probe was not run for that model.
Exploratory: the probes' descriptive columns (NA-11), not hypotheses.
The anchor's readings are uninformative (near-constant outputs after one epoch); its rows are reported, not read as nulls.
[‡]span-02: the S5 canary fired a STOP after the freeze; the operator chose to proceed (PREREG §15, post-freeze NA-11 row).

H Exploratory analyses (post hoc)

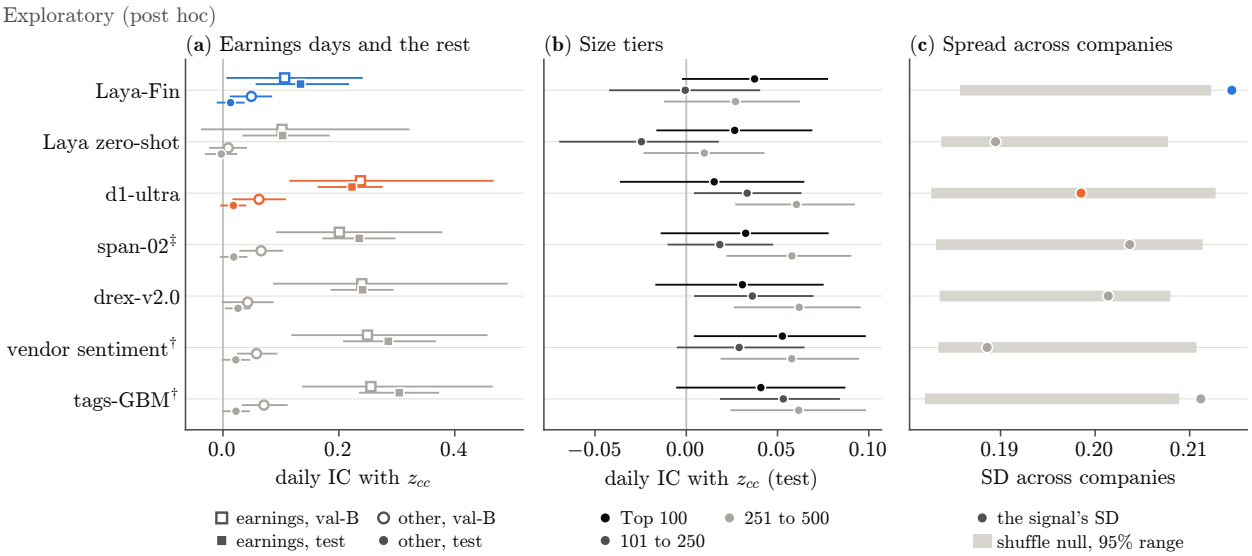


Figure A14: Exploratory (post hoc): where the D0 signal lives. Exploratory (post hoc): requested after unblinding (NA-12 §13); no hypothesis, no claim. (a) Every signal's IC sits on earnings-day events (squares), and there it held from val-B (hollow) to test (filled): Laya-Fin 0.11 and 0.13, d1-ultra 0.24 and 0.22. On other events (circles) it fell, for Laya-Fin from 0.049 to 0.013, and for every other signal too. (b) By size tier on test the API models' IC sits in the two smaller tiers (d1-ultra 0.015 in the Top-100 against 0.060 in the 251–500 tier). The earnings-day ICs of (a) are several times the tier ICs of (b), so the two panels have their own scales. (c) The spread of the per-company rank correlations against a within-company shuffle (gray: its 95% range): Laya-Fin's (0.214, p 0.015) and tags-GBM's exceed it, the others' do not; the test cannot tell company-specific skill from company-specific noise. Not drawn: a one-way test of the daily ICs on the calendar month finds no month effect on test (p 0.72 for Laya-Fin and 0.44 for d1-ultra). † Vendor-label baselines: a possibly look-ahead-contaminated upper bound. ‡ span-02: the S5 canary fired a STOP after the freeze and the operator chose to proceed (PREREG §15).

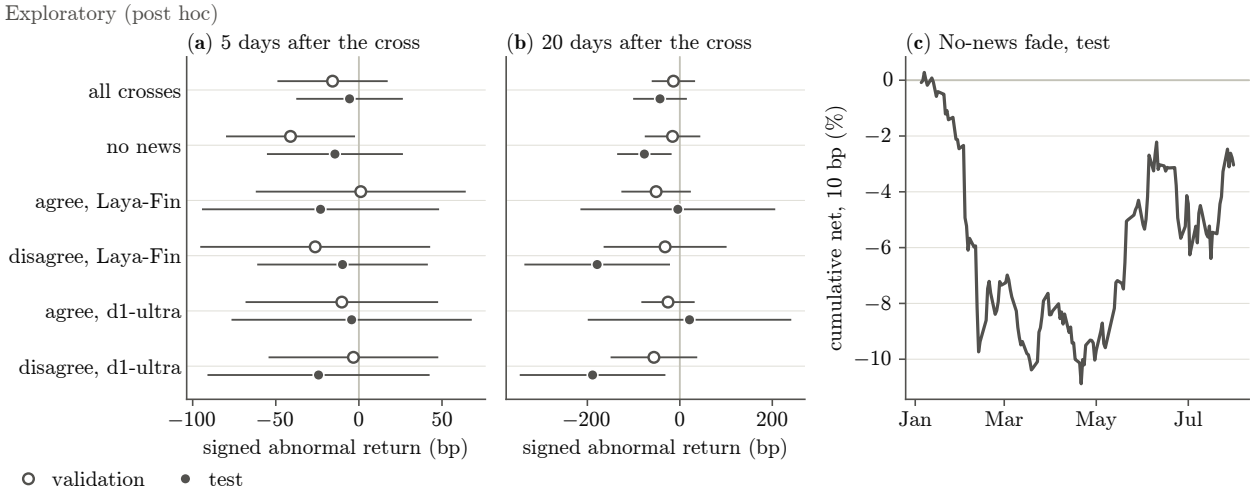


Figure A15: Exploratory (post hoc): news and a moving-average cross. Exploratory (post hoc): the operator's MA20/MA50 cross add-on (NA-13 §9), designed on validation and run once on test; no claim. (a, b) Mean signed abnormal return over 5 and 20 days after a cross, by group (95% intervals, ±1.96 Newey–West SE), validation (hollow) against test (filled); the two panels have their own scales, since 20-day returns spread wider than 5-day ones. The validation reversal of crosses without news (−41.1 bp, t −2.08) did not replicate on test (−14.4 bp, t −0.69). (c) The tradable no-news fade on test, cumulative net at 10 bp: net Sharpe −0.46 (gross 0.98, after 3.85 on validation, where its design was chosen).

Table A33: Exploratory: the IC by GICS sector.

Model, kind, sector	IC z_{cc}	t	Events	IC z_{on}	IC z_{id}
Laya-Fin, by sector, Communication Services	0.049	0.70	524	−0.031	0.035
Laya-Fin, by sector, Consumer Discretionary	0.063	2.22	1,381	0.032	0.023
Laya-Fin, by sector, Consumer Staples	0.052	1.18	934	0.070	−0.026
Laya-Fin, by sector, Energy	−0.046	−0.85	531	−0.009	−0.082
Laya-Fin, by sector, Financials	0.045	1.47	1,798	0.012	0.036
Laya-Fin, by sector, Health Care	0.043	1.51	1,999	0.034	0.030
Laya-Fin, by sector, Industrials	−0.030	−1.25	2,408	−0.003	−0.025
Laya-Fin, by sector, Information Technology	0.036	1.41	2,587	−0.021	0.035
Laya-Fin, by sector, Materials	−0.043	−0.81	435	−0.068	−0.097
Laya-Fin, by sector, Real Estate	0.100	1.20	431	0.043	0.124
Laya-Fin, by sector, Utilities	−0.035	−0.56	720	0.074	−0.113
Laya-Fin, leave one out, Communication Services	0.020	1.61	13,224	0.013	0.012
Laya-Fin, leave one out, Consumer Discretionary	0.022	1.76	12,367	0.012	0.012
Laya-Fin, leave one out, Consumer Staples	0.017	1.32	12,814	0.017	0.007
Laya-Fin, leave one out, Energy	0.021	1.72	13,217	0.014	0.012
Laya-Fin, leave one out, Financials	0.016	1.13	11,950	0.010	0.007
Laya-Fin, leave one out, Health Care	0.016	1.32	11,749	0.009	0.006
Laya-Fin, leave one out, Industrials	0.030	2.32	11,340	0.016	0.020
Laya-Fin, leave one out, Information Technology	0.011	1.00	11,161	0.026	0.001
Laya-Fin, leave one out, Materials	0.021	1.79	13,313	0.016	0.012
Laya-Fin, leave one out, Real Estate	0.020	1.63	13,317	0.016	0.010
Laya-Fin, leave one out, Utilities	0.024	1.99	13,028	0.018	0.014
anchor, by sector, Communication Services	0.033	0.61	524	−0.102	0.058
anchor, by sector, Consumer Discretionary	0.008	0.22	1,381	0.060	−0.037
anchor, by sector, Consumer Staples	−0.095	−2.06	934	−0.028	−0.098
anchor, by sector, Energy	−0.003	−0.07	531	−0.003	−0.008
anchor, by sector, Financials	0.060	2.30	1,798	−0.005	0.081
anchor, by sector, Health Care	0.017	0.50	1,999	0.018	−0.003
anchor, by sector, Industrials	−0.018	−0.58	2,408	−0.048	−0.003
anchor, by sector, Information Technology	0.049	2.48	2,587	0.021	0.030
anchor, by sector, Materials	−0.061	−1.17	435	−0.102	−0.089
anchor, by sector, Real Estate	0.004	0.07	431	−0.070	0.067
anchor, by sector, Utilities	−0.037	−0.82	720	−0.128	0.033
anchor, leave one out, Communication Services	0.009	0.99	13,224	0.002	0.006
anchor, leave one out, Consumer Discretionary	0.009	0.83	12,367	−0.002	0.007
anchor, leave one out, Consumer Staples	0.011	1.13	12,814	0.008	0.006
anchor, leave one out, Energy	0.012	1.28	13,217	0.006	0.006
anchor, leave one out, Financials	0.004	0.43	11,950	0.006	−0.003
anchor, leave one out, Health Care	0.006	0.60	11,749	0.002	0.003
anchor, leave one out, Industrials	0.007	0.68	11,340	0.018	−0.006
anchor, leave one out, Information Technology	−0.001	−0.11	11,161	−0.001	−0.004
anchor, leave one out, Materials	0.008	0.89	13,313	0.008	0.002
anchor, leave one out, Real Estate	0.008	0.82	13,317	0.005	0.003
anchor, leave one out, Utilities	0.008	0.78	13,028	0.009	0.000
d1-ultra, by sector, Communication Services	0.103	1.79	524	0.091	0.010
d1-ultra, by sector, Consumer Discretionary	0.087	2.64	1,381	0.080	0.008
d1-ultra, by sector, Consumer Staples	0.069	1.90	934	0.057	0.020
d1-ultra, by sector, Energy	0.120	1.92	531	0.095	0.047
d1-ultra, by sector, Financials	0.044	1.61	1,798	0.092	−0.002
d1-ultra, by sector, Health Care	0.055	1.74	1,999	0.082	0.050
d1-ultra, by sector, Industrials	0.007	0.28	2,408	0.049	−0.026
d1-ultra, by sector, Information Technology	0.034	1.25	2,587	0.063	0.003
d1-ultra, by sector, Materials	0.008	0.16	435	0.064	−0.022
d1-ultra, by sector, Real Estate	0.090	1.08	431	0.055	−0.009

continued on the next page

Table A33, continued

Model, kind, sector	IC z_{cc}	t	Events	IC z_{on}	IC z_{id}
d1-ultra, by sector, Utilities	−0.084	−1.93	720	0.038	−0.124
d1-ultra, leave one out, Communication Services	0.040	3.26	13,224	0.074	−0.001
d1-ultra, leave one out, Consumer Discretionary	0.035	2.71	12,367	0.067	−0.002
d1-ultra, leave one out, Consumer Staples	0.039	3.16	12,814	0.076	0.001
d1-ultra, leave one out, Energy	0.040	3.20	13,217	0.075	−0.001
d1-ultra, leave one out, Financials	0.041	3.45	11,950	0.071	0.004
d1-ultra, leave one out, Health Care	0.030	2.20	11,749	0.067	−0.010
d1-ultra, leave one out, Industrials	0.040	2.85	11,340	0.075	0.002
d1-ultra, leave one out, Information Technology	0.043	3.72	11,161	0.081	−0.002
d1-ultra, leave one out, Materials	0.040	3.22	13,313	0.075	0.000
d1-ultra, leave one out, Real Estate	0.039	3.17	13,317	0.075	−0.001
d1-ultra, leave one out, Utilities	0.043	3.54	13,028	0.078	0.002

Exploratory (post hoc): requested after unblinding; no hypothesis, outside every family, no claim. The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A34: Exploratory: news and a moving-average cross; the validation result did not replicate.

Group	Mean (bp)	t	Days	Gross Sharpe	Net 10 bp
validation, all crosses, 5 days	−15.8	−0.94	123	—	—
validation, all crosses, 20 days	−13.9	−0.58	108	—	—
validation, no news, 5 days	−41.1	−2.08	122	—	—
validation, no news, 20 days	−15.8	−0.52	107	—	—
validation, agree, Laya-Fin, 5 days	1.1	0.04	110	—	—
validation, agree, Laya-Fin, 20 days	−51.3	−1.34	97	—	—
validation, disagree, Laya-Fin, 5 days	−26.3	−0.74	107	—	—
validation, disagree, Laya-Fin, 20 days	−31.9	−0.47	96	—	—
validation, agree, d1-ultra, 5 days	−10.3	−0.35	106	—	—
validation, agree, d1-ultra, 20 days	−25.6	−0.87	93	—	—
validation, disagree, d1-ultra, 5 days	−3.3	−0.13	103	—	—
validation, disagree, d1-ultra, 20 days	−56.1	−1.17	93	—	—
validation, the no-news fade	—	—	127	3.85	1.83
test, all crosses, 5 days	−5.6	−0.34	140	—	—
test, all crosses, 20 days	−42.9	−1.45	125	—	—
test, no news, 5 days	−14.4	−0.69	138	—	—
test, no news, 20 days	−76.7	−2.55	123	—	—
test, agree, Laya-Fin, 5 days	−23.1	−0.63	123	—	—
test, agree, Laya-Fin, 20 days	−4.3	−0.04	110	—	—
test, disagree, Laya-Fin, 5 days	−9.8	−0.38	117	—	—
test, disagree, Laya-Fin, 20 days	−178.8	−2.23	103	—	—
test, agree, d1-ultra, 5 days	−4.4	−0.12	119	—	—
test, agree, d1-ultra, 20 days	20.9	0.19	107	—	—
test, disagree, d1-ultra, 5 days	−24.3	−0.71	121	—	—
test, disagree, d1-ultra, 20 days	−188.7	−2.35	106	—	—
test, the no-news fade	—	—	144	0.98	−0.46

—: the groups have mean returns and the tradable fades Sharpe ratios.
Exploratory (post hoc): requested after unblinding; no hypothesis, outside every family, no claim.
MA20/MA50 crosses; signed abnormal return over 5 or 20 days by group; the tradable no-news fade.

Table A35: Exploratory: the A3.5 market-level replication.

Item	OOS R^2	p	Sessions
SPY's daily return on d1-ultra's market reading	0.021	0.309	144

Exploratory (PREREG A3.5: optional, exploratory); fit on the validation sessions, evaluated on test.

I When the D0 signal works (exploratory, post hoc; NA-13b)

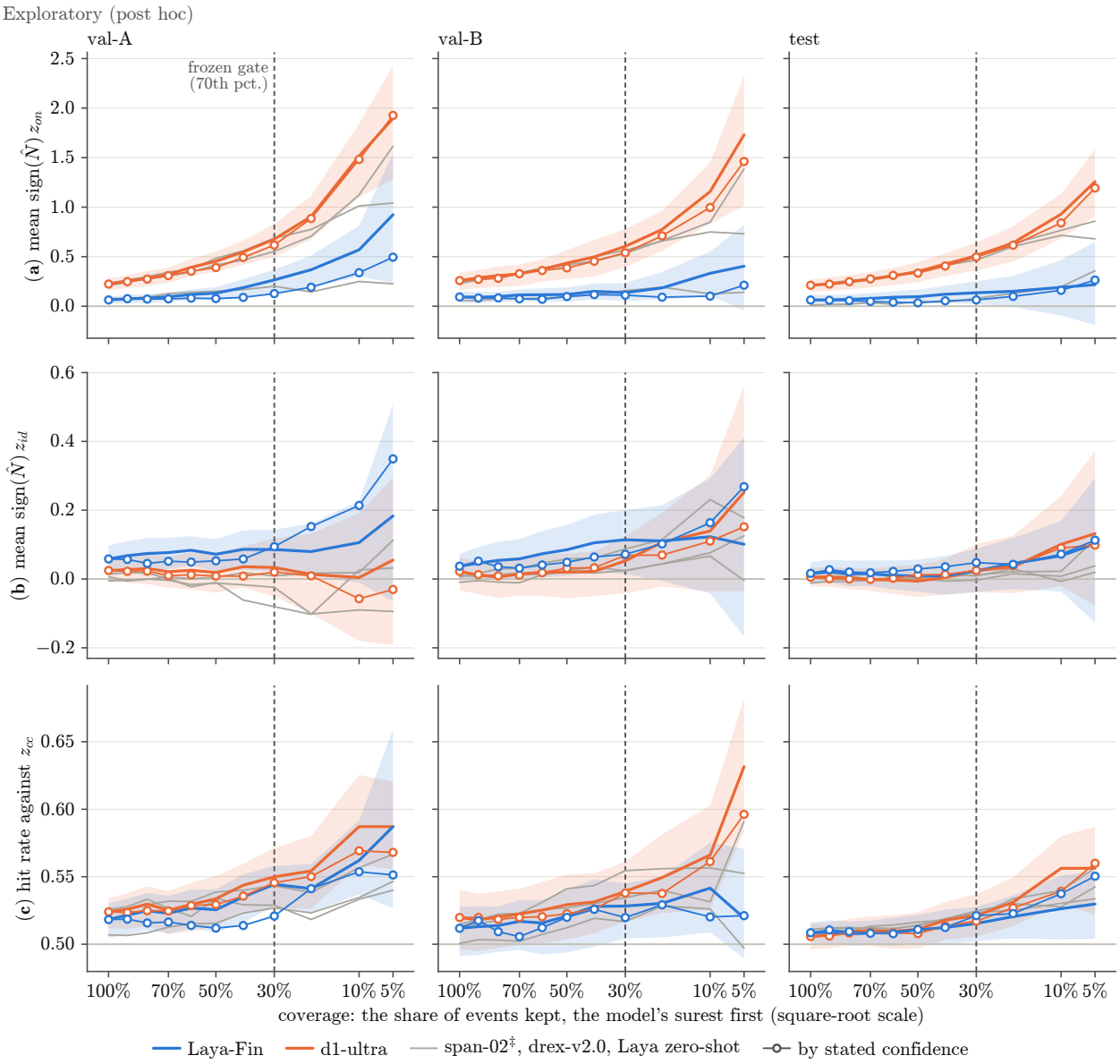


Figure A16: Exploratory (post hoc): risk-coverage curves of the pre-open reading. Exploratory (post hoc): NA-13b, requested after unblinding; no hypothesis, no claim. Its thresholds come from val-A. The operator's assistant's first look by band of conviction, which used the test, came before this figure and shaped it (the band-check table in the appendix); no threshold is chosen from the curves. Each model's D0 events are ranked by its conviction $|\hat{N}|$ (solid lines) or, for Laya-Fin and d1-ultra, by its stated confidence (thin lines, hollow dots); a coverage keeps the events at or above the val-A quantile of the ranking. Rows: (a) the mean signed overnight move sign($\hat{N}$) z_{on}, (b) the mean signed move after the open sign($\hat{N}$) z_{id}, (c) the hit rate against z_{cc}; columns val-A, val-B and test; bands are 95% bootstrap intervals; the dashed line is the frozen confidence gates' coverage (their val-A 70th percentile). Conviction concentrates the overnight move, which the open prices: d1-ultra's surest 10% carry 0.93 typical moves on the test against 0.21 for all its events. After the open little is left: 0.10 (95% interval −0.02 to 0.24) for d1-ultra's surest 10% and 0.07 for Laya-Fin's; neither model's test interval excludes zero at any coverage. span-02's and drex-v2.0's val-A points come from NA-06's 3,000-event stage-2 sample. d1-ultra is model-specific (§1.3). ‡ span-02: the S5 canary fired a STOP after the freeze and the operator chose to proceed (PREREG §15).

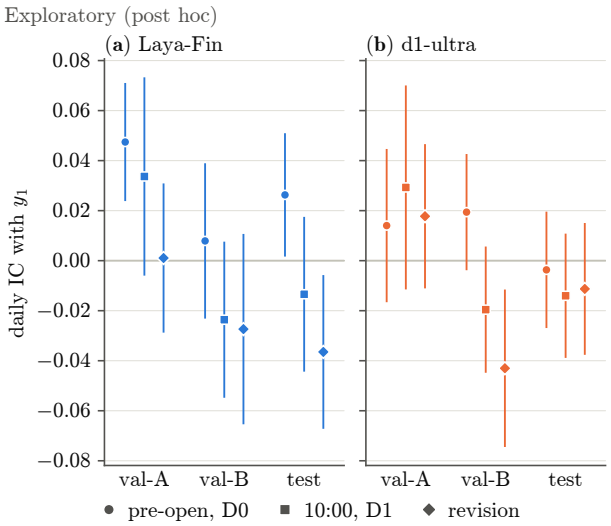


Figure A17: Exploratory (post hoc): the 10:00 reading against the pre-open one. Exploratory (post hoc): NA-13b, requested after unblinding; no hypothesis, no claim. The operator's assistant computed the four ICs on all three splits before NA-13b; they are reproduced exactly. The daily IC with y_1, the move from 10:00 to the close, of the pre-open reading $\hat{N}_{\mathrm{D0}}$ (circles), of the 10:00 reading $\hat{N}_{\mathrm{D1}}$ (Call B at D1; squares) and of their revision $z(\hat{N}_{\mathrm{D1}}) - z(\hat{N}_{\mathrm{D0}})$ (diamonds) for (a) Laya-Fin and (b) d1-ultra on val-A, val-B and the test, with 95% intervals (± 1.96 Newey–West SE). Watching the open adds nothing: on val-B and the test the 10:00 reading's IC is negative for both models (test: Laya-Fin -0.013, d1-ultra -0.014), and so is the revision's (Laya-Fin on test -0.036, t -2.33); only Laya-Fin's pre-open reading keeps a positive test IC (0.026, t 2.09). The first bar, the moves after it and the pre-registered D1 results are in the appendix table on watching the open. With 36 statistics in that table, about two with $|t| > 2$ are expected by chance. d1-ultra is model-specific (§1.3).



Figure A18: Exploratory (post hoc): the conditioning study, validation against test. Exploratory (post hoc): NA-13b, requested after unblinding; no hypothesis, no claim. Thresholds and medians come from val-A; the selection used val-A and val-B only, by the operator's rule fixed before the test; the test was evaluated once. Each row: the difference inside minus outside of the primary outcome $\text{sign}(\hat{N})\, z_{id}$ (the move after the open) for (a) Laya-Fin and (b) d1-ultra on val-A (hollow circles), val-B (hollow squares) and the test (filled), with 95% intervals (± 1.96 |difference/t|). Shaded rows were selected on validation ($t \geq 1$ in val-A and in val-B, pooled $t \geq 2$). None held on the test: their test differences run from -0.06 (Laya-Fin's E9, the strongest on validation, pooled t 3.51) to $+0.04$ (d1-ultra's E7 agreement, t 1.01), every Benjamini–Hochberg q of the family is at least 0.79, and no pair is confirmed. (seen): seen on the test before the study, never selected. † Vendor labels produced retrospectively, possibly look-ahead-contaminated. E7's three-model agreement has no val-A run; its rank average is the ensemble's sign against the model's own on the same events; E9 is Laya-Fin's alone. d1-ultra is model-specific (§1.3).

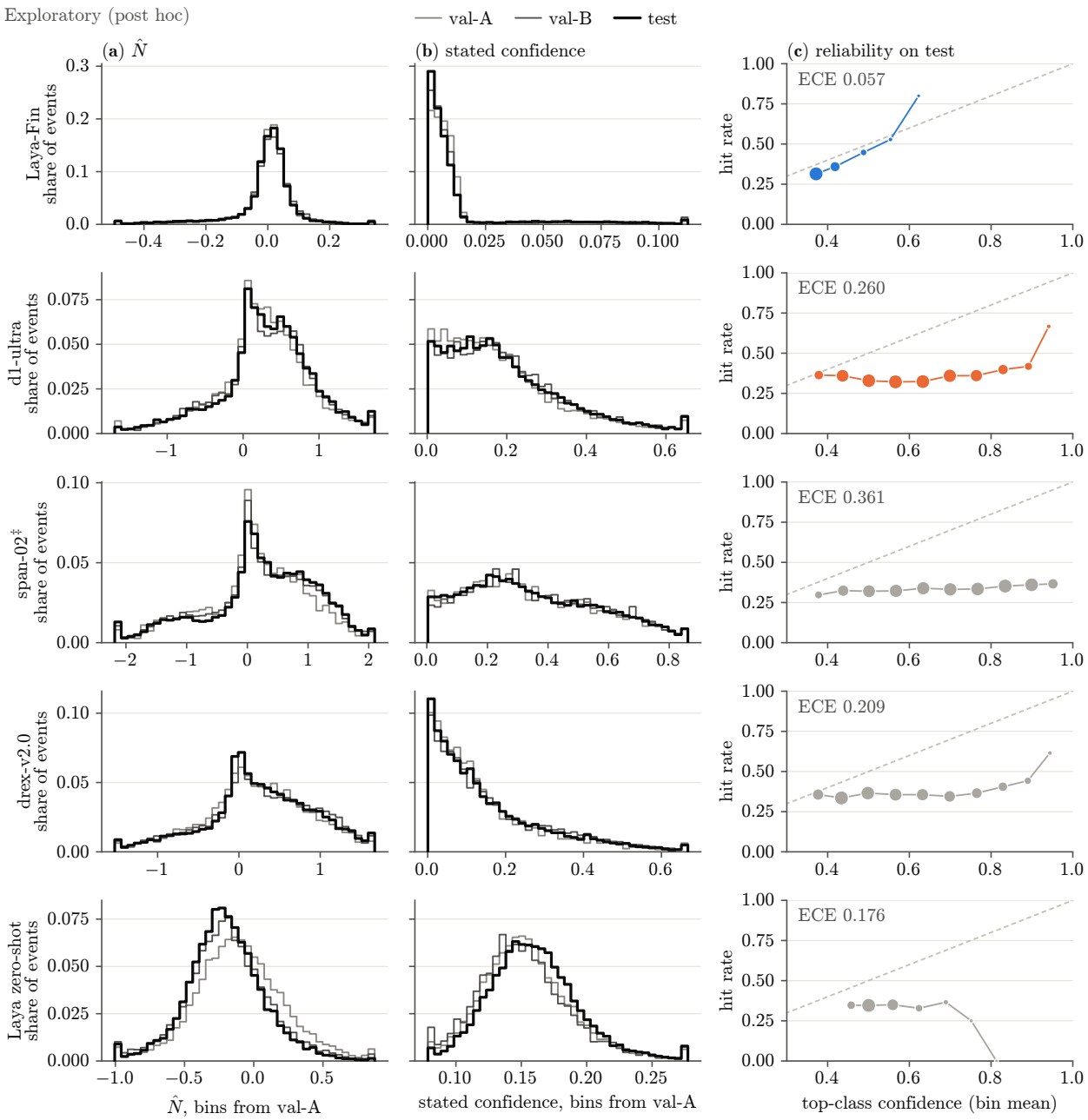


Figure A19: Exploratory (post hoc): the decision models' outputs beside their reliability. Exploratory (post hoc): NA-13b, requested after unblinding; no hypothesis, no claim. Its thresholds, medians and bins come from val-A, and the test was evaluated once. For each model, the share of its D0 events in each of 40 bins of (a) $\hat{N}$ and (b) the stated confidence, $1 - H/\log 3$ of Q1's three classes (the measure of the frozen gates), on val-A, val-B and the test; the bins span the val-A 0.5th to 99.5th percentiles and the end bins hold the values beyond. (c) NA-12's reliability diagram on the test (Figure A4): the hit rate of Q1's top class against e_5 by bin of the top-class probability, the confidence of §11.2's ECE, not the entropy confidence of (b); dot area grows with the events in the bin, the dashed diagonal is perfect calibration. Laya-Fin reads small: its $\hat{N}$ has an SD of 0.11 on test against 0.61 for d1-ultra, and its stated confidence stays below 0.073 for 95% of its test events; it is calibrated (ECE 0.057) where d1-ultra is not (0.260). The distributions barely move from val-A to the test, except Laya zero-shot's $\hat{N}$ (median −0.11 on val-A, −0.21 on test). span-02's and drex-v2.0's val-A histograms come from NA-06's 3,000-event stage-2 sample. ‡ span-02: the S5 canary fired a STOP after the freeze and the operator chose to proceed (PREREG §15).

Table A36: Exploratory (post hoc): ranking by conviction concentrates the overnight move, which the open prices; the move after the open stays within its 95% interval of zero at every coverage on the test (NA-13b).

	val-A			val-B			Test		
Model, coverage	z_{cc}	z_{on}	z_{id}	z_{cc}	z_{on}	z_{id}	z_{cc}	z_{on}	z_{id}
Panel A: mean signed move $\text{sign}(\hat{N})\, z$ *(typical daily moves)*									
Laya-Fin, 100%	0.088	0.064	0.058	0.083	0.093	0.037	0.043	0.063	0.016
Laya-Fin, 30%	0.231	0.265	0.086	0.183	0.137	0.114	0.083	0.136	0.024
Laya-Fin, 10%	0.459	0.569	0.106	0.351	0.332	0.123	0.172	0.191	0.068
d1-ultra, 100%	0.133	0.223	0.025	0.155	0.259	0.022	0.102	0.212	0.005
d1-ultra, 30%	0.366	0.677	0.034	0.369	0.604	0.053	0.273	0.508	0.025
d1-ultra, 10%	0.799	1.516	0.004	0.754	1.157	0.140	0.592	0.930	0.101
Panel B: hit rate, the share with $\text{sign}(\hat{N}) = \text{sign}(z)$									
Laya-Fin, 100%	0.518	0.498	0.516	0.512	0.513	0.511	0.509	0.505	0.505
Laya-Fin, 30%	0.544	0.517	0.522	0.528	0.527	0.527	0.515	0.519	0.501
Laya-Fin, 10%	0.562	0.500	0.520	0.542	0.542	0.539	0.526	0.525	0.510
d1-ultra, 100%	0.524	0.521	0.514	0.520	0.523	0.507	0.506	0.524	0.501
d1-ultra, 30%	0.550	0.575	0.515	0.539	0.535	0.504	0.521	0.553	0.502
d1-ultra, 10%	0.587	0.616	0.501	0.566	0.577	0.517	0.556	0.587	0.523

Exploratory (post hoc): requested after unblinding; no hypothesis, outside every family, no claim. NA-13b, the operator's study of when the D0 signal works (PREREG §15, NA-13b row): thresholds from val-A. The operator's assistant computed the first of these statistics on all three splits, the test included, before NA-13b (the band check of Table A41 and the first four ICs of Table A37); they shaped the study, they are reproduced exactly here, and no threshold is chosen from them.

Coverage: the D0 events whose conviction $|\hat{N}|$ is at or above its val-A quantile at one minus the coverage (100%: every event); events outside the embargo; moves in typical daily moves. The 95% intervals (stationary block bootstrap), every coverage, the other models and the ranking by the stated confidence are in the table's CSV and Figure A16; the daily IC inside the covered set is in the CSV only.

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A37: Exploratory (post hoc): the 10:00 reading adds nothing to the pre-open one; on val-B and the test the ICs of the 10:00 reading and of the revision with y_1 are negative for both models (NA-13b).

	val-A		val-B		Test	
Model: statistic	Value	t	Value	t	Value	t
Panel A: watching the open (NA-13b)						
Laya-Fin: IC, pre-open reading $\hat{N}_{\mathrm{D0}}$	0.047	3.94	0.008	0.50	0.026	2.09
Laya-Fin: IC, 10:00 reading $\hat{N}_{\mathrm{D1}}$	0.034	1.66	−0.024	−1.48	−0.013	−0.85
Laya-Fin: IC, the revision	0.001	0.07	−0.027	−1.41	−0.036	−2.33
Laya-Fin: IC, the first bar G_{D1}	0.030	1.53	−0.034	−1.59	−0.016	−1.03
Laya-Fin: mean, first bar against	−0.010	−0.35	0.000	−0.01	0.045	1.73
Laya-Fin: mean, first bar with	0.111	3.74	−0.037	−1.18	−0.001	−0.04
d1-ultra: IC, pre-open reading $\hat{N}_{\mathrm{D0}}$	0.014	0.90	0.019	1.64	−0.004	−0.31
d1-ultra: IC, 10:00 reading $\hat{N}_{\mathrm{D1}}$	0.029	1.41	−0.020	−1.52	−0.014	−1.11
d1-ultra: IC, the revision	0.018	1.21	−0.043	−2.68	−0.011	−0.84
d1-ultra: IC, the first bar G_{D1}	0.030	1.53	−0.034	−1.59	−0.016	−1.03
d1-ultra: mean, first bar against	−0.039	−1.52	0.039	1.32	0.008	0.35
d1-ultra: mean, first bar with	0.078	3.15	−0.013	−0.38	−0.036	−1.50
Panel B: pre-registered D1 results (NA-12, NA-13; test, for reference)						
Laya-Fin: over/under b at D1	—	—	—	—	0.229	—
Laya-Fin: over/under c at D1	—	—	—	—	0.013	—
Laya-Fin: κ at 10:00	—	—	—	—	0.273	—
Laya-Fin: λ at C_t from D1	—	—	—	—	−0.011	—
Laya-Fin: S-D1 J, gross Sharpe	—	—	—	—	0.76	—
Laya-Fin: S-CONV from 10:00, gross Sharpe	—	—	—	—	1.61	—
d1-ultra: over/under b at D1	—	—	—	—	−0.018	—
d1-ultra: over/under c at D1	—	—	—	—	0.015	—
d1-ultra: κ at 10:00	—	—	—	—	0.610	—
d1-ultra: λ at C_t from D1	—	—	—	—	−0.013	—
d1-ultra: S-D1 J, gross Sharpe	—	—	—	—	−0.28	—
d1-ultra: S-CONV from 10:00, gross Sharpe	—	—	—	—	0.03	—

Exploratory (post hoc): requested after unblinding; no hypothesis, outside every family, no claim. NA-13b, the operator's study of when the D0 signal works (PREREG §15, NA-13b row): thresholds from val-A. The operator's assistant computed the first of these statistics on all three splits, the test included, before NA-13b (the band check of Table A41 and the first four ICs of Table A37); they shaped the study, they are reproduced exactly here, and no threshold is chosen from them.

Multiple comparisons: Panel A holds 36 statistics (six for each model and split; the first bar is the same for both models), so about two with $|t| > 2$ are expected by chance.

Panel A: the daily IC with y_1, the move from 10:00 to the close (PREREG §5.6), of the pre-open reading $\hat{N}_{\mathrm{D0}}$, of the 10:00 reading $\hat{N}_{\mathrm{D1}}$ (Call B at D1), of the revision $z(\hat{N}_{\mathrm{D1}}) - z(\hat{N}_{\mathrm{D0}})$ and of the first bar's move G_{D1}, with its Newey–West t (5 lags); then the mean of $\mathrm{sign}(\hat{N}_{\mathrm{D0}})\, y_1$ over the events whose first bar moved against, or with, the pre-open reading, with a day-clustered t. Events outside the embargo; the first bar is the same for both models.

Panel B: the pre-registered D1 estimates on the test (NA-12, NA-13): the over/under coefficients b and c (Table A13), κ at 10:00 (Table A16), λ at the close from D1 (Table A17) and the annualized gross Sharpe ratios of S-D1 J and of S-CONV from 10:00 to the close (Table A25). —: Panel B holds the test estimates only; their standard errors and p are in those tables.

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A38: Exploratory (post hoc): the conditioning study; none of the pairs selected on validation held on the test, and no pair of the family is confirmed (NA-13b).

	val-A		val-B		Validation		Test, evaluated once				
Condition	Diff.	t	Diff.	t	Pooled t	Selected	Diff.	t	BH q	BH q, unseen	Confirmed
Laya-Fin											
E1 earnings day [seen on test]	0.225	0.72	0.167	0.50	0.87	no	−0.330	−1.58	0.935	—	no
E1 quantified†	−0.006	−0.17	−0.098	−1.18	−1.16	no	−0.008	−0.15	0.935	0.924	no
E1 press wire†	0.020	0.57	0.008	0.15	0.46	no	−0.013	−0.45	0.935	0.924	no
E1 imprecise timestamp†	−0.012	−0.32	0.008	0.11	−0.04	no	−0.060	−1.26	0.935	0.924	no
E2 new, not known	0.033	0.82	0.055	1.18	1.43	no	0.024	0.85	0.935	0.924	no
E2 numbers, not narrative	0.061	1.63	0.150	2.96	3.34	yes	−0.040	−0.98	0.935	0.924	no
E2 scheduled	0.032	1.03	0.012	0.25	0.77	no	−0.008	−0.26	0.935	0.924	no
E2 company-specific	0.027	0.70	0.164	1.95	2.05	no	−0.024	−0.41	0.935	0.924	no
E2 confirmed	−0.012	−0.35	0.136	2.83	2.04	no	−0.007	−0.23	0.935	0.924	no
E2 impact horizon above median	−0.020	−0.56	−0.007	−0.18	−0.51	no	0.033	1.15	0.935	0.924	no
E2 uncertainty below median	−0.015	−0.51	−0.139	−3.37	−2.98	no	0.014	0.43	0.935	0.924	no
E2 mixed signals below median	0.016	0.39	−0.008	−0.14	0.11	no	−0.067	−2.20	0.935	0.924	no
E2 materiality above median	0.074	2.31	0.088	2.49	3.40	yes	−0.016	−0.49	0.935	0.924	no
E2 P(no material effect) below median	0.073	2.61	0.079	2.02	3.17	yes	−0.037	−1.20	0.935	0.924	no
E3 latest article before 08:00 [seen on test]	−0.018	−0.51	−0.102	−1.85	−1.83	no	0.018	0.54	0.935	—	no
E3 04:00 to 08:00, not 08:00 to 09:15	−0.022	−0.55	−0.105	−1.66	−1.69	no	0.030	0.79	0.935	0.924	no
E4 Friday	0.021	0.35	−0.075	−1.46	−0.67	no	−0.007	−0.15	0.935	0.924	no
E4 Monday	−0.003	−0.07	0.034	0.70	0.48	no	−0.008	−0.20	0.935	0.924	no
E4 busy news day (ratio ≥ 1.5)	0.046	1.17	0.191	2.51	2.75	yes	0.014	0.40	0.935	0.924	no
E4 news count above median	0.058	1.33	−0.004	−0.08	1.06	no	0.029	0.87	0.935	0.924	no
E4 earnings count above median	0.090	2.18	0.015	0.33	1.65	no	−0.007	−0.20	0.935	0.924	no
E4 FOMC, CPI or payrolls day	0.052	0.54	0.096	1.04	1.14	no	−0.062	−1.06	0.935	0.924	no
E4 VIX at $t-1$ above median	0.006	0.14	−0.001	−0.01	−0.10	no	0.114	1.50	0.935	0.924	no
E4 |SPY overnight| above median	0.012	0.28	0.028	0.61	0.59	no	−0.008	−0.23	0.935	0.924	no
E5 size tier 251 to 500	−0.001	−0.02	0.091	1.74	1.49	no	−0.007	−0.23	0.935	0.924	no
E5 dollar-volume tier 251 to 500	−0.040	−1.10	0.037	0.64	−0.04	no	−0.013	−0.44	0.935	0.924	no
E5 low usual coverage	0.043	1.32	0.058	0.97	1.50	no	0.021	0.35	0.935	0.924	no
E5 σ above median	0.003	0.11	0.018	0.43	0.42	no	0.047	1.30	0.935	0.924	no
E5 direction agrees with z_{m20}	0.025	0.56	0.001	0.02	0.35	no	−0.006	−0.13	0.935	0.924	no
E5 price-move text	−0.003	−0.05	0.011	0.14	0.08	no	0.045	0.82	0.935	0.924	no
E6 conviction top 30% [seen on test]	0.038	0.77	0.127	2.55	2.35	no	0.004	0.09	0.942	—	no
E6 Laya-Fin's magnitude above median	0.057	1.68	0.120	2.79	3.22	yes	−0.016	−0.48	0.935	0.924	no
E6 d1-ultra's confidence above median	0.019	0.55	−0.014	−0.31	0.09	no	−0.049	−1.73	0.935	0.924	no
E7 Laya-Fin and d1-ultra agree	0.003	0.06	−0.028	−0.52	−0.37	no	−0.016	−0.50	0.935	0.924	no

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Table A38, continued (Laya-Fin)

	val-A		val-B		Validation		Test, evaluated once				
Condition	Diff.	t	Diff.	t	Pooled t	Selected	Diff.	t	BH q	BH q, unseen	Confirmed
E7 three API models agree	—	—	0.086	1.65	—	no	−0.020	−0.53	0.935	0.924	no
E7 rank average of the two models	−0.028	−1.96	−0.010	−0.61	−1.74	no	−0.001	−0.04	0.966	0.966	no
E8 $\lvert V\rvert\,\sigma$ above the hurdle	0.097	0.91	0.182	1.63	1.79	no	0.051	0.73	0.935	0.924	no
E9 numbers and material	0.074	1.44	0.208	3.40	3.51	yes	−0.058	−1.00	0.935	0.924	no

Table A38, continued (d1-ultra)

	val-A		val-B		Validation		Test, evaluated once				
Condition	Diff.	t	Diff.	t	Pooled t	Selected	Diff.	t	BH q	BH q, unseen	Confirmed
d1-ultra											
E1 earnings day [seen on test]	−0.173	−0.56	−0.125	−0.36	−0.65	no	0.268	1.33	0.935	—	no
E1 quantified†	−0.063	−1.55	−0.062	−0.80	−1.43	no	0.012	0.23	0.935	0.924	no
E1 press wire†	0.054	1.41	−0.058	−1.40	−0.07	no	−0.016	−0.49	0.935	0.924	no
E1 imprecise timestamp†	−0.025	−0.64	−0.129	−1.82	−1.89	no	−0.014	−0.32	0.935	0.924	no
E2 new, not known	0.070	1.57	−0.063	−1.60	0.10	no	0.047	1.58	0.935	0.924	no
E2 numbers, not narrative	0.056	1.34	0.022	0.42	1.16	no	0.033	0.75	0.935	0.924	no
E2 scheduled	0.004	0.13	−0.011	−0.23	−0.12	no	−0.021	−0.66	0.935	0.924	no
E2 company-specific	−0.004	−0.11	0.066	0.85	0.74	no	−0.103	−1.60	0.935	0.924	no
E2 confirmed	0.059	1.58	0.038	0.90	1.73	no	0.027	0.94	0.935	0.924	no
E2 impact horizon above median	0.010	0.28	0.025	0.51	0.59	no	−0.021	−0.60	0.935	0.924	no
E2 uncertainty below median	−0.065	−1.61	−0.052	−1.33	−2.09	no	0.005	0.15	0.935	0.924	no
E2 mixed signals below median	−0.037	−0.86	0.078	1.41	0.59	no	0.055	1.60	0.935	0.924	no
E2 materiality above median	0.040	1.29	0.037	0.92	1.53	no	0.023	0.69	0.935	0.924	no
E2 P(no material effect) below median	0.062	1.67	0.040	0.97	1.85	no	−0.012	−0.39	0.935	0.924	no
E3 latest article before 08:00 [seen on test]	0.002	0.05	0.018	0.38	0.34	no	0.029	0.81	0.935	—	no
E3 04:00 to 08:00, not 08:00 to 09:15	0.038	0.92	0.048	0.90	1.28	no	0.012	0.29	0.935	0.924	no
E4 Friday	−0.018	−0.44	−0.142	−2.50	−2.19	no	−0.034	−1.08	0.935	0.924	no
E4 Monday	0.026	0.59	0.047	0.98	1.13	no	−0.006	−0.20	0.935	0.924	no
E4 busy news day (ratio ≥ 1.5)	−0.040	−0.94	−0.095	−1.31	−1.61	no	−0.014	−0.38	0.935	0.924	no
E4 news count above median	0.042	1.13	0.102	1.82	2.31	yes	0.021	0.65	0.935	0.924	no
E4 earnings count above median	0.022	0.61	−0.010	−0.20	0.14	no	0.031	1.06	0.935	0.924	no
E4 FOMC, CPI or payrolls day	−0.053	−0.69	−0.054	−1.04	−1.14	no	0.052	1.03	0.935	0.924	no
E4 VIX at $t-1$ above median	0.016	0.42	−0.024	−0.47	−0.22	no	0.135	3.11	0.790	0.727	no
E4 |SPY overnight| above median	0.029	0.78	−0.051	−1.06	−0.40	no	0.007	0.24	0.935	0.924	no
E5 size tier 251 to 500	−0.018	−0.42	−0.056	−1.31	−1.24	no	−0.005	−0.16	0.935	0.924	no

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Table A38, continued (d1-ultra)

	val-A		val-B		Validation		Test, evaluated once				
Condition	Diff.	t	Diff.	t	Pooled t	Selected	Diff.	t	BH q	BH q, unseen	Confirmed
E5 dollar-volume tier 251 to 500	−0.079	−2.10	−0.055	−0.99	−2.01	no	−0.003	−0.10	0.942	0.933	no
E5 low usual coverage	−0.074	−1.90	−0.048	−0.90	−1.85	no	−0.030	−0.46	0.935	0.924	no
E5 σ above median	0.056	1.72	0.011	0.22	1.15	no	0.016	0.43	0.935	0.924	no
E5 direction agrees with z_{m20}	0.039	0.85	0.018	0.30	0.76	no	−0.007	−0.17	0.935	0.924	no
E5 price-move text	−0.013	−0.25	−0.013	−0.15	−0.26	no	−0.057	−1.10	0.935	0.924	no
E6 conviction top 30% [seen on test]	0.007	0.13	0.082	1.74	1.22	no	0.023	0.60	0.935	—	no
E6 Laya-Fin's magnitude above median	0.052	1.56	0.049	1.05	1.78	no	−0.016	−0.46	0.935	0.924	no
E6 d1-ultra's confidence above median	−0.055	−1.36	0.027	0.66	−0.49	no	0.010	0.33	0.935	0.924	no
E7 Laya-Fin and d1-ultra agree	0.101	2.02	0.073	1.45	2.46	yes	0.039	1.01	0.935	0.924	no
E7 three API models agree	—	—	−0.015	−0.30	—	no	0.029	0.81	0.935	0.924	no
E7 rank average of the two models	0.001	0.08	0.025	1.16	0.98	no	0.016	1.00	0.935	0.924	no
E8 $\lvert V\rvert\,\sigma$ above the hurdle	0.013	0.19	0.193	2.90	2.15	no	0.060	1.24	0.935	0.924	no

Exploratory (post hoc): requested after unblinding; no hypothesis, outside every family, no claim. NA-13b, the operator's study of when the D0 signal works (PREREG §15, NA-13b row): thresholds and medians from val-A, the conditions selected on val-A and val-B by a rule fixed before the test, the test evaluated once.

Diff.: the inside-minus-outside difference of the primary outcome $\text{sign}(\hat{N})\, z_{id}$ (typical daily moves): at the event level the mean of the daily differences with a day-clustered t; at the day level (Friday, Monday, the day's news and earnings counts, macro days, VIX, SPY's overnight move) the difference of the sessions' means with Welch's t; for the rank average the mean daily difference to the model's own sign. Selected on validation by the operator's rule, fixed before the test: positive with $t \geq 1$ in val-A and in val-B and a pooled $t \geq 2$ (the three-model agreement: val-B alone, $t \geq 2.5$). The test, once: two-sided p, Benjamini–Hochberg at $q = 0.10$ over the family and over the family without the conditions seen on test (unseen); confirmed: selected, the same sign on the test and BH $q < 0.10$. [seen on test]: earnings day, latest article before 08:00 and conviction top 30%, which the test had shown before this study; reported, never selected. †Vendor field: a label the vendor produced retrospectively, possibly look-ahead-contaminated. —: the three-model agreement has no val-A run (span-02 and drex-v2.0 have no val-A production run), so no pooled t; a condition seen on test has no unseen q.

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A40: Exploratory (post hoc): the tradable versions; on the test none beats its placebo (NA-13b).

		Sharpe ratio				Names			
Book	Exit	Gross	5 bp	10 bp	Break-even	Long	Short	Placebo	DSR
Test: Laya-Fin									
E2 numbers, not narrative	C_t	−0.73	−3.15	−5.54	−1.5	17.4	14.1	41	<0.001
	$t{+}1$	−1.62	−3.13	−4.63	−5.3	17.4	14.1	15	<0.001
E2 materiality above median	C_t	0.19	−3.24	−6.63	0.3	22.7	19.9	66	<0.001
	$t{+}1$	−1.29	−3.42	−5.55	−3.0	22.7	19.9	14	<0.001
E2 P(no material effect) below median	C_t	−0.57	−3.93	−7.27	−0.8	30.3	24.7	38	<0.001
	$t{+}1$	−1.24	−3.34	−5.42	−3.0	30.3	24.7	18	<0.001
E4 busy news day (ratio ≥ 1.5)	C_t	0.35	−2.77	−5.88	0.6	30.2	23.7	64	<0.001
	$t{+}1$	−0.86	−2.47	−4.07	−2.7	30.2	23.7	28	<0.001
E6 Laya-Fin's magnitude above median	C_t	0.67	−2.75	−6.15	1.0	33.0	28.6	67	<0.001
	$t{+}1$	−0.52	−2.44	−4.35	−1.4	33.0	28.6	37	<0.001
E9 numbers and material	C_t	−0.71	−2.45	−4.18	−2.0	10.7	11.2	31	<0.001
	$t{+}1$	−2.38	−3.58	−4.77	−9.8	10.7	11.2	5	<0.001
E8 $\lvert V\rvert\,\sigma$ above the hurdle	C_t	0.88	−0.38	−1.63	3.5	4.9	7.3	73	<0.001
	$t{+}1$	1.47	0.74	0.01	10.0	4.9	7.3	85	<0.001
Test: d1-ultra									
E4 news count above median	C_t	0.36	−2.81	−5.25	0.5	133.9	37.4	60	<0.001
	$t{+}1$	0.82	−1.20	−3.17	2.0	133.9	37.4	77	<0.001
E7 Laya-Fin and d1-ultra agree	C_t	0.62	−2.84	−6.32	0.9	48.2	14.7	66	<0.001
	$t{+}1$	−0.49	−2.39	−4.30	−1.3	48.2	14.7	30	<0.001
E8 $\lvert V\rvert\,\sigma$ above the hurdle	C_t	0.05	−1.66	−3.36	0.2	11.8	8.2	54	<0.001
	$t{+}1$	−0.70	−1.83	−2.95	−3.1	11.8	8.2	30	<0.001
Val-B, in-sample for the selection: Laya-Fin									
E2 numbers, not narrative	C_t	5.43	2.52	−0.44	9.3	18.2	13.2	100	<0.001
	$t{+}1$	4.00	1.87	−0.26	9.4	18.2	13.2	97	<0.001
E2 materiality above median	C_t	5.50	0.69	−4.13	5.7	24.9	20.0	100	<0.001
	$t{+}1$	2.83	−0.27	−3.37	4.6	25.3	20.3	88	<0.001
E2 P(no material effect) below median	C_t	4.60	−0.14	−4.86	4.8	29.2	21.4	98	<0.001
	$t{+}1$	4.44	1.12	−2.19	6.7	29.2	21.4	98	<0.001
E4 busy news day (ratio ≥ 1.5)	C_t	5.99	3.44	0.84	11.6	16.8	12.6	100	0.001
	$t{+}1$	5.05	3.46	1.85	15.7	16.8	12.6	100	0.005
E6 Laya-Fin's magnitude above median	C_t	5.56	0.65	−4.29	5.7	32.8	25.7	100	<0.001
	$t{+}1$	3.41	0.38	−2.65	5.6	33.4	26.1	94	<0.001
E9 numbers and material	C_t	6.78	4.23	1.65	13.2	12.8	11.1	100	0.004
	$t{+}1$	4.12	2.27	0.41	11.1	12.8	11.1	98	<0.001
E8 $\lvert V\rvert\,\sigma$ above the hurdle	C_t	1.28	−0.09	−1.46	4.7	4.1	5.6	76	<0.001
	$t{+}1$	−0.22	−0.85	−1.49	−1.7	4.1	5.6	46	<0.001
Val-B, in-sample for the selection: d1-ultra									
E4 news count above median	C_t	2.25	−2.39	−6.71	2.4	117.4	41.6	78	<0.001
	$t{+}1$	2.32	−0.22	−2.76	4.6	117.4	41.6	80	<0.001
E7 Laya-Fin and d1-ultra agree	C_t	1.69	−2.20	−6.09	2.2	46.8	15.4	78	<0.001
	$t{+}1$	3.12	1.05	−1.02	7.5	47.4	15.6	91	<0.001
E8 $\lvert V\rvert\,\sigma$ above the hurdle	C_t	2.92	0.66	−1.61	6.4	8.8	7.7	95	<0.001
	$t{+}1$	3.94	2.57	1.20	14.4	8.8	7.7	96	0.002

Exploratory (post hoc): requested after unblinding; no hypothesis, outside every family, no claim. NA-13b, the operator's study of when the D0 signal works (PREREG §15, NA-13b row): thresholds and medians from val-A, the conditions selected on val-A and val-B by a rule fixed before the test, the test evaluated once.

The books of the selected pairs, E8 and E9 (NA-13's engine): each session's inside events, long those the model reads up and short those it reads down, equal weight within a side, β-hedged with SPY; entry at the open, exit at the close (C_t) or the next close ($t{+}1$). Annualized Sharpe ratios, gross and net of 5 and 10 bp per side on both legs and the hedge; break-even in bp per side (annualized gross return ÷ 252 ÷ daily turnover); names: the mean longs and shorts of a session with positions; placebo: the gross Sharpe's percentile among 100 random sign assignments of the same events (none reaches the 90th on the test); DSR at 10 bp with the 1,620 trials of the ledger after NA-13b. The val-B books are in-sample for the selection. The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).

Table A41: Exploratory (post hoc): the operator's first look by band of conviction; the mean signed overnight move is largest in the top 10% in every split (NA-13b).

			Mean signed move		
Split, band of $\lvert\hat{N}\rvert$	Events	Hit rate	z_{cc}	z_{on}	z_{id}
Laya-Fin					
val-A, bottom 50%	4,188	0.511	0.034	−0.005	0.044
val-A, 50–70%	1,675	0.497	0.008	−0.068	0.052
val-A, 70–90%	1,675	0.536	0.117	0.114	0.076
val-A, top 10%	838	0.562	0.459	0.569	0.106
val-B, bottom 50%	2,446	0.500	0.017	0.062	−0.022
val-B, 50–70%	1,097	0.508	0.054	0.084	0.033
val-B, 70–90%	1,241	0.521	0.085	0.024	0.109
val-B, top 10%	722	0.542	0.351	0.332	0.123
test, bottom 50%	6,561	0.506	0.036	0.027	0.025
test, 50–70%	2,724	0.503	−0.006	0.031	−0.015
test, 70–90%	2,753	0.509	0.027	0.102	−0.003
test, top 10%	1,710	0.526	0.172	0.191	0.068
d1-ultra					
val-A, bottom 50%	4,188	0.514	0.028	−0.007	0.031
val-A, 50–70%	1,675	0.509	0.045	0.119	−0.003
val-A, 70–90%	1,675	0.531	0.149	0.257	0.048
val-A, top 10%	838	0.587	0.799	1.516	0.004
val-B, bottom 50%	2,449	0.508	0.039	0.038	0.025
val-B, 50–70%	1,115	0.512	0.039	0.141	−0.037
val-B, 70–90%	1,329	0.527	0.192	0.350	0.013
val-B, top 10%	613	0.566	0.754	1.157	0.140
test, bottom 50%	6,201	0.499	0.028	0.042	0.019
test, 50–70%	2,753	0.493	−0.028	0.080	−0.059
test, 70–90%	3,176	0.503	0.111	0.299	−0.013
test, top 10%	1,618	0.556	0.592	0.930	0.101

Exploratory (post hoc): requested after unblinding; no hypothesis, outside every family, no claim. NA-13b, the operator's study of when the D0 signal works (PREREG §15, NA-13b row): thresholds from val-A. The operator's assistant computed the first of these statistics on all three splits, the test included, before NA-13b (the band check of Table A41 and the first four ICs of Table A37); they shaped the study, they are reproduced exactly here, and no threshold is chosen from them.

The operator's assistant's first look, disclosed with the operator's prompt and reproduced: bands of the conviction $\lvert\hat{N}\rvert$ at its val-A 50th, 70th and 90th percentiles; the hit rate against z_{cc} and the mean signed z_{cc}, z_{on} and z_{id} (typical daily moves), pooled over the events of each split outside the embargo.

The API models fail the §8.3 contamination screens, so their results are model-specific (§1.3).